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Power transformers and reactors

Condition assessment of power transformers

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Condition assessment of power transformers

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Executive summary

An asset manager may be responsible for a large fleet of transformers, each of which may have a large amount of information related to the condition of the asset. Consequently, determining which transformers most need attention can be extremely challenging.

Transformer Assessment Indices (TAIs) can be generated by calculating a Transformer Assessment score for each transformer in the fleet, then using the assigned scores to rank the transformers. Assuming the scores are calculated using an appropriate and sensible method, and using timely and accurate data, the asset manager can easily identify those transformers which most urgently need attention or intervention.



The focus of Working Group A2.49 was to investigate information used to derive Transformer Assessment Indices, its' consolidation, and the uses to which the output can be utilized.

Many asset managers currently use a 'Health Index' for this purpose. However, in many cases the index does not provide any indication of how quickly the worst transformers on the list need to be actioned, nor does it provide any indication of the type of action needed i.e. replace, repair or refurbish. Many indices also fail to provide any indication of the confidence that the asset manager should have in the assessments in the index.

Furthermore, these indices have generally been designed to provide an indication of each transformer's 'health.' Many asset managers use their 'health index' to help determine which transformers in their fleet to replace. However, some "unhealthy" transformers can be (relatively) easily repaired and therefore do not need to be replaced. A 'health' index may therefore not be the ideal tool to determine transformer replacement.

Chapters 1 to 7 introduces a process that can be used to assess a transformer and to develop Transformer Assessment Indices to suit the needs of the user. It also introduces the concept of the scoring matrix, which can be developed and used to ensure that scores are allocated to each transformer failure mode or mechanism being assessed in a consist way and with regard to the time scale required for action. A number of different methods of generating both numeric and non-numeric transformer assessment scores are also explained. Annex B provides a number of examples illustrating how to generate different types of Transformer Assessment Indices and how to use the different scoring methods. This section of the technical brochure also discusses:

- Options for dealing with the uncertainty in some information that will no doubt result for any user with a large fleet of transformers
- How to utilize on-line monitors for assessment and in Transformer Assessment Indices
- Issues that an asset manager should consider before acting to correct transformers in poor condition, and
- Methods that can be used to provide an estimate of the probability of failure.

The information and methodology presented in Chapters 1 to 7 can also be applied to assess and develop and assessment indices for other assets.

Chapters 8 to 12 and Annex A deals with the sub-components of a transformer and discusses:

- Failure modes and mechanisms, and
- Methods of diagnosing the failure modes and mechanisms for each sub-component.

This section is mostly existing knowledge. Annex A provides tables of diagnostic information that are formatted to allow assessment using the methods described in the technical brochure. The information is from IEC & IEEE guides, CIGRE publications, CIGRE experts, and other industry experts. Users can use these values as a starting point when assessing a transformer but should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in the scoring matrix.

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1. Introduction – Fundamentals of transformer condition assessment

Power transformers are expensive and critical equipment in power systems and play a significant role in the transmission and distribution of electricity. Although transformers are generally reliable pieces of equipment, failures do occur, and there are many degradation mechanisms operating in components and sub-systems that will ultimately limit the useful operating life.

It is important for transformer users and asset managers to be adequately equipped to assess the condition of a fleet of transformers in service in their utility/organisation as a basis for making critical decisions about operations, repair/rectification of minor failures, refurbishment or replacement. Users and asset managers need to understand all the failure modes of transformers to pinpoint the part of the transformer affected, and to identify appropriate responses. Broadly there may be failures in active parts of transformers or their accessories due to dielectric, mechanical or thermal breakdown. Some sub-components also have their own unique failure mechanisms.

This brochure aims to set out the failure modes, the tests and diagnostic methods that can be used to detect them, and methods of combining the available data into useful information in the form of assessment indices that can form the basis for decision making and intervention prioritisation in transformer asset management.

1.1 The problem for transformer asset managers

Whilst individual network and business requirements vary, transformer asset managers are generally trying to achieve the required levels of safety and reliability from their fleet of transformers at minimum cost. Knowledge of condition is therefore essential for efficient transformer asset management decisions. Without this information only the most basic activities are possible; such as time-based maintenance, replacement before end of life, or repair after failure.

For asset managers, determining the minimum required budgets for maintenance and replacement and determining the most effective and targeted way of spending, is an important task. Often this must be justified to stakeholders and regulators in an increasingly competitive environment.

Good knowledge of plant condition is an important step to good asset management. The relative costs of replacement, failure and condition assessment will also need to be assessed to determine the most economic approach. In general, for large transformers, the potential consequences of failure and the high cost of replacement, make a sophisticated condition and risk-based approach very worthwhile.

1.2 Transformer deterioration and failure modes

To identify transformer condition, it is first necessary to identify the failure modes and failure mechanisms; their consequences; and the interventions that can be implemented to change them. Much work has been done in this area by CIGRE and others, and individual knowledge of the transformer designs and operating conditions is necessary to identify weaknesses. In this technical brochure, the failure mechanisms have been broadly categorised for the active part of the transformer into thermal, mechanical and dielectric. On top of this, all of the other components of the transformer necessary for satisfactory operation need to be considered; in particular bushings (and other terminals), tap-changers, cooling systems, and the tank. The transformer dielectric fluid, (normally mineral oil) has a special significance because it is not only critical to transformer operation and has deterioration mechanisms, but it influences the rate of deterioration of other parts most notably the solid insulation, and it provides a vector for much of the condition information.

Although it is not always the case, when assessing a transformer, it is useful to think of each failure mode as independent and operating in parallel. Any one of the failure modes could, by definition, lead to failure irrespective of the operation of the other failure modes. Additionally, interventions can be thought of as affecting some failure modes and not others. For example, a deteriorated bushing could cause a failure in an otherwise new transformer, changing the bushing affects the bushing failure mode, but not the other modes. This concept is very important when developing transformer assessment indexes.

Although transformers come in a very wide variety of types and sizes, it is possible to provide a common set of tools and definitions to define and characterise the deterioration and failure mechanisms that are important.

1.3 Condition assessment

The essence of condition assessment is to identify the indications that can be used to determine (and quantify where possible) the extent of the degradation of the components and sub-components of the transformer. Much work has been done looking at individual mechanisms of degradation and failure, together with methods for detection and quantification. This brochure aims to bring this work together to provide a framework for overall assessment of transformer condition, which can then be used for asset management. The appropriate condition assessment regime will very much depend on the size and importance of the transformer, the availability of indicators, and the operating and business environment. Where information for detection of degradation is not available, it may be appropriate to use indirect indicators such as loading and age, and family related issues to assist with the transformer assessment.

Although the assessed condition of a transformer will be critical in asset management decisions, other factors are also generally considered. Refer to *6 Criticality and other condition considerations* for an overview of some of the other issues that should be considered before replacing a transformer or proceeding with any other major interventions.

Users must also promptly respond to correctly set alarms and act upon bad, or suspected erroneous test results or extreme observations. Ignoring a bad result may lead to catastrophic failure soon. **The user need not wait for full condition assessment to act if smoke is observed from transformers!**

1.3.1 Fleet screening

A detailed assessment of a power transformer involves extensive testing (electrical as well as the dielectric fluid), intrusive inspection and collection/compilation of other relevant data (operational as well as maintenance history including commissioning data). This data is then analysed by an expert based on relevant standards and expert knowledge. Before embarking on detailed assessment which is time consuming in terms of outage & availability of transformers in the grid, users may consider a screening assessment of the entire fleet to identify the problematic transformers for further testing and assessment prior to considering major asset management decisions like repair, refurbishment or replacement. Initial fleet screening may be based on a preliminary inspection, limited tests and other factors like loading, age, known history of manufacturing defects etc. The fleet screening may be automated or semi-automated to simplify the process.

It should be noted that an assessment based on limited available information is not complete, but it may give insight to developing critical problems and is better than NO assessment.

1.4 Fundamentals of Transformer Condition Assessment

Many of the common condition assessment techniques used for transformers assess only the transformer active part. For example, Dissolved Gas Analysis (DGA) from a dielectric fluid sample taken from the main tank monitors the condition of the active part. However, all components and sub-components of a transformer should be considered for a complete assessment. It should be noted that unexpected failures of bushings and on load tap changers (OLTC) can frequently lead to catastrophic failure of the transformer. Asset Managers should therefore consider on-line monitors or off-line testing and inspection for the critical items. Design information, family history, maintenance records and visual inspections can also be used to assess the components and sub-components of a transformer. Refer to Chapters 8-12 for details of the sub-components of a transformer, their failure modes/mechanism and how to assess them.

Assessment of Transformer Active part (magnetic core & windings) can be carried out in three ways - thermal assessment, dielectric assessment and mechanical assessment. Condition assessment techniques like DGA through testing at laboratory or on-line DGA monitors are used to analyse the symptoms of an incipient fault being developing in transformers based on IEEE/IEC standards. Reference may also be made to ISO 18095 - Condition monitoring and diagnostics of power transformers [1].

Thermal failure of transformers basically depends on faster than anticipated ageing due to excessive temperatures resulting from winding hot spot, metal hot spot and hot metal fault insulated with paper.

Dielectric Failure of transformers are a function of the degradation process of solid/liquid insulation due to high thermal or electrical stress, moisture in paper/dielectric fluid, manufacturing defects, and (semi) conductive particle (solid particles) contamination.

Mechanical Failure of transformers depends on the disintegration of the structure in core, windings and solid insulation due to one of many reasons like strong electromagnetic forces from short circuit

current flow through windings; mechanical deformation of windings due to shocks during transportation, erection or during system faults etc; and degradation of solid insulating material.

1.5 Role of Insulating Fluid in Transformer Condition assessment

The insulating fluid plays a key role as an insulating medium that acts as a barrier between windings and the tank and acts as a cooling medium that dissipates the heat generated within the windings and core. It also functions as a preserving media that protects the windings and solid insulation from the moisture and other physical contaminations.

The insulating fluid is often derived from hydrocarbon based “petroleum”. Hence it has a rich hydrocarbon molecular structure. This molecular structure is subjected to electrical, mechanical and thermal stresses during transformer operation. Hence the fluid decomposes and as a result generates hydrocarbon-based gasses. These gases dissolve into the dielectric fluid, and therefore can be measured. Some dissolved gases can escape if there is an escape path to the atmosphere via the conservator without a sealing mechanism. Thus, the dissolved gasses may reveal a fault condition in the transformer, if analysed properly by using techniques like DGA etc.

Materials like vegetable dielectric fluid, synthetic esters and synthetic materials like silicone fluid and other materials, are also being used as insulating fluid. These materials also decompose when subjected to electrical, mechanical and thermal stresses and generates certain type of gasses depending upon their molecular structure.

1.5.1 Dissolved Gas Analysis

Dissolved Gas Analysis (DGA) is a powerful diagnostic technique used to analyse dissolved gasses that are generated during the dielectric fluid and solid insulation decomposition process. Dissolved gas analysis (DGA) has been an industry standard for the detection and determination of faults in transformers for over 30 years. Developed in the late 1960s, DGA has been recognized worldwide as the main tool to prevent catastrophic failures of power transformers.

Insulating liquids used in transformers, high-voltage bushings and other electrical equipment require proper testing and maintenance to keep them in good condition. Liquid insulation that is not in good condition may cause damage to equipment and become a hazard to personnel. Deterioration of insulating liquid results from factors like oxygen, moisture, solid particle contamination, high thermal stresses etc.

1.5.2 Effect of Oxygen on insulating fluids

Moisture contamination is one of the most common causes of deterioration in the insulating quality of fluid. A less rapid but more serious deterioration due to the formation of acids and sludge, is caused by oxidation. In free-breathing transformers, the oxygen supply is virtually unlimited and oxidative deterioration is consequently faster than in sealed transformers. Leaky gaskets constitute a very real hazard due to both oxidation and moisture contamination. The rate of oxidation also depends on the temperature of the fluid; the higher the temperature, the faster the oxidative breakdown.

1.5.3 Moisture in transformers and insulating fluids

Water can be present in dielectric fluid in three states:

- In a dissolved form,
- ‘Bound’ water, where water molecules attach themselves to fibrous particles (including dust and dirt), as well as acids formed during the oxidation process of the fluid. Once it is ‘bound’ to another molecule aside from the fluid itself, it never moves, until filtration takes place; and
- As free water normally found at the bottom of the container holding the insulating fluid.

Demulsification occurs when the moisture dissolved in the fluid reaches the point of saturation. At this stage the phenomena known as “raining” inside the tank occurs. The result is water sinks to the bottom and forms a pool of free water. Emulsified water or water in the free state may be readily removed by filtering or centrifugal treatment. The filtration process can partially remove dissolved water if the filter papers are thoroughly dried before filtration and are replaced frequently.

The amount of moisture which can be dissolved in insulating fluid increases-as the insulating fluid temperature increases. Therefore, an insulating fluid treated at too high a temperature within a transformer, may lose a large percentage of its dielectric strength on cooling because some of the dissolved moisture (extracted from the windings at high temperature) forms an emulsion.

1.5.4 Insulating fluid Deterioration in Transformers

In transformers, sludge can adhere to the surfaces through which heat is normally dissipated; the sludge forms a resistance to the flow of heat from the insulating fluid to the cooling system and from the core and coils to the (cooler) Insulating fluid. If allowed to continue long enough, the sludge may even block the flow of insulating fluid through the cooling ducts. As a result, the transformer insulation gets too hot and is damaged, particularly between turns of the windings. Deterioration of the insulation may eventually lead to short circuits between turns and the breakdown of the transformer. When oxidation progresses to the point where sludge is being precipitated, the first step should be to attempt to remove the sludge from the transformer.

1.6 Transformer assessment indices

An asset manager may be responsible for a large fleet of transformers, each of which may have a large amount of information. Consequently, determining which transformers need attention can be extremely challenging. Many asset managers assign a score to each transformer based on some, or all of the condition assessment information that they have.

Transformer Assessment Indices (TAIs) are generated by calculating a Transformer Assessment score for each transformer in the fleet, then using the assigned scores to rank the transformers. Assuming the scores are calculated using an appropriate and sensible method, and using timely and accurate data, the asset manager can easily identify those transformers which most urgently need attention or intervention.

1.6.1 What is the index for?

It is important to be clear about the question that the index or score is going to be used to answer. The question, 'which transformers on my system are more likely to fail?' is very different to the question 'which transformers should be considered for replacement?'¹ The reason for the difference is that the first question would generally take account of all the possible failure mechanisms whereas the second question could exclude mechanisms that can be corrected by an intervention other than transformer replacement.

The failure modes/mechanisms used to determine the transformer assessment score can be chosen with an intervention in mind. For example, an assessment score used to determine if a transformer should be replaced might not include the condition of bushings if they can be replaced separately. On the other hand, a score used to determine the safety of personnel working in close proximity to the transformer would certainly include the bushings.

Throughout the document a distinction is drawn between failure modes/mechanisms which are underlying conditions that can lead to failure, and diagnostic results indications and other measurements that are used to assess the underlying conditions that are not likely to impede operating functionality. The following figure shows how diagnostic results are used to indicate the progress of a particular failure mechanisms, and how scores for each failure mechanism can be combined into different indices. The diagram is an example only, to highlight the possible relationships between diagnostics, failure modes/mechanisms and indices. When designing an index, users should determine the failure modes/mechanisms relevant to their index, and the information available for assessment. Refer to [Chapter 2. Transformer assessment indices](#) for further information.

¹ A TAI generally ranks transformers based on their assessed condition only. Refer to 6 for details of other issues that should be considered before replacing a transformer.

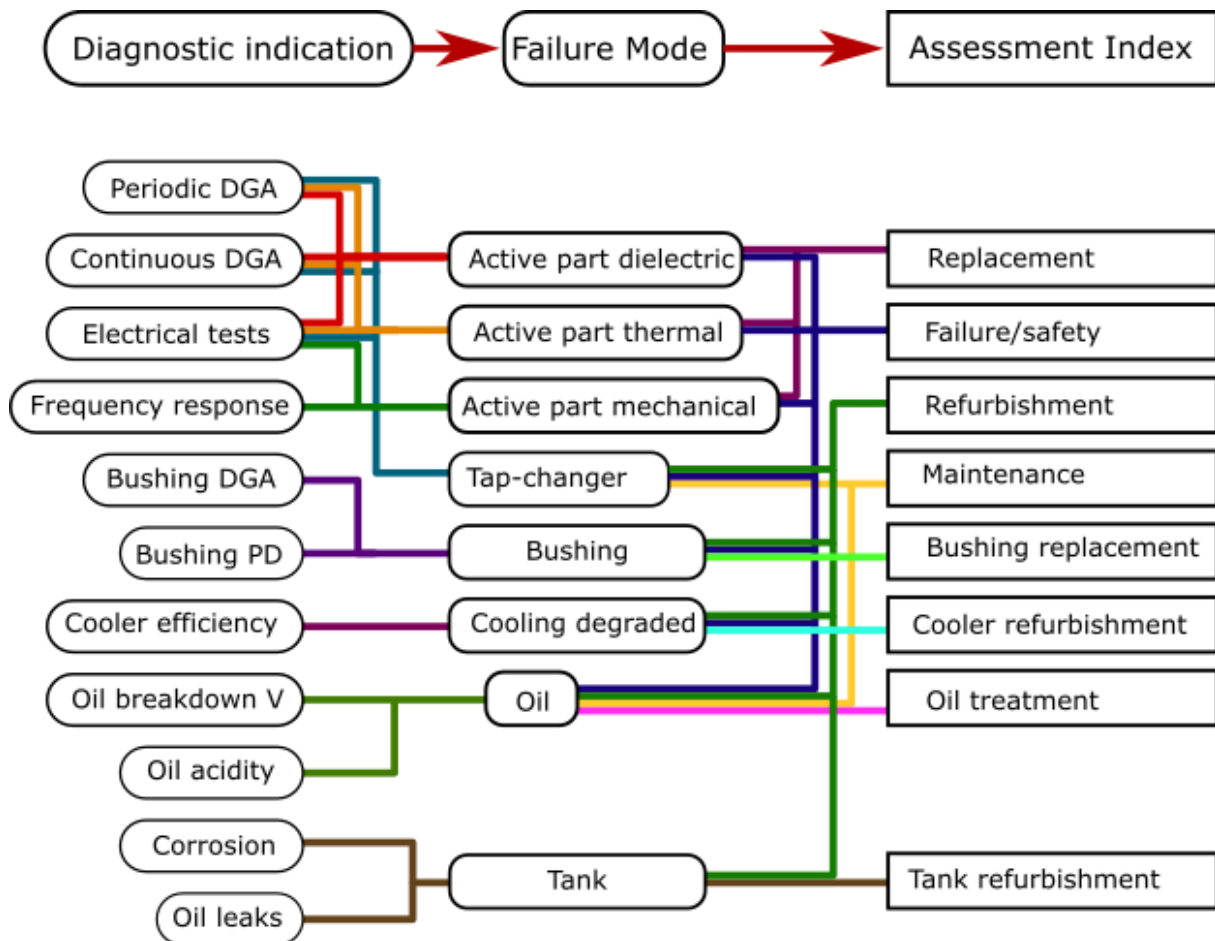


Figure 1-1 Relationship of Failure Modes to Assessment Indices and Diagnostic tools

Many TAIs are designed to consider and prioritise transformer replacement, i.e. a Replacement Index. However, the same general approach can also be used to identify and prioritise other interventions. Each TAI would use a different subset of the available condition information. Refer to section 2.2 Types of Transformer Assessment Indices for more information.

1.6.2 Developing a transformer assessment score

Once a user has determined the purpose of the TAI and the failure modes/mechanism to be included in the index, test data and other available information can be used to assess the included failure mode or mechanisms. It is important to distinguish between failure mechanism that can cause failure e.g. partial discharge, and the indicators used to detect and assess them e.g. the level of a dissolved gas in the dielectric fluid. Although many transformer assessment schemes go directly from the indication to the score, it is hard to correctly interpret and objectively weight the indications without considering the failure mechanisms they indicate.

It is essential that assessments are consistent, so users are encouraged to develop a scoring matrix so that scores can be allocated to each failure mechanism of each transformer consistently. Refer to section 2.3 Steps to develop a TAI for more information.

Once the indicators are attached to a failure mechanism and used to determine the likelihood of failure in chosen timeframe, it becomes easier to combine the progress of different failure mechanisms into an overall transformer assessment score.

A TAI or score is built up from the assessed condition of the failure mechanisms relevant to the index. The condition assessment scores need to be combined in a way that, allows a transformer with a single failure mechanism close to failure to be highlighted, but also for a transformer with multiple small problems to be highlighted above a transformer with fewer small problems. This is not a straightforward process because of differing levels of information and perceived importance for different indications, and many schemes are used or have been proposed.

Probably the most rigorous method of combination is to consider the probability of failure from each failure mechanism and then multiply the survival rates from each failure mechanism together to arrive at the overall survival rate. This is analogous to considering the failure of a chain knowing the failure rate of each of the links. Unfortunately, the state of knowledge of the failure mechanisms and the quality of the indications is not sufficient to apply this method with certainty, but the general approach is effective in fulfilling the combination criteria above.

Weighted average schemes have been used to combine scores, but the choice of weightings is difficult and arbitrary weightings can introduce significant bias. By the nature of the averaging process the more failure mechanisms that are included, the harder it is to allow an individual bad component to have the prominence it deserves overall. *Refer to section 2.4* for further details of methods that can be used to generate a transformer assessment score.

The assessment score will only be as good as the data and information that is used to assess the transformer. In many cases, the data available for scoring will not be current. It is essential that the effect on the score of this missing or low-quality data is understood. In some cases, it will make a trivial difference to the overall score. In other cases, it could make a bad transformer look good; resulting in the transformer being overlooked for critical repair or replacement. Similarly, the missing or low-quality data can make a good transformer look bad. Implementing a “quality score” or some other indication of the quality of the data used (typically called “significance”) may help to understand the quality of data and the reliability or confidence level of a score. *Refer to 4 Dealing with uncertainty in information.*

Users should understand that many numeric scoring systems provide a score which is useful only for ranking transformers. The absolute value of the score may not clearly indicate the overall condition of the transformer or the urgency to address problems with the transformer. However, once a TAI is used to highlight the worst transformers in the fleet, the assessment used to derive the assessment score can be reviewed to determine appropriate actions and the time to act.

1.6.3 Assessment of Failure Mode/Mechanisms

Ideally, all failure modes/mechanisms would be assessed with the most precise and advanced condition assessment techniques. However, in many cases this is not technically or financially possible. Therefore, a selection needs to be made as to how a failure mode/mechanism is assessed. There are multiple ways of assessment and each type of assessment has a corresponding level of accuracy and uncertainty. In principle the following assessment techniques are available, in order of increasing levels of uncertainty:

Table 1-1 Condition Assessment Techniques

A	Advanced condition assessment techniques	Advanced measurements are used to gain information on the actual condition of a part of a transformer. Generally, more advanced techniques provide more accurate information than basic measurements. Chapters in this brochure have identified a number of these techniques to be used for each part of the transformer and each failure mode.
B	Basic condition assessment techniques	Basic measurements are used to gain information on the actual condition of a part of a transformer. It is expected that basic techniques provide a reasonable indication. Chapters in this brochure have identified a number of these techniques to be used for each part of the transformer and each failure mode.
C	Inspections	Visual inspections are used to gain information on the actual condition of a part of the transformer. The detection is limited to degradation that is visible, audible, can be smelt or felt. The inspections are subjective to human interpretations.

D	Logical reasoning	Whenever a measurement or inspection is not available, logical reasoning can be utilised, using indirect information, to gain information about the most likely condition of a part of the transformer, e.g. • Using relevant experiences of brand / types / materials used • Using environmental information and knowledge of physical processes • Using statistical inference of known parameters in other comparable transformers
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In the following chapters, multiple different assessment techniques have been identified for different failure modes/mechanisms. Preferably, and whenever possible, one or more advanced or basic condition assessment techniques should be included. Whenever multiple condition assessment techniques will be used in a failure mode assessment the confidence level of the assessment may increase if all techniques come to comparable results. Similarly, trend analysis techniques using multiple measurements through the transformer's history may provide increased confidence in the failure mode assessment.

Whenever multiple condition assessment techniques have been used to assess a given failure mode, the most accurate data technique should be selected for use in the transformer assessment. A more accurate indicator should overrule less accurate indicators, if indicating the same failure mode. No double-counting should occur. When multiple indicators with the same accuracy have been for the same failure mode, either an average or a worst-case approach can be used.

For example, a Partial Discharge problem can generally be detected from the DGA results, but other types of Partial Discharge e.g. ultrasound measurement could also be used to identify the same problem. The DGA result is typically used as the first indicator. Whenever partial discharging is detected using DGA, a more accurate ultrasound measurement can be used to gain more accurate information on partial discharging. In this case the ultrasound measurement should override the DGA results for partial discharges.

Note however, that some condition assessment techniques may provide information on multiple failure modes. Therefore, the same information can be used to assess multiple different failure modes.

Condition assessment techniques may not always be practical due to the cost or technical limitations. Ideally information from logical reasoning should be included in the assessment of the failure mode/mechanism. Examples of logical reasoning will be discussed in more detail in Chapter 4 Dealing with uncertainty in information.

Users should also consider how often the assessment information needs to be updated. This will be dependent upon the expected speed of degradation of failure modes/mechanism and the cost to implement. Some failure modes show sudden very rapid degradation after prolonged periods of normal operation e.g. degradation of bushings, whilst other failure modes show more gradual degradation, e.g. paper degradation.

1.7 Operational feedback

It is crucial to the entire process that information on degradation and failure is gathered from operational experience. The most valuable information can be obtained from post-mortem forensic examination of transformers when they are scrapped. For most scrapped transformers, it is possible to determine the extent of degradation by observation and testing of critical components, most importantly the solid insulation. If they have failed in service, then the failure mechanism needs to be understood. In any case, the information can be used to help to validate and calibrate the condition assessment methods used and to determine whether there are any previously unrecognised failure mechanisms in play. Please refer to CIGRE TB 735 [2] titled Transformer Post-Mortem Analysis.

2. Transformer assessment indices

An asset manager may be responsible for a large fleet of transformers, each of which may have a large amount of information. Consequently, determining which transformers need attention can be very difficult. Many asset managers assign a score to each transformer based on some, or all of the condition assessment information that they have. The transformers can then easily be ranked according to their score.

This chapter discusses some of the different methods for developing a Transformer Assessment Index (TAI) and details the advantages and disadvantages of the various methods discussed.

Guidance is also provided to allow the asset manager and the transformer specialists to determine and develop the most suitable TAI for their fleet.

2.1 Using Transformer Assessment Indices for asset management decisions

To manage a fleet of transformers an asset manager needs to determine which transformers would most benefit from intervention or action, and then decide the most appropriate type of intervention or action for each transformer.

For example, the asset manager may consider replacement of the transformer, refurbishment or repair. During this process the cost of possible actions or interventions needs to be balanced against the current performance and risks of the transformer, and the improvement(s) expected from an intervention. A TAI is one method that may help identify the transformers which most need intervention or action. A well-designed TAI which has been designed for a specific purpose often suggests an appropriate action for consideration.

Furthermore, other factors need to be considered for each transformer before deciding on plans and priorities for replacement or repair e.g. refer to Chapter 6 *Criticality and other condition considerations* for further details of how these issues can be considered in conjunction with a TAI.

It is also essential that users of a TAI have some understanding of the level of confidence in each assessment that has been undertaken prior to assigning a transformer assessment score to each transformer. In any large fleet, some of the data and information required for assessment of some transformers in the fleet will be missing, inaccurate or out of date, and the user must ensure that assessments based on such low-quality data are highlighted. If the data used for assessment is incomplete or incorrect, a transformer that needs urgent intervention may not be highlighted by the TAI. Refer to Chapter 4 *Dealing with uncertainty in information* for a detailed discussion of this topic.

Users must also be aware that a TAI (particularly one that is poorly designed) may hide, or mask bad results. For example, consider a transformer which is generally in good condition but has a bushing in poor condition. In many TAIs, particularly those that produce a simple numeric output, the total score could show that the transformer is in average or good condition. The bad bushing result has been masked by the good scores assigned to other failure modes being considered.

Some users believe that the masking effect can be easily overcome by using weighting factors. If the bushing score was heavily weighted, a bad score would result for a transformer with a bad bushing, although other failure modes being considered have good scores. The problem is that the masking effect is now exaggerated for other failure modes which have not been heavily weighted. Problems (not related to the bushing) which may lead to failure and require urgent attention are further masked by the weighted system. The urgency associated with a problem can be easily lost in a poorly designed TAI.

Users need to carefully consider the effect of masking when choosing a scoring system and designing their TAI. Furthermore, users must understand that a TAI is not intended to be used for “Alarm Management.” When a bad test result is received, or there is any other indication that a transformer needs urgent attention, the problem should be addressed regardless of the TAI score.

2.1.1 Fleet screening or detailed assessment

Some asset managers choose to use an index for Fleet Screening; a method where they use readily available information to do a quick assessment of each transformer in their fleet and then rank the transformers based on their scores. The assessment process may be automated or semi-automated to simplify the process. The transformers that are identified through this process would then, most

probably, be subjected to a furthermore detailed assessment before any major decisions were made to confirm the outputs of the Fleet Screening.

Other asset managers may have a transformer specialist perform a very detailed and thorough analysis of each transformer and then assign transformer assessment scores. Major decisions could then be made based on the transformers' score. If an appropriate and sensible scoring system is used, a score derived through this more thorough method could help to support a justification to management and/or a regulator to replace or refurbish a transformer.

Other methods combine both above-mentioned methods, where a basic screening is performed with readily available information but may also use smart algorithms and additional data when it is available. Although this may give a better result than a more basic fleet screening TAI, a full assessment should generally be performed before major decisions are made.

2.2 Types of Transformer Assessment Indices

TAIs are generated by calculating a score for each transformer in the fleet, then using the assigned scores to rank the transformers. If the scores are calculated using an appropriate method and using timely and accurate data, the asset manager can easily identify those transformers which most urgently need attention or intervention.

It must be noted that many numeric scoring systems result in a value that allows transformers to be ranked; but the score itself is often arbitrary. In these situations, a user should not assume that a transformer that has a score of 84 is twice as good (or bad) as a transformer with a score of 42. Nor should they assume that the remaining life or the probability of failure of one transformer is twice that of the other. In most cases, the number only provides a means of ranking the transformers.

However, some users will utilise statistical techniques to try to estimate the probability of failure or the remaining life from the transformer's score. In this case, the TAI may be scaled to represent the parameter being estimated.

A TAI can be designed to identify the transformers most in need of the asset manager's attention or intervention. Such an index could be referred to as a Reliability Index, or a Health Index.² When using this type of TAI to identify transformers most in need of attention, the most appropriate type of intervention should then be determined by further technical analysis, consideration of the effectiveness and the costs associated with the interventions being considered, and detailed consideration of the criticality and other issues. Refer to *6 Criticality and other condition considerations*.

Alternatively, users can develop TAIs for specific purposes. Indices can be developed to determine the transformers that could most benefit from repair, refurbishment, or replacement. The purpose of the index would help identify a possible intervention for consideration. For example, refurbishment could be considered for transformers highlighted by a "Refurbishment Index." Note that criticality and other issues still need to be considered as per Chapter 6 *Criticality and other condition considerations*.

Note that a well-designed index and scoring system will generally identify those transformers which most urgently need attention. The timescale for action will be dependent upon the timescales assigned in the scoring matrix used to assess the failure modes and failure mechanisms considered as part of the index.

It is essential that users understand (and communicate to all other users) the purpose of the index, and that the index is not used for any other purpose without careful consideration. For example, a Reliability / Health Index³ is used to identify transformers that are most likely to fail. An asset manager should not simply use their transformer replacement budget to replace the top 10 transformers on the list. It is possible that some of the transformers on the list can be easily and economically repaired, whilst other transformers further down the list need replacement.

2.2.1 Reliability / Health Index

This index is used to rank transformer based upon likelihood of failure or expected time to failure, and those that have degraded from their original condition. This index ranks all transformers that require

² Note however that Health Index is a commonly used term that has been used by many users to describe indices, which may not directly relate to the health of the transformer.

³ Refer to *Section 2.2.1*, which explains why this type of Index does not directly relate to the reliability of the transformer.

any type of intervention, based on a sense of technical urgency. Users need to identify the most appropriate intervention for transformers that are identified as needing attention.

Note that a Reliability or Health Index will not generally result in a score that directly relates to the reliability or health of the transformer. If the index is well-designed, it will assist in ranking the transformers so that those that are less healthy or less reliable are highlighted.

2.2.2 Replacement Index

This index is used to identify the transformers that most need to be replaced. The scores for each transformer are based on:

- Irreversible failure modes and / or
- Single or multiple reversible⁴ failure modes which in combination, cannot be economically repaired. This can only be achieved by some consideration of economic factors during the design of the index.

Users will need to decide if they wish to include components such as bushings, tap changers and insulating fluids in their Replacement index. These components can be repaired or replaced, although some users may consider it more economical to replace the transformer, particularly if more than one of these components is in bad condition.

2.2.3 Repair Index

This index is used to rank transformers that would most benefit from repair or non-essential maintenance. The scoring of transformers is based upon reversible failure modes which can be corrected by repair or by performing maintenance. Users may need to do a more detailed financial and condition assessment on transformers highlighted by this type of index, before proceeding with repairs to ensure that the transformer can be effectively and economically repaired. A Replacement Index could also be used in conjunction with a Repair Index to ensure that transformers that have been identified as needing repair do not also have irreversible failure modes or have not already been identified as being uneconomical to repair.

Users could also consider using a Reliability index in conjunction with a Replacement Index to identify transformers that need repair. Any transformer that is identified as being unreliable that does not need replacement, should be considered for repair.

2.2.4 Refurbishment Index

This index is used to rank transformers based upon their need and / or technical feasibility for refurbishment. This index focusses, typically on multiple reversible failure modes or failure modes which involves excessive cost interventions and lifetime extension. It is possible that transformers identified may not be economical to repair, and further analysis will lead to an economic decision to replace these transformers rather than repair them. A well-designed refurbishment index will however ensure that those transformers which have well developed irreversible failure modes are not identified as being suitable for refurbishment. Refer to Chapter 3 *Refurbishment index* for information about how to develop a Refurbishment Index.

2.2.5 Composite Index

This approach combines the outcomes of the replacement and refurbishment or repair requirements. It includes a technical assessment and generic financial considerations to allow it to decide the best solution, considering the maintainability of failure modes in the transformer. This approach combines the different types of indices in one index, e.g. highlighting maintainable indices and non-maintainable indices in one composite index. The index identifies the reversibility of a failure mode and suggests replacement or maintenance. A composite index may also include generic financial considerations to allow it to decide the best solution. These financial discussions are beyond the scope of this Technical Brochure.

⁴ "Reversible failure modes" refers to failure modes that can be corrected or partially corrected by repair or replacement of a component or sub-component. e.g. failure modes associated with a bushing could be considered reversible as the bushing could be replaced. Failure modes which are corrected by repair are often not fully reversed.

2.2.6 Mitigation Index/Score

Whilst typical indices assess the state of an asset, the approach can be adapted to assess the technical feasibility of a refurbishment or repair. When a repair or refurbishment score $RS_{Original}$ indicates that a transformer should be considered for refurbishment or repair, the user can then calculate $RS_{Mitigated}$, by using the same index and scoring system to calculate a score based on the assumed condition of the transformer after the intervention has been successfully completed.

Some users may consider $RS_{Mitigated}$ to help determine if the repair or refurbishment is justified i.e. the proposed repair or refurbishment would not be considered unless $RS_{Mitigated}$ falls below a predetermined threshold.

A Mitigation score can also be calculated as follows: ⁵

$$MS = RS_{Original} - RS_{Mitigated}$$

Ideally a higher value for MS indicates that the repair or refurbishment has made a more significant difference to the condition of the transformer. However, the value of a mitigated score will be somewhat dependent upon the scoring system used. Many numeric scoring systems result in a value that allows transformers to be ranked; but the score itself arbitrary. Users must understand the scoring system to understand the relevance and value of using mitigated scores.

2.3 Steps to develop a TAI

The basic steps to develop a TAI are listed below:

1. Determine the purpose of the Transformer Assessment Score and Index
2. Identify the failure modes to be included in the TAI
3. Determine how each failure mode will be assessed
4. Design a calibrated system for categorising failure modes (scoring matrix)
5. Calculate a TAI Score for each Transformer

It should be noted that some users, particularly those with smaller fleets, or those performing a detailed assessment on a single transformer, may not want or need an index to allow transformers to be ranked. In this case Steps 1-4 can be followed. This will result in the relevant failure modes of each transformer being assessed and categorised in a consistent way.

2.3.1 Determine the purpose of the Transformer Assessment Score and Index

The design of a TAI is highly dependent on its purpose. Therefore, the first step in the development of a TAI is to determine its purpose.

Users should decide which categories of transformers need to be identified and what business decision need to be supported. For example, a user may want a TAI that identifies those transformers that should be prioritised for replacement; and may wish to use the results from the TAI to support the business case for replacement of these transformers. Users should also decide if the TAI is to be used for Fleet Screening or Detailed Assessment. Refer to *section 2.1.1*.

For examples of TAIs that may suit users' needs, refer to *section 2.2*. Users may also choose to develop multiple indices for different purposes e.g. a Replacement Index and a Refurbishment Index.

The purpose of the TAI should be carefully discussed and agreed with its users and asset managers. The purpose of the index should generally be closely aligned with the decisions to be made by the asset manager.

2.3.2 Identify the failure modes to be included in the TAI

When the purpose of the TAI is understood and agreed, the failure modes to be included in the index can be determined. The failure modes to be included will generally be aligned to the relevant business drivers. For example, a user may choose to include only information indicating the condition of non-reversible failure modes of the main tank into a TAI developed to highlight the transformers that should most urgently be replaced. This would be appropriate if the user generally chooses to repair or replace

⁵ This formula assumes that the scoring system assigns smaller numbers to transformers in better condition i.e. the improved score after mitigation will result in a lower number.

bushings, tap-changers or other components of the transformers in poor condition, rather than replace the transformer.

Users must, however, very carefully consider the failure modes that are to be included in their TAI. Users may choose not to include reversible failure modes in a Replacement Index. Although the user would generally replace components or repair the reversible failure modes, it may not be economical to do so if the transformer has multiple reversible failure modes, or if the transformer has a limited remaining life. For example, a user may not find it economical to replace a tap-changer and HV bushings, unless the transformer is otherwise in good condition.

There are two general options for dealing with this situation:

- a. Design the Replacement Index such that reversible failure modes are considered, but only have a significant effect on the assessment score when:
 - There are multiple reversible failure modes that need to be addressed; or
 - A reversible failure mode is obviously not economical to repair; or
 - The overall condition of the transformer is poor (i.e. low residual value) so that addressing reversible failure modes is not economical
- b. Do not include reversible failure modes into the replacement index; but otherwise identify transformers that have active reversible failure modes that require intervention and determine if these interventions can be economically justified. Transformers that require interventions that cannot be economically justified should be added to the list of transformers that are prioritised for replacement.

Although this example deals with problems associated with a Replacement Index, similar problems may occur when developing other TAIs. In general, this can be resolved by developing a more complex TAI e.g. a composite TAI or using one or more simple TAIs to develop a preliminary result which may need further analysis.

2.3.3 Determine how each failure mode will be assessed

Ideally all failure modes included in a TAI would be assessed with the most accurate and advanced condition assessment techniques. However, for technical and financial reasons, this is generally not possible. Therefore, users need to determine how each failure mode is to be assessed.

Users should design the TAI to ensure that relevant failure modes are not assessed multiple times. When multiple sources of data and information provide information about a failure mode, the user can use the multiple sources to assess that failure mode, generally with a higher level of confidence in the assessment. The multiple sources of data should generally not result in multiple assessment scores for a single failure mode.

Assessment techniques each have a corresponding level of precision and uncertainty. Note however that measurements of high accuracy or precision are not always necessary. For example, touching the radiators of a heavily loaded transformer and finding that one radiator of the bank is much cooler than the others, is a good indication of a cooling system problem. Although such a measurement is imprecise, it gives a good indication of a problem that should be corrected. Such qualitative information can be used on its own or used in conjunction with other information when determining a score for some failure modes associated with the radiators, if the scores are to be used to generate an index for Fleet Screening.

Chapters (8-12) identify the possible condition assessment techniques for each failure mode and have included a calibrated scoring system. In many cases, different assessment techniques have been identified for a failure mode.

Whenever multiple condition assessment techniques will be used in a failure mode assessment, the confidence level of the assessment may increase if all techniques come to comparable results. Similarly, trend analysis techniques i.e. using multiple measurements through the transformers history, may provide increased confidence in the failure mode assessment.

Whenever multiple condition assessment techniques are used to assess a failure mode, the TAI design should select the most accurate data to be used to assess the failure mode. Additional information can be used to gain more confidence in the failure mode assessment, but “double counting” of the same failure mode should be prevented.

For example, a Partial Discharge (PD) problem may be detected from the DGA results. When PD is detected from the DGA results, users may choose to perform a more accurate UHF measurement to confirm the presence and the severity of the PD problem. In this case the UHF measurement would probably be used to assess the failure mode. The user however has a higher level of confidence in PD assessment, since PD problem is also shown in the DGA result. This will affect any quality score assigned to the overall assessment.

As another example, the transformer loading guide (IEEE C57.91 [3] or IEC 60076-7 [4]) can be used to estimate the loss of life of paper due to its historical thermal load. Similarly, furan analysis can be used to assess the paper loss of life. Both techniques still have elevated levels of uncertainty. Therefore, both assessment techniques can be used in the failure mode assessment either using an averaging or worst-case approach. Knowing that the furan result and the loading information both indicate loss of paper life, the user has a higher level of confidence in the assessment.

The TAI therefore needs to be carefully designed. Generally, the following principles should be followed:

- The more accurate indicator should be used as the primary indicator for the assessment of a failure mode. “Double counting” should not be allowed.
- When multiple indicators with similar accuracy are available for a single failure mode, either an average system or a worst-case approach can be used. The choice will be dependent on the purpose of the TAI.
- Users need to also determine how to manage situations where multiple indicators for a failure mode result in a very different assessment. The worst-case result could be used, but the assessment would probably be accompanied by an indication that the user should have a low confidence in the result; due to the uncertainty around the assessment. A poor assessment might result in the transformer being highlighted to the asset manager, but the indicator of low confidence needs to result in the asset manager investigating further before reacting.

Note: Condition assessment techniques may provide information on multiple failure modes e.g. DGA, therefore the same information can be used to assess multiple different failure modes. In such cases the user will need to carefully consider how to include the information in the assessment to prevent double counting.

Design Information, family history, loading information, data from site inspections, and inductive reasoning may be useful if more precise or accurate information is not available. Any assessment based on this information would ideally be based on experience and would be subjective. Users should ensure that these subjective assessments are consistent, so a guide or scoring matrix should be developed to ensure consistency. Refer to *4 Dealing with uncertainty in information*. This information can also be useful to improve the confidence in an assessment where other information is used for the primary assessment. Again, double counting should be avoided in the design of the TAI.

To achieve a reasonable level of confidence in the TAI, the availability of condition assessment information for the transformer population should be considered. There **may** be little value in including a failure mode into the TAI if reasonable indicators of the failure mode are not available for a significant proportion of the transformer population. This will again be dependent upon the purpose of the TAI and its design. For example, if a user has recently introduced a new test technique, and all transformers in the fleet will be progressively tested over the next 5 years, the failure mode indicated by the new test could be included in the TAI. Users would however need to determine how to assess those transformers that have not yet been tested with the new technique.

2.3.4 Design a calibrated system for categorising failure modes (scoring matrix)

Each failure mode being considered should be assessed and the effect on the transformer, assuming no remediation is carried out, should be determined. It is essential that the scoring of failure modes is consistent, so a scoring matrix should be developed prior to allocating scores. By using the scoring matrix, all the failure modes that are considered will have a similar score if they are likely to have a similar effect on the transformer. The scoring matrix can have any number of levels, as long as each level has a label and a definition. Levels will often relate to the urgency that problems should be

addressed to prevent failures. However, given the lack of precision associated with the assessment of most failure modes, a matrix with more than five levels would probably be of little value.

An **example** scoring matrix developed by the working group is used throughout this brochure. This matrix effectively has five levels. (The 6th level - labelled F - is not used when generating a TAI but is noted to consider very-short term failure criteria) The five levels, which relate to five different condition levels, are each allocated a colour from Green to Red. Letters (A to E) or any other label can also be used.

The user must also carefully consider and define the time scales that they wish to use in their matrix. The purpose of the matrix is to ensure consistency when scoring failure modes/mechanisms. It is therefore essential that the terms used in the example matrix, such as “as soon as possible,” “short term,” “medium term” etc. are defined by the user. An engineer who determines that a failure mode/mechanism needs to be addressed within 2 years, needs to understand if the failure mechanism should be assessed as Pink or Red.

Note that if any failure mode is assessed as Level F, the Transformer should be removed from service. A TAI is not intended to assist in short-term decision making.

The mitigated score (2.2.6 Mitigation Index/Score), however, could be calculated to assist in determining if the transformer should be scrapped or repaired.

Table 2-1 Example of a Scoring Matrix

F	De-energize as soon as possible. Don't return to service until problem is repaired. Component is at end of life.
E	Very Poor condition – high likelihood of failure. Component is near end of life. Repair or replacement as soon as possible is recommended. De-rating or restricted operation of the transformer may be appropriate, and operation under extreme conditions may not be appropriate until replacement is possible.
D	Poor Condition. Repair or replacement should be considered within the short term. Reliable operation may be impaired or compromised. Performance or component may be causing deleterious effects. Consider review of rating and operating condition.
C	Acceptable condition with significant signs of aging or deterioration. Reliable operation expected for medium term – but consider condition-based maintenance if applicable.
B	Good condition. Some signs of aging or deterioration are evident. Reliable operation expected for a lengthy period.
A	Minimal Signs of ageing or deterioration. As new condition.

User can use the example matrix developed by the Working Group (after defining the time scales), or they may also develop their own matrix with any number of levels. It is however essential that there is a clear definition of each level or category to ensure that scores for each failure mode being assessed can be applied consistently.

The matrix above has been used throughout the document. However, it has been varied in some sections so that the descriptions are more specifically aligned to the failure modes and mechanisms associated with some of the sub components of the transformer.

Note that to calculate a total transformer assessment score it may be necessary to assign a numerical score to each category. Refer to section 2.4

2.3.5 Calculate a TAI Score for each Transformer

Once all relevant failure modes have been assessed and assigned to a category on the scoring matrix, a method is required to calculate an overall Transformer Assessment score. There are multiple ways to generate an overall score. The method chosen will depend on the purpose of the TAI and the individual user's needs.

When designing the scoring system, the following points should be considered:

1. The scoring system should allow all transformers in a fleet to be ranked, such that those which are the highest priority for action or intervention are easily identified.
2. The scoring system should ensure that transformers with a failure mode that requires urgent attention are easily identified. That is, a bad failure mode should not be masked.
3. The scoring system result should be easily interpretable by any user, with reference to the purpose of the TAI.
4. The scoring system should be transparent and reproducible.

Designing a method for calculating transformer assessment scores such that the above conditions are achieved can be challenging. If for instance an averaging scoring system is used where the overall assessment score is the average or weighted average score of all failure mode scores, an urgent failure mode may be hidden if the transformer is otherwise in good condition. A non-linear scoring system (i.e. low scores for good and average conditions) will help minimise the effect of masking.

The following section provides an overview, with advantages and disadvantages, of scoring systems discussed in the working group.

2.4 Methods of Calculating a Transformer Assessment Score

There are multiple ways to present the final result of a TAI. Many users expect their TAI to produce a single number to allow transformers within the index to be easily ranked. Other users prefer the output presented as simple colour to indicate the overall state of each transformer. The colour code enables quick fleet assessment and dashboard functionalities. The output from a TAI may be both a number and a colour; perhaps a number that represents the “average condition” of the transformer and a colour to indicate the worst-case failure mode assessment.

A number of methods used for calculating a score are described Table 2-2.

Table 2-2 Methods of calculating a Transformer Assessment Score

#	Description	Advantages	Disadvantages
1	Summation of individual failure mode scores: $TAI = \sum_{i=1}^N S_{FMi}$ S_{FM} is the score of an individual failure mode. A simple set of linear or non-linear scores would generally be added to the scoring matrix. Weighting can also be added as per method #2.	Simple algorithm Transparent Weightings can be added if required.	Poor condition assessments may be masked so that score may provide an optimistic indication of transformer's condition. Using non-linear scoring may help to prevent masking Scores do not generally reflect urgency.
2	(Weighted) average: $TAI = \frac{\sum_{i=1}^N W_{FMi} \cdot S_{FMi}}{\sum_{i=1}^N W_{FMi}}$ S_{FM} is the failure individual mode score W_{FM} is the weighting per failure mode N is the total number of failure modes A simple set of linear scores or non-linear scores would generally be added to the scoring matrix. Weighting factors are applied to failure modes that the user wants to highlight.	Transparent Weighting allows some failure modes to be highlighted.	As above Weighting of failure modes, may mask further mask some failure modes.

#	Description	Advantages	Disadvantages
3	Non-linear mathematical approach: $TAI = \sum_{n=0}^{k-1} x_n i^n$ i is the number base / radix, which is equal to or greater than the number of failure modes included in the TAI x_n is the number of failure modes per category k is the number of categories included in the failure mode assessment n is the counter in this formula. Refer to the example later in this section for further explanation.	Masking of worse scoring failure modes is prevented.	More complex scoring system The scoring results can be more difficult to interpret If weighting factors are also used, a slight modification of the formula would be needed to prevent masking.
4	Numerical Score using estimated failure probabilities A probability of failure, based on test results data and other assessment information, can be estimated for each of the failure mode. A score can then be calculated $TAI = 1 - ((1 - \text{est. PoF}_{FM1}) \times (1 - \text{est. PoF}_{FM2}) \times (1 - \text{est. PoF}_{FM3}) \times \dots \times (1 - \text{est. PoF}_{FMn}))$	The TAI score can be scaled if required Highlights single advanced failure modes and properly combines several less advanced failure modes for overall comparison Works well provided each failure mode probability, or score is on the same scale even if it is not a true probability.	It is generally only possible to estimate a very approximate probability for each failure mode Method might tend to indicate an unjustified level of precision.
5	Worst case approach $TAI = \text{worst}(S_{FM})$ S_{FM} is the score of an individual failure mode It is also possible to indicate the number of failure modes which have been assessed with the worst-case score. Score = Red 3 (transformer has 3 failure modes that have been scored as Red).	Simplest algorithm Transparent Worst case failure mode is highlighted.	Weighting of failure modes only possible if the number of failure modes assessed with the worst-case score is included.
5a	Hybrid Score The worst-case score can be used in conjunction with one of the numerical scoring methods described above. For example, a simple summated score can be combined with a Worst-Case score. Scores would be of the form: Tx 1 = 64 Red Tx 2 = 64 Orange It is clear that Tx 1 needs urgent attention, although its numeric score is the same as Tx 2.	Combines two simple scoring methods The simple numeric score gives an indication of the overall condition of the transformer, and the worst-case score highlights the worst failure mode of the transformer.	A transformer with a single advanced failure mode, cannot be distinguished from a transformer with multiple advanced failure modes.
6	Count per category The TAI is shown as a set of numbers, rather than an individual scalar value. The number of failure modes assessed as being in each category is shown. For example, using the five-colour matrix shown in Table 2-1, a score for a transformer with 12 failure modes assessed might be as follows: 	Very good visibility of the total transformer health assessment score Masking of worse scoring failure modes is prevented Weightings are possible if required, and will not mask problems.	The TAI is not a single number but a set of numbers. This may make representing the result on a dashboard, or in other simplified forms more difficult.

#	Description	Advantages	Disadvantages
7	Machine learning: The index does not use a predefined formula to calculate the assessment score, but instead uses modern data analytic techniques, where smart algorithms e.g. neural networks, analyse the condition data and failure data.	These techniques may find new correlations between condition indicators and failure modes, potentially enhancing the index quality.	Complex algorithms are required Large volumes of data are required (including failures) Not transparent. Results would need to be validated to ensure that the machine has 'learnt' correctly.

2.4.1 Use of weighting factors

Users often apply weighting factors to ensure that problems of greater consequence are highlighted over lesser problems. For example, a dielectric fluid leak and a HV bushing may both urgently need attention, but it is likely that the asset manager would want to see the bushing problem highlighted above the dielectric fluid leak. This could be achieved by applying a lower weighting factor to the dielectric fluid leak, and a higher weighting factor to the bushing.

The problem with weighting any failure mode assessment score is that it may further mask or hide problems; although this will be dependent upon the scoring system chosen. For example, if a transformer in generally good condition has a bad dielectric fluid leak, and a simple summation of failure mode scores is used to calculate the transformer assessment score, it is probable that a good score will result for the transformer; if the dielectric fluid leak has a low weighting factor. The urgency to address the bad dielectric fluid leak is lost.

For an index that considers only the failure modes that are the critical drivers to the decision process (e.g. an index that considers only the critical failure modes that will be used to determine replacement) weighting factors may not be necessary.

Weighting factors may become relevant when an index includes failure modes that are not critical to the decision process, but still need to be considered. For example, some users may choose not to include bushings in a replacement index, as bushings, tap-changers and dielectric fluid leaks are not generally critical drivers to transformer replacement, since these problems can be mitigated by the repair or replacement of the component. However, it may not be economical to replace the bushings and tap-changer and to repair dielectric fluid leaks. Therefore, these failure modes may be included in the replacement index with a weighting factor applied, so that only their cumulative effect will push the transformer further towards the top of the replacement index. Note that with weighting factors applied, a transformer with a bushing that urgently needs replacing, may not be highlighted for replacement. It is therefore essential, that this failure mode is identified using another index, or through direct interpretation of test results, and addressed where appropriate.

If weighting factors are used, the user should take care when assigning values. Even with a non-linear scoring system, a heavily weighted good score could hide a problem that requires urgent attention if that failure mode has a low weighting. If a user believes that some failure modes have a significantly lower consequence than others, it may be appropriate to eliminate the less significant failure modes from the index. Alternatively, a scoring system that does not hide failure modes should be considered.

If weightings are to be applied, they should be applied consistently. Some examples of aspects to be considered when determining weighting factors include failure consequence, cost to repair, and time required to restore the transformer to working condition. A matrix may be developed to assist in determining the most appropriate weighting factors for each failure mode being considered. The values in *Table 2-3* are arbitrary but allow weightings to be applied consistently.

Table 2-3 Example of a weighting matrix

Weighting Factor	Criteria
1.5	This failure mode could result in a failure that will take more than 3 months to repair or cost more than 50% of the replacement cost of the transformer.
1	This failure mode could result in a failure that will take between 1 week and 3 months to repair with an estimated cost of 10% - 50% of the replacement cost of the transformer. OR This failure mode will lead to accelerated ageing of the transformer which result in premature replacement of the transformer or a premature failure.
0.75	This failure mode could result in a failure that will take less than 1 week to repair or cost more less than 10% of the replacement cost of the transformer.

2.4.2 Examples of Scoring Methods

In the following examples it is assumed that a simple index is based on 4 failure modes: $FM_1 - FM_4$. Each failure mode has been assessed using a scoring matrix with the following non-linear scores assigned for use in examples 1 – 3.

Note that the scores allocated to the five categories are only for generating a numeric assessment score for the transformer and to allow ranking of the transformers. A non-linear scoring system has been chosen to help ensure that transformers with one or more failure modes / mechanisms assessed as “Red,” have a higher transformer assessment score. Users can assign their own linear or non-linear scores to the matrix.

10 Red (E) FM1	4 Pink (D) FM4	2 Orange (C) FM3	1 Yellow (B)	0 Green (A) FM2
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Table 2-4 Examples of Scoring Methods

	Method	Calculation	Interpretation
1	Summation	$TAI = FM1 + FM2 + FM3 + FM4$ $TAI = (1 \times 10) + (1 \times 4) + (1 \times 2) + (0 \times 1) + (1 \times 0)$ $= 16$ $TAI_{max} = 40$	The failure mode that is red should be addressed urgently, but this is not obvious from the score, since a combination of less urgent failure modes could give the same score.
2	Average	$TAI = \frac{(10 + 0 + 2 + 4)}{4} = 4$ $TAI_{max} = 10$	The failure mode that is red should be addressed urgently, but this is not obvious from the score, since a combination of less urgent failure modes could give the same score.
3	Non-linear (ignore scores assigned in the matrix)	$TAI = 1 * 4^0 + 0 * 4^1 + 1 * 4^2 + 1 * 4^3 + 1 * 4^4$ $TAI = 1 + 0 + 16 + 64 + 256 = 337$ $TAI_{max} = 4 * 4^4 = 1024$	This method ensures that bad results cannot be masked. A score above 256 can only be achieved with at least 1 failure mode assessed as being red.

4	Numerical Score using estimated failure probabilities	Estimated Failure Probabilities (for this example) ⁶ are: FM1 – Red = 0.1 (10%) FM4 – Pink = 0.03 (3%) FM3 – Orange = 0.01 (1%) FM2 – Green = 0.002 (0.2%) $TAI = 1 - (1-0.1) \times (1-0.03) \times (1-0.01) \times (1-0.002)$ $TAI = 0.1375$	Although not immediately obvious, TAI scores for transformers with one or more failure modes that are Red, will have a higher score than transformers which do not have Red failure modes. Although the methodology is the same as that used to calculate a Probability of Failure, it cannot be assumed that the score is indicative of the expected PoF, since the probabilities for each failure mode were only estimates. This methodology is only intended to allow ranking of transformers.
5	Worst case	$TAI_{\max} = \text{Red or E}$	The score highlights that at least one Failure mode has been assessed as being in the Red category
5a	Hybrid Score	$TAI = 1E \text{ or } 1 \text{ Red}$	This score indicates the number of failure modes assessed as Red
6	Count per category		The score clearly shows the number of failure modes in each category. Scores can be easily ordered by the number of failure modes in Category E, then D etc.
7	Machine learning	Not provided	

2.5 Limitations of a Transformer Assessment Score and Index

Users should be aware of the following potential problems when using scores and indices.

2.5.1 The score may hide major problems

If the transformer's total assessment score is either the sum or average of the scores from each of the transformer's failure modes, the overall transformer assessment score may mask a bad failure mode. If most of the failure mode scores are good and only one, or a small number of, failure mode scores are bad, the overall assessment will hide or mask a problem in the transformer that needs to be addressed more urgently. For example, a transformer in overall very good condition, except for a single oil impregnated paper bushing which is in poor condition, may have a good overall assessment score.

If this type of scoring system is used it is essential that the major transformer failure modes are identified and assessed by analysis of the individual test results or condition information, as the overall assessment score may mask problems which need to be addressed urgently.

Another option is to use a "worst case" scoring system in conjunction with the average score, to ensure that major problems are highlighted. Of course, this method may also have limitations.

2.5.2 The score has limitations

It is essential that users understand what information is used to determine the transformer assessment score. A user may develop a scoring system which scores only the active part of the transformer. This means that components such as bushings and tap-changers would not be included in the assessment. All users must understand the limitations and have some other method to ensure that those transformers with bushing or tap-changers in poor condition are identified. It is essential that the purpose and limitations of the model are clearly defined and communicated to all users of the index.

⁶ The estimated failure probabilities have been chosen only to illustrate the methodology. Users should estimate their own failure probabilities based on knowledge of their fleet and historical data. To improve the accuracy of the ranking users should try to ensure that the relativity of the probabilities assigned are realistic.

2.5.3 Quality of the data

The quality of the assessment score will not only be a function of the design of the TAI, but will also be a function of the accuracy, completeness and timeliness of the data. In many cases, the data available for scoring will not be current. For large fleets, the testing of transformers may be spread over a period of several years. For example, dielectric fluid samples for DGA may be taken annually, but bushings might only be tested every 5 years during a scheduled outage. This means that some test results may be several years old⁷. For some transformers, the last test may not have been completed as scheduled e.g. transformer not available for an outage. In this case, the data will be even older. Furthermore, some tests results may be incorrect and need to be discarded.

It is essential that the effect on the score of this missing or low-quality data is understood. In some cases, it will make little difference to the overall score. In other cases it could make a bad transformer look good; resulting in the transformer being overlooked for critical repair or replacement. Similarly, the missing or low-quality data can make a good transformer look bad.

Implementing a “quality score” or some other indication of the quality of the data used (typically called “significance”) may help to understand the quality of data and the reliability or confidence level of a score. Refer to *4 Dealing with uncertainty in information*.

If users implement any kind of scoring system (particularly an automated scoring system), it is essential that they consider data quality. The system must include data validation to ensure that data that is obviously incorrect is not used and that missing data is treated appropriately, as detailed in 4. Furthermore, the automated system should also calculate a quality or confidence level score. A transformer with a good assessment score based on low quality data, might actually be a transformer with a major problem!

2.6 Examples – How to Develop a Transformer Assessment Index

For detailed examples showing how to develop a TAI refer to Annex B.

⁷ Users who want a high level of confidence from their TAI should consider the frequency of testing and inspection necessary to ensure that the data used for assessment is current. Quality scoring should reflect the rate of change of degradation of the failure mode being considered.

3. Refurbishment index

Asset Managers will sometimes decide to perform maintenance or repair work on a transformer that is not immediately necessary for the continued reliable operation of the unit. In this chapter, this work is referred to as refurbishment. This chapter describes how indices can be used to help Asset Managers determine which transformers should be considered for refurbishment work. Previous chapters have described how indices can be used to identify transformers that are in poor condition and should be considered for replacement or maintenance. These indices are not appropriate for determining which transformers should be considered for refurbishment work.

Some users believe that an intervention, prior to significant degradation or major component failure, is the most economical option in asset availability, cost control and residual value.

This work is normally done for one of two reasons.

1. To eliminate the need to perform this work in the future. This will possibly eliminate the need to arrange another outage of the transformer and the requirement to have a work team return to the site. Refurbishment is indeed an opportunity to perform works on transformers on one occasion rather than numerous maintenance interventions.
2. To extend the life of the transformer. For example, if a transformer has a high moisture level and acidic dielectric fluid, it is unlikely that these problems will cause an immediate failure of the transformer. However, if left uncorrected, they will accelerate the degradation of the cellulose insulation, possibly reducing the transformer's life. The Asset Manager (after an economic assessment) may choose to remove the excess moisture and to arrange replacement or treatment of the dielectric fluid as part of a maintenance strategy that could extend the transformer's life.

It is not within the scope of this Technical Brochure to determine what maintenance interventions should be performed during a refurbishment or to determine the economic justification for the refurbishment. Users should refer to CIGRE Technical Brochure "Guide for Transformer Maintenance" (TB 445) [5] as well as C57-140™ IEEE Guide for the Evaluation and Reconditioning of Liquid Immersed Power Transformers [6], for information regarding maintenance tasks that should be considered.

Some failure modes of a transformer can be readily corrected. For example, a failed cooling fan can be easily replaced. However other failure modes are not economical to correct. For example, badly degraded cellulose insulation could only be corrected by replacement of the windings. Similarly, many internal faults would require de-tanking, removal of the top yoke of the core and possibly replacement of part of the winding. This work would generally not be considered economical; particularly if the transformer has other defects that need to be corrected. Some failure modes may need more detailed consideration. For example, major repairs or replacement of a tap-changer may be relatively simple if suitable parts or a replacement unit is readily available. However, in some cases, major modifications to the tank may be required to fit a replacement tap-changer if a similar unit is not available.

3.1 Application of the transformer assessment index to identify eligible candidates for refurbishment

A simple way to identify transformers suitable for refurbishment consists of using a Transformer Assessment Index (TAI) Index and appropriate filters.

Filters can be applied to failure modes that cannot be easily repaired. The filters can be used to exclude transformers from the ranking list that score pink or red colours for these chosen failure modes. This approach assumes that a consistent and correct categorisation of the failure modes has been applied to the transformers. A single well advanced and non-repairable failure mode that is not correctly detected can be sufficient to nullify the potential refurbishment benefits and will be a major reliability risk for the transformer following the refurbishment.

Refurbishment is typically applicable to transformers that have not revealed any significant abnormality of non-repairable components. Both technical and economic aspects need to be considered.

3.1.1 Failure modes to be used as a Filter

Filtering of specific failures modes should be consistent with the fleet characteristics and user's objectives. When a transformer reveals critical defects for the chosen failure mode then it should not be considered for refurbishment. Deriving a complete list that satisfies all identified defects is impossible, but the user can refer to the guidance below to select failure modes that should be used for filtering to prevent a transformer being considered for refurbishment:

- Failure mode has no repair solution available
- Repair of failure mode is unlikely to be cost-effective
- Damage is irreversible

A fundamental driver for the decision to refurbish is the active part condition, especially the cellulose and solid insulation that often determines the remaining life of the transformer. Dielectric, mechanical and thermal defects in the winding and core typically require too much effort to be cost-effectively repaired. Most users would not consider a refurbishment if the active part was in poor condition and the transformer therefore had a limited remaining life.

A list of common failure modes to filter would include: design issues, compatibility issues of old components, core insulation, core fault, winding fault, dielectric fault, thermal fault, advanced paper degradation, winding movement, corrosive sulphur and major sludging issues that are likely to affect the flow of insulating fluids. There are numerous others that the user can add to the list drawing on a user's experience from their own fleet.

3.1.2 Failure modes to include in the Refurbishment Index

Final candidates for refurbishment are generally transformers that:

- require significant maintenance work (particularly in relation to accessories)
- show normal or reduced ageing and degradation of the active part compared to expectations based on loading and operating conditions
- do not have other failure modes that are likely to significantly limit the life of the refurbished transformer.

This approach assumes a consistent determination of the repairable conditions. Deriving a complete list that satisfies all identified cases is difficult, but the user can refer to the guidance below to select relevant failure modes:

- failures modes related to components and accessories that can be replaced cost-effectively
- degradation that can be reversed or corrected (in contrast to irreversible damage)

A list of common failure modes that should be considered during a refurbishment include: failure modes related to bushings, pumps, fans, tap-changers, control equipment, moisture, clamping pressure, degraded dielectric fluid, corrosion, paint, dielectric fluid leaks and gaskets.

Repairable failure modes can be divided into two categories; those that will degrade or age the transformer, and those that may have an effect on the operation or function of the transformer but do not necessarily accelerate ageing.

Other failure modes need only to be considered during a refurbishment if they are scored as red (i.e. need attention in the short term to minimise the risk of failure)

A TAI can be used for refurbishment decisions to identify transformers for refurbishment based on their technical assessment. Criticality, economic considerations, losses etc. should also be considered. Refer to 6 *Criticality and other condition considerations*. Furthermore, a detailed economic assessment which considers the estimated cost of repair (refer to CIGRE TB 248 Guide on Economics of Transformer Management) and the expected remaining life of the refurbished unit should be carried out before a transformer is refurbished.

3.2 Application of the TAI to estimate the potential benefits from refurbishment

After obtaining a list of eligible candidates ranked by their overall condition and derived using the filters detailed above, a mitigated index ranking can be established considering all of the failure modes that can be repaired and effectively corrected. This index can be compared to the original TAI (before refurbishment work). This comparison enables the estimate of interventions required and the derived benefits from the refurbishment and allows the user to identify the transformers where the difference/gap between these two indices is high. However, the value of a mitigated score will be somewhat dependent upon the scoring system used. Many numeric scoring systems result in a value that allows transformers to be ranked; but the score has no real meaning. Users must understand the scoring system to understand the relevance and value of using mitigated scores. Refer to *section 2.2.6*.

4. Dealing with uncertainty in information

All assessments of transformers include a non-avoidable level of uncertainty. This level of uncertainty is based on the fact that degradation may not act in a predictable manner and that transformer degradation is not perfectly assessed by the known condition indicators. The user of a transformer assessment or index should always consider the basic level of uncertainty of their assessments. This level of uncertainty is influenced by the data quality used in the assessment. If too much confidence is given to the assessment result, incorrect decisions might be taken, e.g. a bad assessment result may have led to a replacement decision where in reality additional information could have resulted in a better solution.

Unavailability of data or incorrect and/or obsolete data will adversely affect the assessment results and increase the level of uncertainty. Therefore, an indication of the uncertainty should be part of the decision-making process.

It would generally be preferable if a transformer is able to be assessed and an assessment score generated, even if some data is missing or does not match the required attributes. However, a quality score or confidence score should be generated with the assessment so that the Asset Manager understands that they should have limited confidence in the score.

In some cases, it may be appropriate to terminate an assessment if quality data is not available. However, it is essential that transformers that have not been assessed are highlighted to the user, so that they can arrange for quality data to be obtained (e.g. transformers could be re tested) where appropriate.

When a Transformer Assessment Index (TAI) is designed, or an assessment methodology determined, it should initially be assumed that quality data will be available for the assessment of the transformer fleet. However, as it is likely that quality data will not be available for all transformers in a fleet, the user should determine how the TAI manages uncertainty due to inaccurate data or unavailable data.

This chapter discusses techniques to manage unavailable and/or poor or obsolete data. The more accurate the data that is available to feed into the TAI, the more likely the Index will be useful. The designer of the TAI should be aware that lower value assets often have less data available for assessment, leading to a higher level of uncertainty in the TAI result.

A reference to the levels of uncertainty should generally be included with the TAI. The uncertainty requires its own scoring methodology and can be presented with the TAI score. It could also be presented as separate dimension of the TAI score.

Uncertainty can be expressed using an uncertainty index, a colour code, or the TAI score can be shown as a range to represent the uncertainty i.e. falling between a minimum and maximum. If the TAI score is presented using a colour code, the colour intensity can be also used to show the levels of uncertainty.

4.1 Dealing with uncertainty within available information

Uncertainty within available data may have different causes. The following three causes will be discussed in more detail in this section:

- Incorrect data entry, or erroneous or questionable test results
- Uncertainty in the condition assessment
- Aged data

4.1.1 Incorrect data entry

Specifically, if data is entered manually, typing errors or spelling errors may cause incorrect data. To enable correct TAI-results it is recommended that the data entry is validated.

The following table shows types of data validation.

Table 4-1 Data validation methodologies

Type of solution in data validation	Purpose	Level of complexity	Pros and Cons
Comparison of data / table look up	Compares data to a known data set; e.g. raises a flag if an entered substation name does not exist.	Low	Pro: Validates within a preset of valid options Con: small typing errors or language issues may cause unwanted validation failure
Data type / unit	Checks if for example the data in a date field is entered as yyyy-mm-dd.	Low	Pro: always good format of data Con: no value checking, only the type of data format
Digit check	Checks the number of digits or characters	Low	Pro: simple to install Con: Limited validation capabilities
Limits and ranges	Raises a flag if the entered data is outside the possible range, e.g. manufacturing date older than 1886 (first commercially available transformer) is considered incorrect, or the make and brand is spelled correctly according to a predefined list, or measurement values are within a certain range, e.g. power factor must be between 0 and 1.	Medium	Pro: good check for mistyped numerical values, when fall out of predefined ranges Con: no validation within the predefined ranges. Predefined ranges need to be designed carefully. Outliers need to be managed.
Consistency	If the asset under consideration is a transformer, the data attributes should only allow transformer related data	Medium	Pro: Only relevant data is accepted Con: No content validation
Chronological order	Check if for example an inspection data is not before the manufacture data	Medium	Pro: Very useful in date and time sensitive data Con: Only valid in case of dates and time
Cross reference	Checks if data is consistent with other data attributes. For example, the actual load of the transformer cannot exceed a given range depending on voltage class.	Medium	Pro: Very useful as an enhanced range validation Con: requires the reference data attributes to be accurately available
Incremental data	Checks if incrementally entered data is not exceeding an expected value. A maximum number of operations can be calculated based upon historical data. If the counter reading of a new reading is exceeding the expected number a validation warning could be triggered	Medium	Pro: Very useful as an enhanced range validation in repeatable data attributes Con: requires historical data

4.1.2 Uncertainty in the condition assessment

The accuracy of condition assessment techniques varies between the different techniques. This means that each condition assessment technique does have a certain level of uncertainty. The user and designer of the TAI should be aware of this effect. Small uncertainties may be ignored, having little effect on the decision. But higher uncertainties may affect the decision. In the case of high uncertainties, the decision maker needs to be informed, enabling them to adjust the intervention decision. Failure modes can be assessed using multiple techniques, as discussed in later chapters which describe the different failure modes and condition indicators. The likelihood of detection may

vary based on the method. Therefore, the techniques with a higher likelihood of detection will have a lower level of uncertainty. If needed, additional techniques should be used to perform a more accurate condition assessment.

A few examples are:

- Partial discharges may be detected by DGA, however DGA will most likely not provide detailed information on the location and type of PD-activity. Therefore, an intervention is usually not performed on DGA results only. A second, potentially more accurate, PD-measurement (e.g. UHF) is recommended whenever DGA shows PD-activity.
- Manually performed inspections will have a subjective interpretation with inherent levels of uncertainty. The inspection regime should be designed to minimize the uncertainty, by calibration, training, use of qualified inspectors, and approved 3rd party service providers such as laboratories. Still, in some cases, a level of uncertainty could be included in the TAI for subjective condition indicators.

A simple approach calculates the TAI with the minimum, maximum and average value of the uncertain input parameter and then shows these results to the decision maker e.g. by using a boxplot. This works fine in situations with only one or two uncertain parameters. In the case where more parameters have uncertainty a more advanced TAI is suggested using statistical techniques to show the levels of uncertainty. These techniques interpret data attributes as distributions, rather than single data points (scalars), and use more advanced techniques to calculate the final results. An example that can be used is Monte Carlo simulations, as discussed in *section 4.2.5. and 4.5.2.*

4.1.3 Uncertainty due to old data

A condition assessment provides information of the asset's condition at the time of measurement or assessment. Degradation is however continuing after the assessment. It should be recognized that a test report being old, does not inherently mean it is out of date. It only becomes out of date if the underlying circumstances change so that the test result no longer reflects the current status of the transformer. However, it is generally impossible to determine this without repeating the test. Therefore, the assumption is made that over time, the test result becomes less and less likely to represent the current condition of the transformer. In other words, the uncertainty of the assessment result increases in time. Whenever using older condition assessment data in the TAI the decision maker should preferably be informed, that older data is used, so they can take appropriate actions.

Bouzeghou and Peralta [7] Introduced terminology for describing the different aspects of timeliness. They differentiate between currency (the time since the measurement was made) and obsolescence (how much the underlying value has changed since the measurement was taken). With regard to the whole database, they define a freshness rate, which is the percentage of records which are not obsolete. Finally, they define a timeliness period as being an application-specific length of time in which a piece of data may be considered not obsolete.

Sparling and Catterson [8] show how this work can be applied in the cases of a transformer where there is uncertainty about whether or not a test result is obsolete. The currency can be calculated from the known date of a given test, but the challenge lies in determining whether that test result is obsolete given the time elapsed since it was taken. The timeliness period (as defined by Bouzeghou and Peralta) can be used to help assess the likelihood of obsolescence.

As an example, two DGA tests a year apart may show low absolute values of key gases and very little change within the year. In this case, it would be reasonable to assume that the test report is unlikely to become obsolete within the next year, and that the likelihood of the test result becoming obsolete increases with each subsequent year. However, the timeliness of a report is highly dependent on the test type and test result. If the two DGA reports show a rapid rate of change over the year, the timeliness period would shorten significantly, and the chance of obsolescence one year later is high.

Table 4-3 suggests a method of scoring or evaluating the uncertainty in the TAI score. If multiple data is missing or out of date (obsolete), it should result in a low level of confidence in the TAI. If weighting factors have been used in the TAI, similar factors should be used in the Data Quality Score, as used in the TAI.

Whenever using a more advanced TAI, statistical techniques, such as Monte Carlo analysis, can be used to show the levels of uncertainty due to old data. Here data attributes are interpreted as distributions rather than single data points (scalars). In this technique, the uncertainty in the distribution (spread) increases with the age of the condition assessment result. To illustrate this technique, an example is provided taking the DC resistance of a transformer winding.

Example: The actual DC winding resistance, measured during the routine test, is likely to stay constant in time, but may increase due to degradation mechanisms. Contacts in the winding circuit potentially degrade leading to higher contact resistance, or connections may come loose. Given this degradation mechanism, the likelihood of highly increased winding resistance is low, and the winding resistance will not decrease in time. For this example, an exponential distribution can be used when estimating the actual DC winding resistance if the winding resistance measurements are old. Figure 4-1 shows the exponential distribution's shape, providing a shift parameter and the shape parameter β . In this graph the shape parameter is fixed. The shift of the exponential distribution is to be determined by the actual DC winding resistance measurement result. The shape parameter β has to be determined by the level of uncertainty. This can be done by pattern fitting from other available similar data, or by experts' design. The level of uncertainty of the measurement increases with the age of measurement. If a proper DC winding resistance has just been measured the level of uncertainty is rather low, however if many years have passed since this measurement, the level of uncertainty is high. Therefore, the shape parameter β from the exponential distribution describing the measurement data is time dependent. Figure 4-2 shows a three-dimensional graph with the age of the measurement on the third axis. Please note that this distribution describes the most likely behaviour of the input parameter (measurement in the example). It is not related to the failure probability of the transformer.

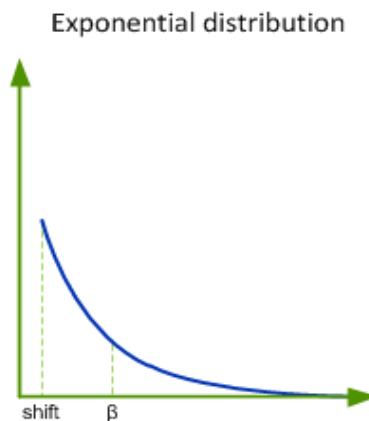


Figure 4-1 Exponential distribution

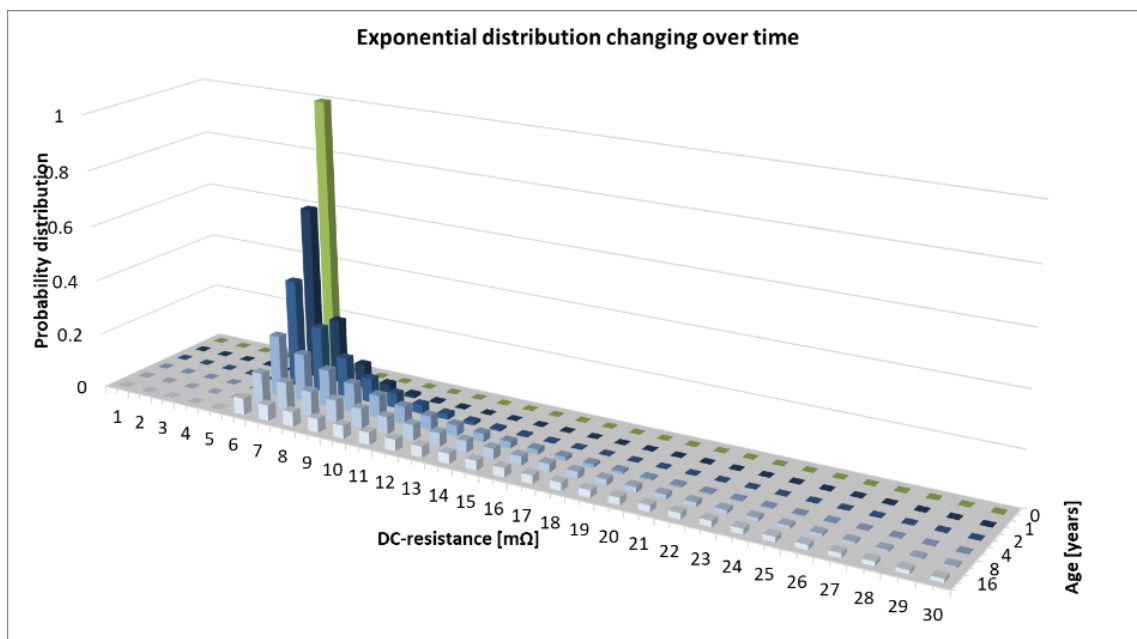


Figure 4-2 Exponential distribution with time dependent shape

4.1.4 Incorrect data

In some cases, data validation shows that the data is simply incorrect, while correct data is not available. Data enhancement actions are preferred prior to running the assessment tool. However, this may not be feasible in all cases. Whenever incorrect data is found, it is recommended to discard the incorrect data and treat it as missing data, with the options provided in *section 4.2*.

As older data is not necessarily incorrect, it may be appropriate to use older where more recent data is obviously incorrect. As discussed in *section 4.1.3*, this data can be used in conjunction with a level of uncertainty depending on the relative age of the data (age of the data compared to average degradation speed).

4.2 Dealing with unavailable Information

Once a transformer assessment index has been developed, the required information is determined by the condition indicators that are included in the design of the TAI. It is unlikely that an engineer will have all the required information readily available to generate a transformer assessment index for an entire fleet of assets. In order to deal with any unavailable information and to still generate a meaningful TAI, the TAI should be robust to prevent it from providing no answer or a false answer. If the index stops and does not provide an answer, the decision maker is still not informed on the correct status and there is a high likelihood that the decision maker is ignoring this transformer, risking unknown bad conditions.

Preferably the assessment index provides a most likely answer including an indicator showing that information is missing and providing an output range or level of uncertainty due to the unavailability of information. This prevents over-confidence in the TAI result when information is missing. When information is missing sensitivity analysis is recommended to assess the impact of the missing information on the assessment index score. This section describes some techniques which can be used if required information is missing.

To perform any analysis at all, a minimum set of information is required. If the available information, related to the transformers under study is too limited, the analysis will not result in a sensible transformer assessment index result. The techniques described in this section, requires sufficient reference information from similar assets or parameters to perform some manner of statistical analysis. It may be that little or no information at all is available or will become available for some parameters. In this case, the engineer should re-affirm the need for the parameter to be included in the assessment index.

The methods mentioned in *Table 4-2*, describe a number of ways in which missing input parameters can be estimated. Using estimates however, includes a level of uncertainty, i.e. estimations give an indication of the possible and even likely values of an input parameter, but the actual value is not known. Therefore, this estimate influences the transformer assessment index result and confidence level. Where a single parameter is not known, the determination of the transformer assessment index and uncertainty of the result can be easily derived using statistical or algebraic calculations. Where multiple parameters are unknown, algebraic calculations often become too complex. Using an iterative method of uncertainty modelling, such as Monte Carlo analysis, may be preferred.

The level of complexity of the design of the transformer assessment index is dependent on its purpose and the level of information at hand. Whenever an asset manager has only a limited number of transformers, it most likely is not worthwhile to pay lots of effort in making a complex transformer assessment index. However, if an asset manager is responsible for vast numbers of transformers, it becomes rather relevant to design a transformer assessment index, which is accurate and provides sufficient granularity even in situations where information is partly not available. Bigger granularity might be required whenever larger transformer fleets are assessed.

Table 4-2 Different options for managing of unavailability of information in assessment index design

	Type of solution for managing unavailable information	Purpose	Level of complexity	Pros and Cons
1	Stop assessment, or ignore missing data and manually assess	Low number of assets e.g. less than 10. The assessment index algorithm cannot be fully automated, and the engineer must assess the health of the transformer manually.	Extremely low	Pro: Simple to design, low cost. Easy to identify missing data. Con: No decision support, risk of missing bad transformers. Manual assessment required.
2a	Use a default value	Low to medium number of assets e.g. less than 100. The default value can vary e.g. Best case, Most likely case, or Worst case. It enables an index result at all times, but may be inaccurate.	Low	Pro: Simple to design, low cost. Con: No decision support, risk of missing bad transformers. No indication when data is missing.
2b	Use a default with a range	A default value is provided including a possible range of that value. The index result now provides a range of possible results, e.g. min, max, and average or median.	Medium	Pro: Index result is provided, range is provided (level of certainty). Con: More complex to design. The range may not be tuned to local circumstances.
3a	Use of statistical inference on limited number of input parameters	High number of assets, e.g. more than 100, with limited unavailability of data. The level of uncertainty is provided for 1 or 2 input parameters. The level of uncertainty is mathematically calculated and shown in the index result.	High	Pro: Level of uncertainty is provided; a range is provided. The range can be adopted to local circumstances. Con: Method is only applicable if only 1 or 2 input parameters are missing. Limited capabilities in adjusting towards individual transformer circumstances. Statistical data is required.
3b	Use of statistical inference on many input parameters	High number of assets with a varying or high number of unavailable data. The level of uncertainty is provided for many input parameters. The level of uncertainty is calculated numerically (e.g. using Monte Carlo simulations) and shown in the index result.	High	Pro: Level of uncertainty is provided; a range is provided. The range can be adopted to local circumstances. Capable of handling many missing input parameters. Con: Limited ability to adjust for individual transformer circumstances.

				Statistical data is required.
4	Imputation using external and local circumstances	High number of assets with a varying or high number of unavailable data varying with local circumstances. Known physical influencing factors are considered in the analysis, allowing a statistical assessment imputing the likely value of unavailable input parameters, including levels of uncertainty.	High	Pro: Level of uncertainty is provided, a range is provided. The range can be adopted to individual transformer circumstances. Con: impact of local influencing factors and statistical data is required.
5	Machine learning	Machine learning can be used to assess the missing input parameters.	Highest	Pro: The systems are flexible and can provide promising results and potentially makes new correlations and insights Con: Requires high volumes of information. Costly to implement.

The above-mentioned methodologies 2 – 5 enhance the dataset and fill in the missing gaps in case data is missing. Options 3, 4 and 5 include capabilities to calculate and show the levels of uncertainty. Option 1 and option 2 are not able to do so. For those options a confidence indicator can be simulated by using a data completeness index. This index calculates the percentage of data points provided. If needed, weights can be added per datatype, allowing higher weighting factors for more critical data and therefore having a bigger impact in the data completeness index. Further details of these methods are given below.

4.2.1 Stop assessment, or ignore missing data and manually assess

These methods are mostly used whenever the number of assets to be studied is relatively low.

When data is missing the engineer has the option to:

- Stop the generation of the transformer assessment index, providing no result for the transformer in question. In this case, the missing data would need to be obtained to allow the assessment to be completed, or
- Manually completing the TAI assessment without the missing data. This may be possible for some TAI designs. For example, a worst-case score can still be generated without data to assess one or more failure modes/mechanisms. However, for other scoring systems the scores would need to be adjusted if all failure modes/mechanisms are not assessed e.g. where the TAI score equals the sum of individual failure mode assessments scores,

This option is only recommended when the fleet comprises a small number of transformers, and the engineer can verify the status of the transformer.

4.2.2 Use a default value

An automated algorithm can always provide an assessment when input parameters are missing by using default values. However, the answer may be misleading, and the use of default values may not be visible to the decision maker when they are analysing the results of the assessment index. In this option, the engineer needs to choose the desired default value. For the case of a best-case default (“Innocent until proven Guilty”), the assessment index will always be biased towards a better-condition score, with a high likelihood of overestimating the quality of the transformer. Winding deformation can be used as an example. Typically, specialized tests like SFRA are used to assess winding deformation. Most utilities will not perform this periodically, therefore this information is not available. As a default, the user can choose “no deformation” until an SFRA measurement results proves differently. When using a worst-case approach (“Guilty until proven Innocent”) the assessment index result will always be pessimistic in the case of missing information, providing an early warning. This approach has a potential risk that the asset manager is going to ignore the early warnings, if the

warning turns out to be false too often. To overcome this the best solution is using the most likely value (expected value, median or average). This however needs an assessment of the input parameter in general to indicate what the expected value would be and may need further consideration of local circumstances and basic statistics. When using this approach, the type of default should be carefully considered.

4.2.3 Use a default with a range

Like the previous option, however, the default is now a range of values providing the minimum, maximum, and if possible, the average or median. The assessment index is now calculated for all values and the results are presented as a range, providing an indication of the possible results and the impact of the unavailability of the information. In a smart design of the assessment index, the ranges can vary based on the type of data and the type of transformer, allowing better alignment of the assessment index default values. This approach is very useful but increases the complexity of the tool. The user needs to consider that this approach requires statistical information. The way the results are distributed within the range is unknown, but this approach can provide a median, minimum and maximum value providing high level insights into the level of uncertainty.

4.2.4 Use of statistical inference on limited number of input parameters

Statistical inference can be used in situations where it is known which input parameters are potentially missing and if the number of missing input parameters is limited to 1 or 2. In these situations straightforward mathematical algorithms can be used to calculate the impact of the uncertainty of the input parameter in the result of the transformer assessment index, provided that the input parameters distribution is known or estimated. In this methodology, the missing input parameters are estimated by a distribution, providing the likelihood of the parameter value. These distributions can be of various designs, e.g. a discrete distribution when only a limited number of values are possible, a normal distribution, a Weibull distribution or exponential distribution. The type of distribution selected is dependent upon the behaviour of the unavailable input parameter, and can be assessed by statistical inference, provided that the information used for the inference is valid for the type of transformer and the input parameter. Please note that this type of analysis cannot differentiate between different local circumstances for individual transformers.

4.2.5 Use of statistical inference on many input parameters

This method uses the same methodology as the previous option, but now the potential number of available input parameters is too high for algebraic calculations, e.g. more than 1 or 2 of unavailable input parameters. In these cases, it becomes too complicated to calculate the impact of the uncertainty to the TAI result. To manage this Monte Carlo simulations can be used. This is a numerical methodology, using random sampling techniques and multiple calculations, to calculate the final assessment index results. This is a more complex methodology, but it provides significant insights into the level of confidence and the expected assessment index result distribution. The value of these more complex tools can be found in situations with many transformers and many or a variable number of missing input parameters.

4.2.6 Imputation using external and local circumstances

This method can be used whenever multiple input parameters are missing and local circumstances for individual transformers are varying and may have a high impact on the transformer assessment index result. In those cases, it is preferred to use imputation techniques to assess the likelihood of an input parameter's value based upon known degradation factors or accelerators. In these cases, individual transformer circumstances are included in the assessment of the input parameter. As an example, if one needs to know the level of corrosion of a transformer radiator, but the inspection information is not provided; one can include the location information to assess the likelihood of corrosion, e.g. outdoor transformers are more susceptible to corrosion if located near the sea shore or polluting industries. It is recommended that this type of data imputation includes levels of uncertainty and is used in combination with the techniques mentioned in option 3.

4.2.7 Machine learning

This method uses modern algorithms e.g. neural networks, to learn the behaviour and failure characteristics of transformers. These techniques may find some new correlations and insights with possibly very positive results, however, to achieve reliable results, a large amount of data is required and potentially a long learning curve is required. In this brochure, these techniques are not discussed in detail.

4.3 Options for showing uncertainty

It is essential that the end user of a TAI has an understanding of the level of uncertainty before making any decisions based on the TAI. There are multiple ways of showing levels of uncertainty, some of which have already been discussed.

The output of the TAI could be expressed as a range to show the uncertainty. The range could show the minimum, maximum and possibly the average value, or the output could be shown as a boxplot. Refer to example 4.4.2.

A separate index can also be provided to give an indication of the level of uncertainty. The index could be presented as a numeric or alphabetical score, a colour code (e.g. Green, Orange Red) or the uncertainty could be shown as a colour shading intensity where the TAI output is already expressed as a colour.

To ensure that any measure of uncertainty is assessed and represented consistently, user should consider developing a data quality scoring matrix. A simple example is provided below.

Table 4-3 Example table of data quality indication

Evaluation of the transformer data quality	
Results	Quality Indicator
All the tests related on the main indicators were performed with the normal frequency and the results obtained were reliable.	10
One or more of the tests were performed more than 6 months ago and less than 24 months ago , after the date corresponding to their normal frequency and the results obtained were reliable.	7
One or more of the tests were performed more than 24 months ago and less than 36 months ago , after the date corresponding to their normal frequency or some of the results are not available or are questionable.	4
One or more of the tests were performed more than 36 months ago , after the date corresponding to their normal frequency, or there is no available results, or many of them are questionable.	0

4.4 Examples

For the following examples, which illustrate the methods discussed in *Table 4-2*, a standard case is used. It is assumed that a simple index is based on 4 failure modes: FM_1 - FM_4 . Three failure modes have been assessed using a scoring matrix. Input for one of failure modes is however missing. We assume the TAI is designed using a linear summation principle: $TAI = \sum_{i=1}^4 S_{FM_i}$. The maximum score of individual failure modes is 25, hence the maximum TAI-score for a good performing transformer is 100.

- FM_1 : Paper degradation: $S_{FM1} = 25$ (DP = 850)
- FM_2 : Dielectric strength: $S_{FM2} = 20$ (DGA Values OK, but BDV=55 kV)
- FM_3 : Bushing: $S_{FM3} = 15$ (tan delta = initial value * 1.5)
- FM_4 : Mechanical structure: $S_{FM4} = \text{unknown}$ (No SFRA measurement)

4.4.1 Stop assessment, or ignore missing data and manually assess

When using this methodology, the ideal situation would be to assess the mechanical structure so that data was available to properly assess the transformer. However, for this example it is considered not feasible to collect that data. If not corrected for the missing data, the score of the TAI would now be 60 (from a maximum score of 75). The scoring needs to be recalibrated such that the maximum score sums up to 100 again. For this example, the corrected score would be

$$\text{TAI} = 60 * 100 / 75 = 80$$

4.4.2 Use a default value

When using a default for the missing information, a standard value will be used instead of the missing data. For this example, the SFRA data is missing. The SFRA result could either be “Good condition” (“Agreement with benchmark, or acceptable phase comparisons”) or “Very poor” (“Signature differences detected”). When using the “innocent until proven guilty” approach the default will apply the value “Good condition” and the score for FM₄ will be 25. The TAI would score 85. When using a “Guilty until proven innocent” approach the value “Very poor” and the score for FM₄ would be 0. The TAI would be 60

When using an average scoring as a default, the designer needs to know the mechanical deformation distribution. They can calculate that by counting the total operational transformers of the population of transformers not having mechanical deformation and compare that with the number of transformers with mechanical degradation. This analysis should be performed with a timeframe long enough to collect sufficient data e.g. 10 years. Assuming that within this timeframe the total transformer population was 25 and two transformers showed mechanical deformation. The average score for FM₄ would then result in 23 and the TAI would be 83.

When using these methods, the decision maker is potentially unaware if data was missing, and overconfidence in the assessment result may occur. By using a default value with a range, the score is not a single value. In this case the range could be provided with the following parameters:

- Min: “Very poor” with $S_{FM4} = 0$
- Average: $S_{FM4} = 23$
- Max: “Good condition” with $S_{FM4} = 25$

The TAI now provides a scoring range which can be shown in a boxplot:

$$\text{TAI}_{\min} = 60, \text{TAI}_{\text{average}} = 83, \text{TAI}_{\max} = 85$$

4.4.3 Using statistical inference

When using this option, the inference is automatically applied by the TAI-algorithm. Using this methodology, the TAI will look into the database of similar asset types and checks the values for the data point in those assets. Consider the example that in an asset base of 100 similar transformers, 80% of the transformers have relevant SFRA results. Ten transformers show differences in SFRA signatures. This will mean that using statistical inference a discrete distribution can be used with the following results:

- Very poor: $10/80 * 100\% = 12.5\%$
- Good: $70/80 * 100\% = 87.5\%$

This results in a TAI result, also presented as a discrete distribution:

$$\text{TAI}=85 (87.5\%); \text{TAI}=60 (12.5\%)$$

When only one or two parameters are missing (and it will always be the same parameters that are missing), these calculations can be performed algebraically in the TAI. But if the number of missing data points is varying or higher, then this becomes more complicated. It is then recommended to use iterative algorithms like Monte Carlo solutions. For this example, it will result in the same result. But whenever multiple inputs are missing, the expected value is calculated, including the complete TAI distribution, providing the level of uncertainty and a sensitivity analysis.

4.4.4 Imputation using external and local circumstances

When using imputation in this example, the likelihood of mechanical deformation can be assessed using information of known mechanical degradation principles, e.g. earth quakes and through faults. Let's assume that the transformer is not located in an earth quake sensitive area, so this will not play a factor for our TAI-design. Through faults can however cause mechanical deformation if the dynamic

forces exceed the design forces of the transformer. Therefore, the number of through faults and the magnitude of the short circuit current can be used to assess the likelihood of mechanical deformation.

Assuming for this example that the rated maximum dynamic short circuit current $I_k' = 50$ kA. For transformer 1, one through fault occurred in its lifetime with a measured short circuit current $I_k' = 25$ kA. This short circuit current is well below the rated value and it can be expected that the likelihood of mechanical deformation is low.

For transformer 2 with the same ratings, two through faults have occurred in its lifetime. The first had an amplitude of $I_k' = 45$ kA and the second had an amplitude of $I_k' = 55$ kA. The actual short circuit currents were close to and exceeding the ratings, therefore this transformer has a higher likelihood of mechanical deformation. Applying a discrete distribution for the two transformers, with an engineer's judgement of the level of uncertainty could provide the following results:

Input Mechanical deformation	Transformer 1	Transformer 2
"Very poor"	5%	80%
"Good condition"	95%	20%

Whenever more knowledge is available more accurate estimations can be provided for the likelihood and the levels of uncertainty.

4.4.5 Machine learning

Machine learning and data analytics techniques can provide good results for this type of analysis; however, a detailed discussion of these techniques fall outside the scope of this brochure.

4.5 Additional examples

4.5.1 Single unknown parameter estimation using statistical inference

If a transformer assessment index is missing one single parameter and the impact of this parameter on the transformer assessment index result is known, relatively simple methods can be used to estimate the transformer assessment index result by using statistical inference.

To use statistical inference on this missing parameter a batch of similar transformers under similar conditions should have known values for the same parameter. This provides a distribution of this parameter. The expected value of this distribution can then be used to calculate the most likely transformer assessment index result. The spread of the distribution can provide the level of confidence of the most likely transformer assessment index result. The following example shows a simplified transformer assessment index with only 1 unavailable parameter, which is assessed using a normal distribution.

- A utility has a fleet of 132/11 kV transformers, assessed using a 4-parameter non-linear transformer assessment index. The transformer assessment index comprises four attributes (core and coils, OLTC, bushings and oil).
- In this fleet, a 40-year old transformer is being assessed. Three out of the four attributes are assessed to be in acceptable ('Orange') condition (Score =10), but the 132 kV RBP (Resin bonded Paper) bushing condition is unknown. Thus
- $HI = 10 + 10 + 10 + ? = ?$ (In this simplified example the assessment index is a sum of four attributes)

The utility has tested a number of similarly aged bushings, by measuring the dielectric dissipation factor (DDF), and has used the data to develop a distribution of similar results, shown below. The results are represented as a normal distribution with parameters $\mu = 1.2\%$ and $\sigma = 0.35\%$.

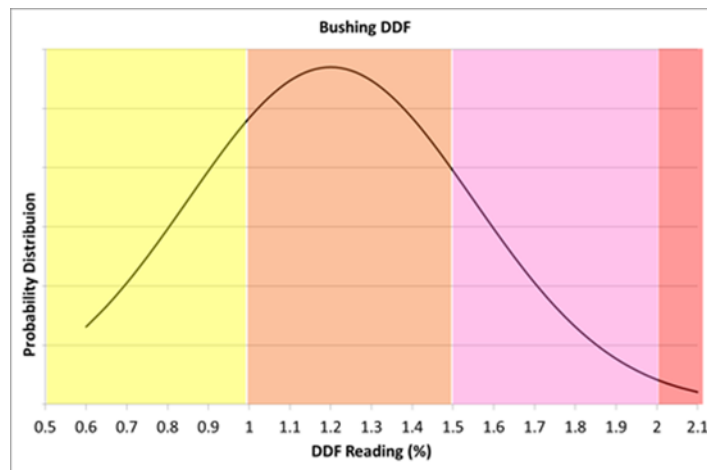


Figure 4-3 Distribution of bushing test results

From the above distribution, the utility can deduce (from the normal distribution) that the most likely bushing assessment index score is 10, with less than 20% likelihood that the bushing score is 25 or greater.

Bushing Condition	Colour Code (Score)	Likelihood	Overall Transformer HI
As-new	Green (0)	28.4%	30
Acceptable	Orange (10)	52.0%	40
Poor	Pink (25)	18.4%	55
Poor	Red (50)	1.2%	80

The utility is now able to develop a transformer TAI, and articulate a degree of uncertainty around the score, e.g. the transformer assessment index score is most likely to be 40 with a 52% certainty.

4.5.2 Example: Multiple unknown parameter estimation

When there is more than one unknown parameter, and the unknown parameters may each be represented by a non-linear distribution, a computational method such as Monte Carlo simulation can be used.

The Monte Carlo simulation takes the following steps:

1. For each unknown variable in the transformer assessment index, use available data from other assets or parameters to develop a suitable statistical distribution.
2. Select the suitable statistical distribution and identify the parameters.
3. Randomly choose input values from the default values and uncertainties for all input parameters from their distributions. A Monte Carlo simulation tool will be managing this.
4. Determine the transformer assessment index results from this dataset through the transformer assessment index model.
5. Randomly choose a second dataset from the distributions and repeat the sequence 3 and 4, and so on.
6. Collect all iteration results and plot it as a distribution, then use the most likely value of the median as the assessment index result and use the spread of the distribution as the uncertainty indicator.

Step 1 in this process is crucial: instead of having a single value for a data-point, a data-point should be represented by a statistical distribution. This distribution should match the expected behaviour of the data-point. Types of typically used distributions are: Normal, Weibull, Lognormal, Exponential or a discrete distribution. The distribution parameters are set by the expected behaviour and the spread is determined by the level of uncertainty. Preferably these parameters are calculated from applicable reference data. If suitable reference data is not available, the engineer's judgement can be used.

Steps 2 to 4 are repeated many times (typically thousands of times). The result of this simulation is a distribution of outcomes for the transformer assessment index results. The expected value of this transformer assessment index result distribution provides the most likely transformer assessment index value. The spread in this distribution represents the level of uncertainty in the transformer assessment index result, due to the uncertainty in the unknown input parameters. To make the simulation work all uncertain input parameters will require a distribution rather than a single scalar value. These distributions can be created using inductive reasoning or statistical inference.

For example, a utility has a fleet of 132/11 kV transformers, assessed using a 4-parameter non-linear transformer assessment index. The transformer assessment index comprises four attributes (core and coils, OLTC, bushings and oil).

The utility is trying to assess the assessment index for an aged power transformer with only two known conditions as the bushing and OLTC condition is unknown. From similar test results, the utility knows that the bushing DDF reading can be accurately represented by a normal distribution

$$\mu = 1.5\% \text{ and } \sigma = 0.35\%,$$

and that based on the operating experience of similar transformers, the OLTC has a 70% likelihood of being in acceptable condition, and 30% likelihood of being in poor condition. The core and coils and the transformer oil are both assessed as being in good condition.

Monte Carlo Simulation Results

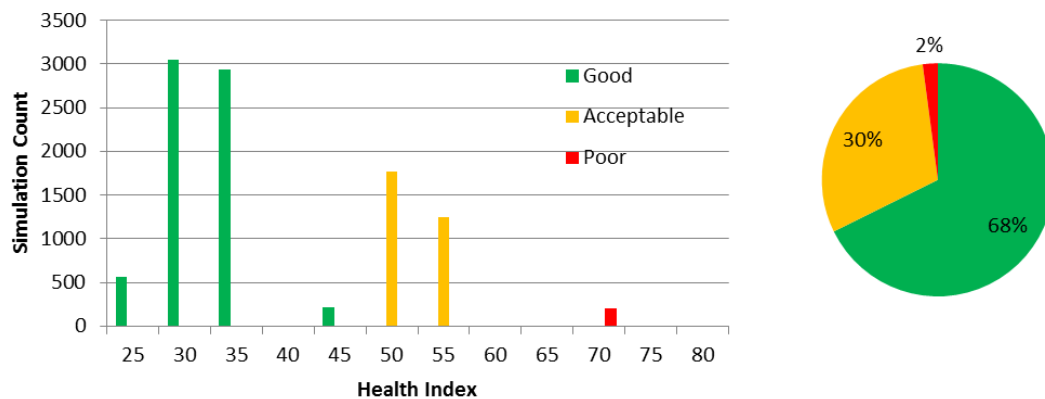


Figure 4-4 Unknown parameter distribution

Using these distributions, a Monte-Carlo simulation can be performed to estimate the Transformer assessment index, as well as the possible variance. From the simulation, a median assessment index of 35 is obtained, which the utility classifies as being in 'Good' condition. However, 32% of the simulation results indicate a possible assessment index result of 'Acceptable' or 'Poor'. The transformer expert may then decide to assign a 'Good' assessment index to the transformer, with a caveat that there is a 32% chance of the assessment index being worse; the transformer assessment index is misqualified 32% of the time.

This is quite a high level of uncertainty due to half of the input variables to the assessment index being unknown. The utility would need to undertake more testing and assessment to increase the certainty of the TAI result.

4.5.3 Using colour intensity indicators

In many cases transformer assessment index results are presented in colour schemes. If using colour coding, one can use the colour intensity to indicate the level of uncertainty. In the following example, Monte Carlo analysis is used to determine the most likely TAI result. The colour intensity code is determined by the number of Monte Carlo iteration results not matching the colour code of the most likely TAI result.

Figure 4-5 shows the transformer assessment index result distribution as "Remaining lifetime" and shows the transformer assessment index results as either Red, Orange or Green.

The concept of misqualification is illustrated by Figure 4-5. Each Monte Carlo iteration result is plotted in the transformer assessment index result distribution, and each result is assigned a colour coding. The expected value now represents the most likely result of the Transformer assessment index, in this

case green. The non-green iteration results are now an indication of the level of uncertainty of the colour green. The larger the number of non-green results in the total distribution, the less certainty of a green TAI score. In the example of

Figure 4-5 the misqualification is approximately 15%. Therefore using Table 4-4, no action is required. The level of misqualification will be presented to the engineer through the intensity of the green colour.

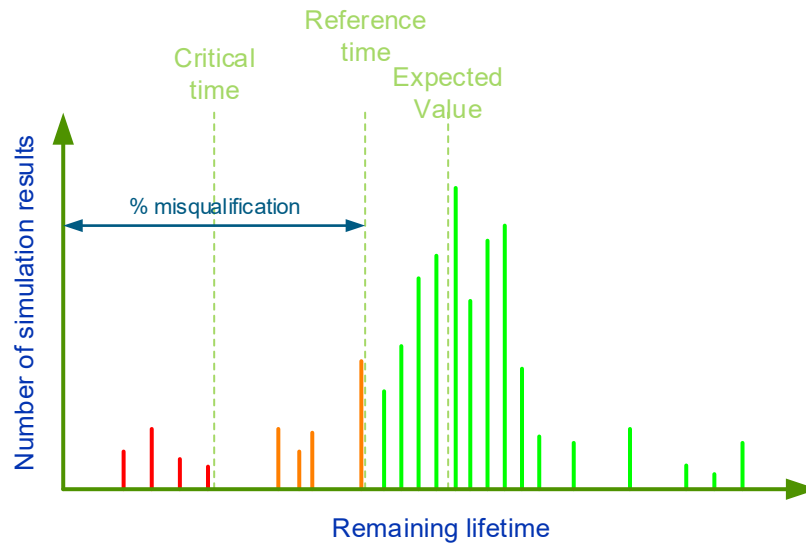


Figure 4-5 Colour misqualification example

Table 4-4 shows that for the 15% misqualification, the certainty is considered “medium,” and this determines the colour intensity of the green result. The final transformer assessment index colour indication is highlighted in the table below with a red outline.

Table 4-4 Colour intensity classification including required actions

Misqualification	Certainty	Colour & Intensity					
<5%	Very High	No action	No action	More Testing	Urgent Action	ImmediateAction	Immediate Action
5% - 10%	High	No action	No action	More Testing	Urgent Testing	Urgent Action	Immediate Action
10% - 25%	Medium	No action	More testing	More Testing	More Testing	Urgent Testing	Urgent Action
>25%	Low	More testing	More testing	More testing	More Testing	More Testing	Urgent Testing

5. The role of on-line monitoring systems

The increasing capability and decreasing cost of on-line monitoring technology for transformers and many of their components, means that it is now feasible to cost-effectively monitor the condition and operation of power transformers in near real time. For the purpose of this Chapter, “On-Line Monitoring” refers to the sampling or measurement, and where appropriate analysis of the information, from a device on or near the transformer, whilst the transformer is operating. For example, on-line systems for dielectric fluid analysis, bushing monitoring or partial discharge measurements.

It is now common practice for many medium and large sized transformers to be fitted with one or more on-line monitors. The complexity of the monitoring system is generally dependent upon the size and criticality of the transformer, and the user’s budget.

Some transformers will be fitted with only one monitoring device to monitor a combination of critical gasses. More complex gas monitors that analyse up to nine dissolved gasses and moisture are commercially available and common on larger or more critical units. Other types of monitoring systems allow the monitoring of other parameters such as moisture content, load, temperature and partial discharge. On-line monitoring devices are also available for bushings and tap-changers.

CIGRE TB 630, “Guide for Transformer Intelligent Condition Monitoring (TICM)” published in September 2015 [9], and IEEE C57.143-2012 (currently under revision) “Guide for Application for Monitoring Equipment to Liquid-Immersed Transformers provides” the reader with in-depth information and recommendations on the application of these on-line monitoring technologies.

5.1 Condition assessment using on-line monitors

On-line monitoring devices can be a valuable tool for the condition assessment of power transformers. Assessment of a transformer fitted with on-line monitors should generally be conducted in accordance with this Technical Brochure. However, the assessment can be updated more frequently as updated information becomes available from the monitoring devices.

Many on-line monitoring devices use common analysis methods to detect faults and assess the condition from the continuously measured data. For example, some of the more complex gas monitors will not only provide information about each of the measured gasses but will also analyse the results using methods such as the Key Gas method or the Duval Triangle. The Key Method is a graphical interpretation technique found in IEEE C57.104-2008 [10], to compare certain key gases found in a sample to faults conditions where these gases are predominating. The Duval Triangle method is another graphical interpretation technique used to identify the nature of a fault [11] [12]. Once the user is satisfied that the methods used by their monitor are appropriate for their transformers and sets any threshold levels, these devices can greatly simplify the process of transformer condition assessment.

It is however essential that the user understands that a complete assessment of a transformer should consider all failure modes. It is therefore necessary to consider the capability, and the limitations of the online monitoring system being used. Not all transformer components can be assessed using an on-line monitor, and for components that are monitored, the monitor may not be able to analyse all condition metrics, and therefore may be unable to detect all failure modes. For example, an on-line DGA monitor that can measure all key gasses is not able to measure particle count or the dielectric strength of the dielectric fluid.

A complete assessment of a transformer can only be conducted using on-line data in combination with other test data and information. The data from the on-line monitor that is used in the assessment can be regularly updated, based on the latest information from the on-line monitors; whilst the other data used for the assessment will be updated only when new data is available, e.g. following the next scheduled transformer outage.

Some critical failure modes of power transformers can be measured and assessed using on-line monitors. Although an assessment based only on the results of on-line monitors will never be complete, it is likely to include some of the more critical failure modes of the transformer; and may still be a very useful tool provided that users are aware of the limitations of such an assessment. For example, a user should be aware that an assessment based on the results of an on-line gas monitor does not give any information relating to any critical failure modes of transformer bushings.

5.2 Use of on-line monitors with a transformer assessment index

Users of on-line monitoring systems who wish to develop indices to help prioritise transformers within a fleet for refurbishment, replacement or other purposes, should develop the index in accordance with this Technical Brochure. In most cases, information about failure modes that are not monitored continuously will also be included in the index so that the on-line data will be used in conjunction with other information about the transformers. The data from the on-line monitors used in the index can be regularly updated, whilst the other data used in the index will be updated only when new data is available. Refer to *Section 5.1*. This method ensures that the assessments used within the index are updated as the information about some of the critical failure modes becomes available.

If a user has a large group of transformers, each fitted with one or more on-line monitors, an index could be developed using only the data from these monitors. Such an index could be continuously updated. Although an index based only on the results of on-line monitors will cover only some of the failure modes of the transformers, it is likely to include some of the more critical failure modes; and may still be a very useful tool. Again, the user must be aware of the limitations of the index.

5.3 Alarms from on-line monitors

One of the major advantages of on-line monitors is the high sampling rate of the parameter being measured (relative to the alternate test methods). This gives the user significant volumes of data to trend and analyse, new insights into the performance of the plant, and the ability to observe high rate-of change events, including rapid deterioration of condition. On-line monitoring systems can be configured to alert the asset operator when incipient failure modes are observed within the transformer that necessitates rapid decision making and possible action. If users respond promptly to these alarms, major failures can often be prevented.

It is important for asset owners and operators to understand the distinction between short-term and long-term monitoring of the transformer. Transformer Assessment Indices are intended to be used to assess the transformer over medium to long time periods, whilst on-line monitors are also useful for monitoring the operation of the transformer in real-time. If an on-line monitoring system brings to attention a condition measurement that indicates a critical failure could be imminent, users should respond quickly and appropriately e.g. de-energising a transformer after receiving a critical gas alarm. Users should not wait until an index or an assessment of the transformer is completed to respond to an alarm (or other information indicating an imminent failure) from an on-line monitor. However, a more detailed assessment of the transformers using all available data, may be appropriate before the asset manager decides how to deal with the problem transformer.

Some users of on-line monitoring devices will also use alternate tests to periodically measure the same parameters. For example, transformer owners may continue with periodic sampling of dielectric fluid for dielectric fluid quality and DGA assessment; even where an on-line monitor is installed. If the different test methods are producing comparable results, the user's confidence in the data increases. However, the different techniques, technologies and accuracy of these assessment methods may lead to different results, and therefore to different interpretations of the defect type or severity. Users would then need to understand the accuracy of the test and sampling methods used to determine which measurements are the most accurate; and then use the more accurate test results for critical decisions.

6. Criticality and other condition considerations

A transformer asset manager needs to make decisions about when and what type of interventions are required, e.g. replacement, refurbishment or additional maintenance. During this process, the asset manager needs to balance the cost of the interventions and the anticipated performance improvement, against the current risk of the transformer. Therefore, the consequence of a transformer failure will often need to be considered in conjunction with the condition assessment. Some of the consequences of a transformer failure are listed below. The list is not intended to be comprehensive but is intended only to highlight some of the issues that may need to be considered.

The consequence of transformer failure is influenced by

- **Possibility of consequential damage or injury from fire or explosion:** Fires or explosions resulting from a transformer failure can damage other transformers and equipment in the substation, but also result in damage to property outside the substation. This is a major issue if the substation is in a high-rise building or in a built-up area. If staff are regularly required to work near the transformer, or if the substation is located near an area with heavy pedestrian traffic, the possibility of severe injury or death should be considered.
- **Possibility of environmental damage:** When a transformer is located near a waterway or some other environmentally sensitive area, the risk of insulating fluids escaping into the environment must be considered. The consequence will depend on the availability of oil containment and the type of insulating fluids in the transformer.
- **Size and type of load interrupted:** Some loads such as hospitals and high-rise buildings with elevators, transit systems etc. are more important than some other types of loads. The number and type of customers likely to be affected will also be an issue. This will potentially affect revenue and the company's reputation.
- **Duration of the possible interruption:** In some circumstances, more critical loads have alternate sources of supply or are provided with back-up generators and uninterruptible power supplies (particularly ITC applications). In other circumstances, there is adequate redundancy in the system to supply the interrupted load, particularly if nearby network assets can be operated at an emergency rating. However, in some other situations, supply cannot be restored until the transformer is repaired or replaced. The duration of the outage will also potentially affect revenue, expenses and the company's reputation.
- **Time to repair or replace the transformer:** Repair will not only be a function of the extent of the damage, but of the availability of spare parts, skilled labour and suitable workshop facilities. Replacement will be determined by contractual arrangements, transformer suppliers and of course, if funding is available. The ability to replace a transformer may also be affected by limited access to the site and transport limitations. Although supply may have been restored to customers, the reliability of the system will generally be compromised until the transformer is returned to service.

6.1 Transformer assessment and consequence of failure

The condition assessment of a transformer or an entire fleet would not normally consider the possible consequences of a transformer failure, as these consequences are related to the location and application of the asset, not to its condition. As stated in [Chapter 2](#), a Transformer Assessment Index (TAI) would not normally consider, or include, the consequence of potential failures.

However, the consequence of a potential failure should generally be considered in conjunction with any condition assessment⁸. If two transformers were in a similar condition, the asset manager would more likely improve the condition of the transformer that would have more serious consequences if it were to fail.

⁸ If a user chooses to use weighting factors when developing a TAI, they may be based on the relative consequences of failure modes being considered e.g. failure modes related to Oil Impregnated Paper bushings generally have a higher consequence than some other failure modes, and this would probably be reflected in the weighting factors chosen. These weighting factors would be applied to all transformers affected by these failure modes regardless of the location or application of individual transformers.

How to assess the consequences of a potential failure and utilise these factors to make asset management decisions is beyond the scope of Working Group A2.49. However, the Working Group agrees that a Criticality (Impact) Index could be developed to help identify those transformers whose continued safe operation is most critical.

6.2 Assessing criticality & developing a Criticality Index

Assessing criticality can be challenging. For example, most users would assume that a transformer supplying a hospital would be very critical. However, the hospital may be supplied from a network with adequate redundancy to carry the load in the event of a transformer failure and the hospital may also have its own backup generation. Users must therefore consider the capability and redundancy of the network. To ensure that the criticality of each transformer is assessed accurately and consistently it is recommended that users develop a scoring matrix prior to any assessment.

Any of the scoring methods used for the development of a TAI (refer to Chapter 2) can be similarly applied to the development of a Criticality Index. The method selected will be dependent upon the user's preferences.

Further details of how to assess criticality or develop a criticality index are beyond the scope of the Working Group A2.49 and this document.

6.3 Economics of Transformer Management

The TAI is a method to identify candidates for consideration, of repair/refurbishment/replacement and/or scrapping, and as highlighted by this section, there are many other considerations in making asset management decision. CIGRE TB 248 "Economics on Transformer Management" describes a methodology that could be used in addition to the TAI, to arrive at a final decision.

7. Estimating a probability of failure

Transformer assessment indices (TAI) are a useful tool to condense a large amount of technical data and result analysis into a score that easily conveys information about overall asset conditions, and to prioritise the selection of assets for further investigation or investment. The subsequent decisions upon where and when to act are often made including an estimate of the *risk* associated with the asset not being able to perform as a result of the estimated condition.

In order to assess the risk, knowledge of two elements are required; the consequence of failure of the asset and the probability of failure (PoF) of the asset at a given point in time. The consequence of a failure is generally well understood by Asset Managers who have detailed knowledge of the role of the asset in the system, the substation design, levels of redundancy and availability of spares, and the safety, economic and social consequence of a fault or failure. Determining the likelihood of failure remains difficult to quantify.

Many transformer asset managers follow the generally accepted practice to proactively repair impaired components once detected, or replace transformers prior to failure, based on the assessed criticality in the system and the consequence should an in-service failure occur. This practice has prevented the study and collection of failure data based on condition, and as a result, the actual remaining life is rarely known.

The TAI alone can be used to derive a risk score that is not a quantified probability but can be sufficient to identify assets that should be prioritised for risk mitigation. For example, identification of assets that a pre-determined reasonable allocation of capital should be assigned to, based on a combination of condition and criticality [13]. However, if a systemic multi-asset investment process is used, a PoF is likely to be required; the TAI could be used to categorize like assets in the fleet into groups, and then assign a risk profile or PoF for each group, which can then be analysed further.

7.1 Statistical Methods

Commonly used statistical methods and models to analyse failure rates such as Weibull Analysis [14] are only suited to population studies, while more sophisticated methods (e.g. accelerated failure time models) could be used to study assets as complex as transformers. Unfortunately, it remains difficult to determine the probability of an individual unit with a specific condition, due to the difficulty in gathering data for transformer behaviour and condition. In addition, whilst the overall probability of failure is a function of all the individual failure modes, linking an individual test result is challenging, as many condition assessment results may not be attributed to a single root cause or condition.

If commonly used statistical methods are to be applied, the analysis should be limited to similar transformers i.e. transformers that are of similar designs, component selection (thus having similar failure modes), with similar operating and environmental conditions [6]. This will derive a failure rate for a subset of a transformer fleet, but it will not be possible to differentiate between transformers in the subset. However, transformers are often bespoke designs, so most transformer fleet operators will not have sufficiently large populations of similar transformers to analyse failure data using commonly used statistical methods.

Where failure statistics are only available for an entire fleet, it should be noted that transformer design tools, construction practices and the materials and components have evolved over time. These nuances and product-specific issues that alter the expected lifetime should be taken into consideration when considering the likely failure rates for subsets of transformers within the fleet. More sophisticated methods are thus needed to estimate a particular transformer's probability of failure.

7.2 Estimation techniques

Some failure mechanisms are a function of time (e.g. cumulated thermal, dielectric or mechanical stresses) whilst others are random with possibly constant or predictable probability in nature (e.g. a severe transient that instantly creates a dielectric failure). However, as power transformers are generally repaired or retired prior to failure when the condition assessment indicates a failure is likely in the near future, the likelihood of failure can be difficult to quantify.

The assessed relative conditions of a fleet of transformers (e.g. from a TAI or PoF) can help identify the different families of transformers that have a higher probability of failure. This information can be useful in determining repair and replacement plans, and the number of spare transformers for reliability and budgeting purposes.

A number of utilities have developed methods to estimate the probability of failure utilising the TAI of their fleet of transformers and the expected failure rate for their fleet. Note that these methods could also be used with failure rates derived from other utilities failure statistics or international surveys. However, users need to ensure that their fleet, their operating conditions and their operating environment are similar to those used to derive the statistical failure rates that they intend to use.

The conversion of a TAI into an estimated Probability of Failure, will only be able to give the likelihood of an event, and is not able to distinguish the possible consequences of the various failure modes. Further work to assess a likelihood of consequence may be required.

However, in the absence of relevant data needed by an appropriate statistical method, an estimated PoF obtained from available data and a commonly used statistical method may assist the Asset Manager in making an informed decision.

Two methods to estimate the PoF are included below. Whichever method is used to assign a probability of failure, the result should correlate with the historic experience of the utility. Any estimation model should be ‘calibrated’ using actual failure data where possible [15] and updated as more data becomes available. It should be made clear that the specific failure rate is an estimation, and a sensitivity analysis on the relevant values should be conducted.

7.2.1 Method 1 – Adjustment of Failure Rate

For large utilities with hundreds of similar types of transformers and a known annual failure rate, a fleet-wide relative ranking of the transformers by condition can be useful in estimating transformer failure probabilities.

This method assumes that the fleet has been consistently operated and maintained for several years, and the asset manager has historical data (with assessed conditions) for the number of in-service failures, and transformers proactively removed from service. Using this information, a reasonably accurate prediction of the number of failures in a future time period can be estimated. In other words, if the fleet has suffered 5-8 failures each year for the past 5 years, and there have been no significant changes in operational or maintenance practices, it is reasonable to expect the same range of failures next year.

In the absence of probability data for each condition category, a relative ratio (based on the experience and judgement of the engineer, as well as considering the construction of the TAI) of the likelihood of failure between categories must be estimated, based on how the condition categories are allocated, and the failure modes covered. A Relative PoF (RPOF) can be estimated from this information.

The Estimated PoF for each condition category can then be approximated by apportioning the expected failures for the fleet across the condition categories relative to the number of transformers in that category and the relative probability (RPOF) of that category resulting in a failure. The PoF for the i^{th} condition category can then be solved by the equation

$$PoF_i = \frac{\text{Annual Failures} \times RPOF_i}{\sum_{k=1}^n \text{Assets}_k RPOF_k}$$

Where there are a total of n condition categories in the TAI.

Consider an electric utility that has 1000 medium and large power transformers and a history of averaging 8 failures in service each year over the past 10 years (annual failure rate of 0.8%). Each of the transformers in the fleet is assessed, and then assigned to one of five condition categories that are used in the utility’s TAI. A series of relative probabilities of failure are assumed, based on the condition thresholds used in constructing the TAI, and the condition of assets at failure (in this example, it is assumed that the probability of failure rises exponentially as condition deteriorates). Applying the equation above allows for the estimation of a probability of failure in a given time period based on condition.

Table 7-1 Estimated Probability Calculation

Condition	# of Transformers	Relative Probability ⁹	Estimated Probability of Failure per annum
As-new	120	0.5%	0.1%
Good	320	1%	0.2%
Acceptable	380	3%	0.6%
Poor	150	10%	2%
Very Poor	30	30%	6.1%

In this example, the utility has assumed that the likelihood of a 'Poor' transformer failing is 10 times that of a transformer in 'Good' condition. After calibration based on the utility's experience, the estimated PoF for a transformer in 'Poor' condition is 2% per annum.

The outcome of this method is dependent on the assumed relative probabilities. If the utility decides to change the relative probabilities, the Estimated Probability will change. Table 7-2 Estimated Probability using Reduced Relative Probabilities includes an example using the same fleet data, but with a different set of relative probabilities; in this case, a linear increase in probability based on condition is applied, rather than an exponential increase used in Table 7-1 Estimated Probability Calculation. Note that the estimated PoF for transformers in Very Poor condition has decreased from 6.1% to 1.5%

Table 7-2 Estimated Probability using Reduced Relative Probabilities

Condition	# of Transformers	Relative Probability	Estimated Probability of Failure per annum
As-new	120	0.5%	0.3%
Good	320	1%	0.6%
Acceptable	380	1.5%	0.9%
Poor	150	2%	1.2%
Very Poor	30	2.5%	1.5%

It can be seen by comparison of Table 7-1 and Table 7-2 that the results can vary significantly depending upon the initial assumptions as there is an infinite number of solutions for this problem. Available failure data should be analysed carefully and used to ensure that the initial estimates of PoF are as realistic as possible. Furthermore, users of estimated PoFs should be aware of the possible significant inaccuracies in the estimates.

7.2.2 Method 2 – Failure/Replacement Probability versus Apparent Age

Using statistical mathematical analysis tools, it is possible to create a PoF vs. age curve using the following available data:

- major failures in service, and
- condition-based preventive retirements.

Recent CIGRE reliability studies (e.g. CIGRE TB 642 [16]) indicate that failures are random in nature and there is no correlation of the PoF with age. However, power transformers are not generally 'run to failure' as they are generally repaired or retired preventively when the condition assessment indicates a failure is likely to occur in the near future. When preventive retirements are included in the calculation, the analysis demonstrates an increase of the PoF with age. Published studies in Canada [17], and Australia [18] have all concluded that PoF is increasing with age when preventive retirements are included in the analysis. Including the preventive retirements in the estimation calibration means that this PoF vs age will mean that the result represents the probability of replacement or failure vs age. The reader should understand that such a probability does not mean that the failure will actually happen in service.

However, a probability of failure estimated from the transformers age alone does not take into account variations of assets of the same age. In order to increase the probability of failure estimation precision, transformers' condition is used [17], [19], [20]. The condition of each transformer from a subset of

⁹ These assumed initial category failure probability values are for example only and not intended to define any standard values for this approach

transformers of a similar age, would generally be different. It is assumed that transformers in poorer condition would have a higher PoF than those in a better condition.

Using the results of transformer condition assessments or a TAI, it is possible to compare a subset of transformers of the same vintage. Transformers that are in a worse condition than the average of the units of the same age, are assigned a higher “apparent age,” which would be higher than their nameplate age. Similarly, transformers in better than average condition are assigned an apparent age which is lower than their nameplate age. A PoF can be estimated for each transformer using the statistically derived age vs. PoF curve, and the assigned apparent age of the transformer.

Transformers of the same vintage are compared in the second step of this approach, so by construction it takes into account the technology advances in design tools and materials. It also links the condition assessment data with the nameplate age inherently including all environmental and electrical stresses that the transformer may have experienced with time.

Of course, the statistically derived Age vs. PoF curve will be incorrect if retired transformers (assumed to be approaching end of life) are actually in good condition. Ideally a detailed forensic analysis should be carried out on retired transformers to ensure that their condition was poor and failure was imminent, to validate inclusion in the statistical analysis.

7.3 Commentary on Operational and External Influences

Users should be mindful of the complexity of calculating or estimating a probability of failure of a given apparatus and of the potential impact of other actions that can be taken by the network operator on the probability of failure in a given period. Transformer operators, upon identification of an asset with poor condition, often apply prudent risk management techniques that influence the probability of failure. In some cases, transformer operators may apply operating restrictions with the objective of mitigating the risk. In these cases, management of system events by other measures will alter the probability of failure of the transformer.

Power systems are complex systems operating in many different conditions across the world. These different network designs and operating environments can influence failure rate; to make things even more complex, these external influences may change over time, including rate of exposure to and magnitude of external events, or change in system fault levels. Whilst not proven, utilities can experience failures when multiple conditions are met (for example, poor insulation condition combined with a voltage transient; high winding moisture content subjected to high loading on a hot day). This should be considered when estimating a probability of failure; the final result will ultimately be determined by the experience, knowledge and judgment of the transformer engineers.

8. Transformer active part

The transformer winding (active part) is subject to many mechanical, thermal and electrical stress cycles during its operating life. Accordingly, this chapter is broken into three distinct parts; thermal, dielectric and mechanical assessment. Each section focusses on condition assessments that are indicative of winding degradation due these stresses.

8.1 Solid insulation degradation assessment

The main thermal-related failure modes of the transformer active part considered in this section are cellulose aging, hot spots (winding, and paper-insulated metal) and bubbling.

It should be noted that some of these factors also affect the likelihood of dielectric and mechanical failures (but in different ways) and are discussed in the next sections. As well, some of these factors are indirectly impacted by the cooling system (i.e. cellulose aging) but the cooling system is separately discussed in a later chapter.

Dissolved gases in oil analysis (DGA), oil quality analysis, electrical tests and furans are the main indicators in the assessment of thermal failure. [21], [22]

Information on the transformer design may also give an indication of the increased risk of thermal-related failure.

8.1.1 Thermal-related failure modes

The main thermal-related failure modes are listed below.

Accelerated cellulose aging

This condition has a DGA signature of elevated or high dissolved carbon oxide gasses, Carbon Monoxide (CO) and Carbon Dioxide (CO₂). Other dissolved gasses may also be present like hydrogen and methane.

Cellulose aging is a complex function of temperature and the presence of oxygen, moisture and sludge. Cellulose will naturally age since the normal transformer operating temperature is well within the cellulose thermal decomposition range.

Cellulose aging is normally measured by degree of polymerization (DP) [23] which is essentially a measurement of the general cellulose hydrocarbon chain length. It is generally accepted by many transformer specialists that cellulose has a DP value of approximately 1200 when new, an approximate value of 1000 after the initial factory high temperature drying and approximate value of 100 - 200 when it has reached end of life.

International standards (IEC & IEEE) define the nominal temperature rise values for power transformers, which should result in transformer cellulose life expectation in the range of 35 to 40 years, depending on the loading and ambient temperatures.

For IEC 60076-2 [24], the “conventional insulation system” is formed by plain kraft paper immersed in mineral oil, which is assumed as thermal class 105 and whose nominal life temperature is 98°C. As the average ambient temperature is assumed to be 20°C, the resultant nominal temperature rise for the hottest point of the windings, so called “hotspot”, is 78K (98°C - 20°C).

For IEEE the “conventional insulation system” is considered to be thermally upgraded kraft paper immersed in mineral oil, having a thermal class of 120 and nominal life temperature of 110°C. Since the considered average annual ambient (highest monthly average) is 30°C, the resultant nominal temperature rise for the hotspot is 80K (110°C - 30°C).

The average permissible winding temperature rise is directly derived from the hotspot limit. In both cases, despite considering different solid insulation materials and different ambient temperatures, the conventional limit for average winding temperature rise is 65K, which is aimed to limit, indirectly, the hotspot temperature.

The top oil limit is defined based on the oxidation stability of mineral oil. For IEEE, where the transformer is supposed to have an oil preservation system limiting the contact of the insulation liquid with the ambient air, the nominal limit for top oil temperature of 105°C. Since the maximum ambient temperature is 40°C, the temperature rise limit is defined as 65K. For IEC, considering that some transformers may be of free breathing type where the insulation liquid is permanently in contact with

ambient air, the nominal limit of top oil temperature is defined as 100°C, subtracting the maximum ambient which is also assumed to be 40°C, the resultant top oil temperature rise limit is 60K.

These two sets of temperature rise limits are assumed to grant the nominal life expectancy for the transformers:

Table 8-1 - Temperature rise limits

	IEEE Standard	IEC Standard
Thermal class of solid materials for the conventional insulating system	120	105
Hotspot nominal temperature rise limit	80°C	78K
Average winding temperature rise limit	65°C	65K
Top oil temperature rise limit	65°C	60K

Assuming an operating condition with a maximum of 0.5% wt/wt of moisture in the solid insulation system and with constant loading that results in the maximum permissible temperature rise and constant maximum ambient temperatures, the Arrhenius curve will report an insulation end of life expectancy after approximately 180,000 hours or 21 years [4]. In practice however, variations in daily ambient temperatures and transformer loading will result in a much longer insulation life expectancy. The insulation life expectancy for a typical substation transformer in a moderate climate can be assumed to be closer to 40 years, while other types of transformers such as generation plant units with constant loading at or near the nameplate value and in hot climates, or at the other extreme, very lightly loaded transformers located in cold climates may have insulation life expectancies that are significantly lower or higher than a typical substation transformer.

Accelerated cellulose aging is then the case where the cellulose is aging at a faster rate than the considered for the nominal life expectancy, which can be a result of a higher temperature (i.e. overloading, higher ambient temperature than planned, higher actual winding hot spot temperature, or higher temperature due to degraded/failed cooling), presence of moisture in the cellulose and increased acidity of mineral oil.

Transformers are required to be designed with a maximum winding hot spot temperature rise of 80°C or 78K rise over ambient temperature. Although transformers are tested with a temperature rise type test, the winding hot spot temperature is not directly measured (unless fibre optic probes are inserted). If the winding hot spot was not calculated accurately then it is possible that the winding hot spot temperature rise is significantly greater than 80°C rise and this could lead to localized accelerated aging of cellulose in that region. Deviations from the temperatures measured by fibre optics are also frequently reported, since the hottest point of the windings (where the fibres should be located) is a theoretical concept, whose precise location is difficult to be predict.

When cellulose has reached the end of its life, paper becomes very brittle and is highly susceptible to failure due to any significant mechanical movement (inrush current, short circuit fault). The aged paper will also have reduced dielectric withstand, making it more susceptible to dielectric failure due to overvoltage or transient voltage or, depending on the degradation level, even at nominal voltage levels. Transformers with a low DP value (less than 200) and high chance of short circuit faults or dielectric stresses will have the highest risk of failure due to cellulose aging.

Different solid insulation materials and the combination of the cellulose based materials with different insulating liquids may result in different thermal classes and, consequently, different sets of temperature rise limits. These values may be obtained from the standards IEEE C57.154 [25] and IEC 60076-14 [26], which both cover transformers using high temperature insulation systems.

Hot metal

This condition has a DGA signature of elevated or high levels of dissolved Methane (CH₄), Ethane (C₂H₆) and Ethylene (C₂H₄), the “hot metal gasses.” Dissolved Hydrogen (H₂) may also be present in elevated quantities due to thermal issues but can be caused by other conditions. Elevated or high levels of carbon oxide gasses may also be present if the hot metal is in contact with cellulose insulation.

International standards (IEC & IEEE) dictate that metal in transformers should have a maximum allowed temperature of 140°C [3], at the normal cyclic loading and for conventional insulating system.

If a metal temperature is greater than 140°C, the risk of some gas generation in oil must be considered.

Most causes of hot metal gas generation are not harmful to the transformer however, the gas generation can mask other issues (or the hot metal has paper involved which is the next discussed failure mode) so it is imperative that hot metal gassing issues are investigated and resolved.

Hot metal issues can be due to improperly controlled leakage flux (heating of the clamp structure, tank or core outer packet), improperly isolated/grounded metal parts, excessive circulating currents, or overheated tap-changer contacts. All of these issues can have temperatures ranging from 150°C to 700°C and each temperature range will show a differing signature of hydrocarbon gases.

There are cases of “stray gassing oils” which can produce gases in transformers at temperatures from 105°C. This was observed as production of H₂, CH₄ and C₂H₆, very often in practice seen as rise of C₂H₆ from 50-100 ppm, up to few hundreds of ppm. There is also a special case in moderately overheated transformer cores that generate small amounts of hydrogen gas at core hot-spot temperatures as low as 110-120°C. Higher core hot-spot temperature will have higher rates of gassing. This special case has been shown to occur in overexcited transformers in high ambient conditions [27], IEC 60599 Section 4.3 [28] and [29].

Hot metal fault covered by paper

This condition has a DGA signature of hot metal gases (mainly ethylene, methane and ethane) and carbon oxide gases. If a metal hot spot is covered by paper, this is a far more serious condition and heavily increases the likelihood of a transformer failure. This could occur in an overheated bushing connection, lead joint, winding exit lead or in the winding. Each of these conditions could burn up the paper and cause increased risk of dielectric failure.

Bubbling

Bubbling occurs with the faster increase of temperature of solid insulation in comparison to the oil may lead to a non-equilibrium of the moisture in the materials eventually resulting in the formation of micro water bubbles in the windings. This can cause partial discharge activity and eventual dielectric breakdown of the insulation. The bubble evolution temperature is a function of the oil gas content (worse if saturated) and moisture content (worse with higher moisture content). Thus, with increased load, ambient temperature, moisture content and hot spot temperature, the bubble evolution temperature may be less than the permissible hot spot temperature.

8.1.2 Test and diagnostics

Below are the main tests and transformer information to diagnose the thermal-related failure modes.

Dissolved gas in oil (DGA)

The below is a very limited overview of DGA and other more detailed standards should be referred to IEEE C57.104 [10], IEC 60599 [28]. The main gas signatures are:

- Carbon Oxides - As mentioned above, as cellulose ages under normal conditions, it will produce moderate levels of dissolved Carbon Oxide gasses but if the carbon oxides are very high then this can indicate excessive cellulose aging. Per IEEE C57.104¹⁰ [10], the normal carbon monoxide and carbon dioxide limits are 350 ppm and 2500 ppm respectively. In addition, the CO₂ to CO ratio should be in the range of 5 – 10 for cellulose aging (lower ratios would indicate paper involved in a fault). The general levels of dissolved Carbon Oxide gasses in oil are a function of the relative quantities of cellulose materials such as paper insulation, wood and pressboard used in the construction of the transformer as well as the quality of oil. If possible, it is best to compare similar manufacturers and types of transformers to determine what Carbon Oxide gas levels are normal or elevated.
- Hot Metal Gassing - This is indicated by hydrocarbon gassing (methane, ethylene and ethane) with ethylene being the key gas. However, methane or ethane might be produced in greater quantities depending on the temperature of the hot metal. Acetylene is normally not produced in hot metal gassing as it requires a temperature close to 1000°C.¹¹
- Carbon Oxide gassing + Hot Metal Gassing - This is indicative of a hot metal covered in paper which as noted above is the most serious thermal risk of failure.

¹⁰IEEE C57.104 is planned for revision to significantly change these carbon oxide limits and ratios.

¹¹ Acetylene can be present in low concentration.

- Core gassing – This has a special gas signature of hydrogen and methane produced in the ratio of approximately 7 (H₂ / CH₄) [27].

For different insulating liquids (i.e. ester oils), the standard IEEE C57.155 [30] should be verified, which may require some adjustments on the values of ppm previously mentioned.

Furans analysis

Furans are a group of chemical compounds that includes 2-Furfuraldehyde and other related substances that are formed when cellulose degrades. A high furans content or a high production rate at the same or lower operating temperatures¹² may indicate a high rate of paper degradation. The furans value can be used to infer the Degree of Polymerization (DP) value and hence estimate the remaining life in the cellulose, provided sufficient time has passed (up to 6 years), following any oil treatment or replacement. – see CIGRE TB 323 and 494 [22], [21]

Another emerging chemical indicator of cellulose remaining life is methanol, see References [31] [32] [33]. In the same manner as furans, the methanol is a by-product of paper degradation and its evolution is correlated with reduction of the paper life expectancy.

Oil quality properties

Oil quality properties that can be used to diagnose the thermal risk of failure are:

- Acidity and Interfacial tension (sludging tendency) – The higher the acid number, the more acid that is in the oil. Acid will also enter the cellulose and increase the aging of the cellulose. The reduction of the IFT is caused by the presence of polar contaminants in mineral oil, typically by-products of oil oxidation. A combination of high acidity and low IFT is critical for mineral oil. For esters the limits of acidity will be different due to the lower corrosiveness of the longer molecular weight acids. Due to the higher polarity of ester liquids in comparison to mineral oils, the IFT test gives no meaningful information, presenting low to no variation along with fluid degradation. In general, the sludge formation in esters is negligible.
- Oxygen – High oxygen in oil (saturated oxygen in oil is approximately 25,000 ppm) will accelerate cellulose aging. The paper degradation rate in oxygen rich liquids may be increased in 2 to 5 times. [22], [34]
- For ester liquids, polymerisation can occur, leading to a rise in viscosity. This can be severe when oxygen is available.

Moisture in cellulose

It has been shown that the rate of paper degradation (aging), will double with every 1% increase in moisture content in the paper insulation above 0.5% wt./wt., which is the optimum moisture content in paper [35]. The moisture in cellulose thus affects the aging of the celluloses and also the bubbling evolution temperature (high moisture lowers the temperature). The moisture in cellulose can be estimated / measured per below:

- Inferred from moisture in oil – The most common method used to estimate the insulation moisture level is to calculate the relative moisture saturation value of the oil using the tested moisture PPM and the measured sample temperature with relative saturation formulae or curves such as the Oommen curves [36]. The PPM levels of moisture dissolved in the oil can vary over a wide range due to changes in the moisture equilibrium between cellulose insulation and oil which are affected by changes in ambient and oil temperatures.

The relative saturation method can provide more consistent results especially when the transformer is maintained in a constant thermal condition for several days. It should nonetheless be noted that this method must be used with caution due to the absence of thermodynamic equilibrium in an operating transformer. Sensors must be carefully located in an area of the transformers where there is an adequate flow of oil.

- Dielectric Frequency Response – The measured frequency response can be used to estimate the average moisture content in solid insulation between windings. It can also measure between windings to ground but accuracy and reliability of estimation depends of type of tested transformer, number of windings and chosen measuring connection.

¹² The partitioning of furans between paper and oil changes with operating temperature.

- Sample blocks – Pressboard samples can be removed from the transformer and the moisture in cellulose can be measured in a laboratory using one of the established methods.
- Dew Point – Dew point measurement can be used to estimate the moisture in cellulose using curves of dew point temperature versus ambient temperature. As the method measures the moisture content of the surface of the materials, it is more representative of the thin insulation regions such as the conductor insulation.

Insulation Power Factor (PF) (or Dielectric Dissipation Factor (DDF) or tangent delta

The DDF / PF of the major (main) insulation can indicate moisture in the windings. Values below 0.5% are typically considered normal for mineral oil transformers. However, the most valuable information comes from the power factor variation which is the comparison of the measured values to the earlier or initial values, since different transformers may have a characteristically higher insulation power factor value than compared to typical cases.

Winding DC resistance

High winding resistance can indicate a bad lead joint or OLTC contact resistance variation which could be causing hot metal gassing. The measured winding resistance should not differ more than about 1% compared to the factory test report, if the winding temperature at measurement conditions is corrected to the factory conditions. Difference between phases should usually be less than 2-3%. [5]. The windings should be tested in all tap positions to detect high resistance joints in tapping leads. Prior to making a conclusion though, the measurement circuit and the contact resistance for the measurement devices must be carefully assessed to ensure reliable and comparable results.

Thermal scans of the transformer

Transformer thermal scans can indicate tank, bushing connections or bus duct overheating which could be the cause of hot metal gassing. Thermal scans can also show if there are cooling issues which could be causing higher oil temperatures.

Core and core structure grounding check

For transformers equipped with external core and/or core structure grounding, the testing of the insulation resistance should be verified. The failure mode of this insulation component can be detected with the measurement of insulation resistance.

Corrosive sulphur

The presence of *potentially* corrosive sulphur compounds in the transformer oil under *appropriate* thermal conditions *and winding materials (bare or un-enamelled copper)* can lead to the development of copper sulphide [37] [38]. The presence of corrosive sulphur compounds in the transformer oil can lead to the decrease of dielectric integrity of the active part, by the formation of metallic sulphides in operating transformers. Depending on the type of sulphur compounds in the oil, different thermal conditions and construction metals present, silver, copper, and/or winding paper will be affected by sulphides deposition.

8.1.3 Design information

Paper type

The cellulose paper type will have a large impact on the rate of cellulose aging. In particular, kraft paper has a much higher rate of cellulose aging than thermally upgraded cellulose for a given operating temperature. Thermally upgrading of paper may reduce the ageing rate by a factor of 1.5 to 3. The ageing acceleration from the presence of water is seen to be strongly reduced for upgraded paper, while ageing with access to oxygen seems to be less influenced. It also appears that thermally upgraded paper does not produce furanic compounds to the same extent that normal kraft paper does. In addition, it can be important to note if the windings do not have cellulose paper covering and hence would not have a cellulose end of life. This could occur if the windings are insulated with Nomex, netting tape or enamelled bare copper.

Oil preservation system

The type of oil preservation system (sealed or free breathing) has an impact on the DGA interpretation (gases will dissolve differently) and on the amount of moisture and oxygen in the oil (which affects the cellulose aging and fluid thermal degradation).

Temperature profile of the transformer

Historical transformer temperature data (top oil or winding temperature) can be correlated with the instantaneous loading and ambient temperature to assess the proper thermal performance of the transformer. This can show if the transformer temperature has exceeded the standard limits which could then be assumed to have had an effect on the cellulose aging.

Core overheating calculation

For the special case of core overheating causing hydrogen/methane gassing, this risk can be determined by calculating the core hot spot temperature. This is a function of the core excitation (note that it may be greater than 110% for generator transformers), number of core cooling ducts, core geometry and top oil temperature. If the core hot spot temperature is greater than 130°C in certain conditions, than core gassing can occur. See IEEE Transactions on Power Delivery [27].

The core overheating is characterized by 1 – 3 ppm per day of gas generation (H₂ and CH₄) with an H₂/CH₄ ratio in the 6 – 8 range.

Transformer winding hot spot recalculation

If the transformer design is available, the winding hot spot temperatures can be recalculated using present day transformer design tools. This can especially show more accurate winding hot spot temperatures (and higher values) if a 3D Finite Element Method (FEM) calculation of the leakage flux was not available at the time of the original design. See CIGRE TB 659 “Transformer Thermal Modelling” [39] for more information.

Temperature rise type test report

This report has winding average temperature rise which is based on many assumptions with indirect measurements. The hot spot temperature is not measured (unless fibre optic sensors are installed in the proper windings locations).

8.1.4 Failure mode assessment

Dielectric condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

Below is a summary of the information recommended for a basic fleet assessment for the transformer active part - solid insulation degradation.

- Loading profile and ambient temperature registers
- Dissolved gas in oil test
- Oil quality results including moisture and acidity
- Estimated or measured moisture in cellulose
- Furans analysis
- The DDF / PF of the major (main) insulation
- Measured DC winding resistance
- Corrosive Sulphur in oil test

Below are optional items that can be included if possible:

- Calculated winding hot spot temperature
- Calculated bubbling evolution temperature

8.1.5 Scoring Matrix for thermal Assessment

The generic scoring matrix illustrated in 2.3 has been adapted to more easily allow the assessment of thermal related failures.

	Scoring matrix for thermal assessment
F	Transformer should be removed from service due to significant and serious test results indicating actual breakdown of insulation.
E	Indications show end of life aged insulation with multiple tests showing very high risk of failure
D	Indications show significantly aged insulation with additional test results showing high risk of near term failure
C	Indications show aged insulation with increased risk of dielectric failure
B	Indicators show small deviation from new, however follow normal degradation behaviour. Normal operation is expected for long term.
A	Normal - All test results normal: no trace of acetylene; normal levels of other gases and no indication of problems from electrical tests.

8.2 Dielectric assessment

The transformer active part dielectric risk of failure is based upon the partial or complete breakdown of the insulating material of both the solid and liquid insulation. A transformer's primary function is to change the voltage from one circuit to another circuit. The transformer will have windings and components at different voltages including ground voltage. This results in dielectric stresses which must be controlled using insulation materials. When the dielectric system fails there will be partial discharges and faults. Partial discharge is defined as a localized electrical discharge that only partially bridges the insulation between conductors and which can or cannot occur adjacent to a conductor [40]. A fault occurs when the insulation completely breaks down and the voltage between the electrodes collapses to zero and fault current flows until the transformer is disconnected from the system.

Insulation systems in power transformers can be broken into four categories:

Major insulation – This is the insulation between windings and from windings to ground.

Minor insulation – This is the insulation within a winding.

Core insulation – This is the insulation within the core and insulation from the core to ground.

Insulating liquid – This is the dielectric quality of the insulating liquid. Refer to the oil chapter for more information on dielectric quality measurements.

8.2.1 Dielectric failure modes

The main dielectric failure modes are listed below:

Failure of minor insulation

Failure of the minor insulation is likely to cause serious damage to the winding and is likely to require transformer replacement.

- Turn to turn failure in winding

The individual turns in a transformer winding will be at a different power frequency voltage than the adjacent turns based on the voltage rating of the winding and the number of turns. Under impulse conditions, the turn to turn voltage will be based on the voltage distribution in the windings. The impulse voltage distribution can be improved by the use of static rings at the winding ends, winding geometry and special winding designs such as interleaving and floating conductors which increases the series capacitance of the winding. A turn to turn insulation failure will cause short circuit current in

that winding location and the whole winding will be destroyed by either electromagnetic forces or by consequential insulation breakdowns due to gas generation in the failure location. A turn to turn insulation failure will also change the voltage distribution in the remainder of the winding often leading to complete breakdown of the insulation between adjacent windings or between the winding to ground potential.

- Disc to disc failure in winding

Disc to disc failure in a winding is typically found in the high voltage windings where groups of turns are wound together as a disc and each disc is separated by insulation spacers to provide dielectric clearance as well as space for oil flow for cooling purposes. Both ends of the disc will have a crossover where the conductor is brought from one disc to the next to continue the winding. There is very little disc to disc stress at this point, however at the opposite end of the disc there will be a high voltage potential between adjacent discs which could lead to partial or complete breakdown.

Failure of major insulation

Failure of the major insulation or tracking/partial discharge is likely to require transformer replacement or major work.

- Inside winding to core

Normally the inner winding of a transformer wound with concentric winding construction is the lower voltage winding. This minimizes the insulation space required to isolate the inner winding from the core leg which is at ground potential. However, there is a potential difference between the inner winding and the core which can cause the insulation to break down partially or completely. This failure mode often follows a through fault event in which the inner winding buckles at some point reducing the clearance to the core leg.

- Winding to winding (high to low) failure (major insulation)

Windings in transformers are at different voltages. The voltage stress between windings is controlled by a combination of solid insulation barriers and oil ducts created by insulation spacers. The dielectric stress at the ends of the windings can be higher due to the stress enhancement at the edges of the windings. This is often controlled by using contoured insulation material. The winding to winding insulation system must be designed for the voltage stresses found during normal operation at line frequency and under impulse and switching surge conditions. Winding to winding insulation can partially breakdown (partial discharges) or completely breakdown.

- Winding to top yoke failure (major insulation)

Transformer windings must be insulated from the grounded core material. Many windings have the highest potential at one end of the windings which results in a voltage stress between the end of the winding and the core and its associated clamping structure. Just as for the winding to winding space, this insulation system is often made from solid insulation barriers separated by insulation spacers. Typical failure modes are creep discharges along the surface of the barriers and spacers or oil breakdown between the barriers. These result in partial or complete breakdown of the windings to ground.

- Winding lead to ground failure (major insulation)

The winding lead will be at the highest voltage. This lead needs to be insulated dielectrically as well as supported mechanically. Often for extra high voltage transformers, a series of concentric insulation barriers separated by spacers is utilized as the "lead exit" insulation system. As in the winding to winding and winding to ground insulation systems, there will be dielectric stress both along the surface of the insulation (creep) and in the oil space between the barriers which can either partially breakdown or breakdown completely.

- Phase to phase failure (major insulation)

For three phase transformers there will be a potential difference between the winding of one phase and the winding of an adjacent phase. This will create a dielectric stress that must be accounted for. Typically, a solid insulation barrier is placed between phases. For higher voltage transformers, a system of barriers and spacers is utilized. As with other insulation systems, there will be creep and oil gap stresses. Creep stresses are particularly high at the ends of the windings.

- Winding to tank failure (major insulation)

For both single phase and three phase transformers there will be a potential difference between the outmost winding and the grounded tank wall. This will create a dielectric stress that must be accounted for. Depending on the voltage of the outer winding and the design, a solid insulation barrier is placed between the winding and the tank. For higher voltage transformers, a system of barriers and spacers is utilized. As with other insulation systems, there will be creep and oil gap stresses. Creep stresses are particularly high at the ends of the windings.

Failure of core insulation

Failure of or arcing/sparking/partial discharge in the core insulation may require transformer replacement, major work or corrective maintenance depending on the nature of the fault.

- Core to clamp insulation

In order to minimize circulating currents within the core, the core is typically grounded at only one single point. An insulation material is placed between the steel clamping material and the core. Although the potential difference is small in comparison to the major insulation spaces (high to low and to ground), a dielectric stress does exist which can breakdown partially or completely. Complete breakdown of core clamp insulation normally does not cause a transformer to trip offline however it can lead to carbon generation (which can lower oil and winding insulation dielectric strength) and high rates of gas generation. Circulating currents can burn the core lamination insulation leading to an increase in circulating currents and eventually can lead to complete breakdown due to the high generation of gasses and subsequent lowering of the strength of the insulation system.

- Core lamination insulation failure

Each lamination in the core has insulation material on the surface which provides electrical insulation to adjacent laminations. This limits the circulating currents within the core. This insulation material can breakdown which increases the circulating current resulting in higher losses. Failure of the core lamination material does not result in a transformer fault except in extreme cases (see failure mode - core to clamp insulation) but will increase core losses and gasses.

Failure of liquid insulation

The liquid insulation in a power transformer must withstand the dielectric stresses subjected to it during normal and abnormal operation. When the dielectric strength of the liquid becomes too low due to moisture, contamination or aging, it can either partially or completely breakdown. It is important to test the liquid insulation to assess its condition as well as to identify any gasses that could have been produced by partial or complete breakdown. The liquid should also be tested for stray gassing, as the increased production of gasses can interfere with transformer diagnostics. The one advantage of the liquid insulation system is that it can be processed or replaced. Thus poor liquid insulation quality is not included in this section and instead it is considered in Oil Quality in Chapter 10.

Both IEC and IEEE have transformer oil maintenance guides which give recommendations for oil reprocessing or replacement based on analytical tests of oil removed from the transformer [41] [42].

8.2.2 Test and diagnostics

Dissolved Gas Analysis (DGA)

DGA is an important indicator of dielectric breakdown failure (considered essential). The main gas signatures are:

Hydrogen- Hydrogen is the key gas for partial discharges in oil or gas. Typically there will be methane gas present as well.

Carbon Oxides – Dielectric breakdown inside the solid cellulose insulation material will generate carbon monoxide and carbon dioxide.

Sparking Gasses: Partial discharges of the sparking type (IEC 60599 D1 [43]) will create hydrocarbon gasses

Arcing Gasses: Complete breakdown of the insulation material, particularly from winding to winding or winding to ground will result in an electrical arc with high levels of acetylene as well as other hydrocarbon gasses and hydrogen.

Oil physical properties:

Dielectric Strength –The dielectric strength of the oil needs to be sufficient to maintain the proper insulation strength of the oil/paper system [44].

Dissipation factor –The oil should be tested at 25 and 100 degrees centigrade to ensure that it has no contaminants and will provide sufficient insulation strength [45].

Moisture Content – The insulating performance of the oil is strongly influenced by the moisture levels [46].

Dielectric dissipation (power) factor The dielectric dissipation (power) factor of the main insulation between windings and windings to ground can indicate moisture and/or contamination in the winding to winding or winding to ground insulation system. Measured values below 0.5% are considered normal however the measured values should be compared to earlier measured values as some transformers can have a characteristically high insulation power factor value. Generally, measured values that are greater than 0.5% are considered deteriorated insulation and values greater than 1% are considered very poor. (Section similar to above, Insulation power factor (or Dissipation Power Factor or Tangent Delta)

Excitation current

Excitation current is a measure of the core characteristics but can also be influenced by shorted or open turns. If the excitation current is less than 50 mA, the difference between the two higher currents for a three phase transformer should be less than 10%. If the excitation current is greater than 50 mA, then the difference should be less than 5%.

Turns ratio

Turns ratio is a measure of the effective voltage transformation from one winding to the other. The turns ratio should be within 0.5% of the calculated turns ratio based on the nameplate [47]. Turns ratio problems can be an indicator of shorted or open turns.

Dielectric frequency response (DFR)

DFR is similar to an insulation dissipation factor test except that it is measured over a wide frequency range. The purpose of this test is to estimate the moisture content in the solid paper insulation. The conductivity of the oil as well as other contaminants in the insulation system will also influence the results.

Insulation resistance

Insulation resistance is a DC test sensitive to high moisture or contamination in the solid insulation. The DC insulation test is also used to confirm that there is only one core ground (the intentional ground is disconnected for the test and a very high insulation resistance should be measured between the core or clamp and ground). The polarization index is determined by the ratio of the resistance after ten minutes to the resistance after one minute. This ratio can also indicate moisture and/or contamination.

Partial discharge

When the dielectric system fails there will be partial discharges and faults. Partial discharge is defined as a localized electrical discharge that only partially bridges the insulation between conductors. Because partial discharges are caused by dielectric stress they need to be measured when at a minimum, operating voltage is present. This test is done off-line by applying voltage to the transformer by an external power supply or on-line at operating voltage.

Frequency response of stray losses (FRSL)

The leakage reactance is measured over a frequency range of 400 to 15 Hertz. The plot of leakage reactance to frequency is compared for all phases. If one phase is different, this can be caused by shorted strands in a multi-strand winding.

Bubble evolution temperature

The temperature at which a bubble in the oil can evolve is calculated with curves against moisture in cellulose (See C57.91.2011 [3] and CIGRE TB 349 [35]). Essentially high moisture in cellulose will give lower temperatures at which a bubble may form which effectively limits the overload capability of the transformer. If this bubble evolution temperature is exceeded by a higher hot spot temperature in

the transformer, then the transformer has a higher risk of dielectric failure caused by the bubble evolution.

8.2.3 Failure mode assessment

Dielectric condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

8.2.4 Scoring Matrix for thermal Assessment

The generic scoring matrix illustrated in Section 2.3 has been adapted to more easily allow the assessment of dielectric related failures.

	Scoring matrix for dielectric assessment
F	Severe partial discharge fault in main tank, likely to lead to imminent failure. Electrical tests indicating actual breakdown of insulation.
E	Indications show arcing/sparking/tracking or partial discharge fault of the major insulation is getting worse and indicating failure on short term.
D	Indications show steady arcing/sparking/tracking or partial discharge fault but not getting worse with time, indicating potential failure on medium term.
C	Indications show dormant or intermittent arcing/sparking/tracking or partial discharge fault e.g. Core insulation deterioration
B	Indicators show small deviation from new, however follow normal degradation behaviour. Normal operation is expected for long term.
A	All test results normal: no trace of acetylene; normal levels of other gases and no indication of problems from electrical tests.

8.3 Mechanical assessment

A mechanical assessment of the transformer active part considers the structural integrity of the core and coil assembly, its alignment and position within the tank, and its ability to withstand the electromagnetic forces exerted within that structure when the transformer endures a severe overcurrent event. A mechanical assessment of the active part should not overlook the following componentry where present:

- Leads and connections to terminations and tapchangers (including electrostatic shields, shrouding barriers, supporting cleat work, barrier boards)
- Auxiliary equipment such as voltage transformers, current transformers, fault limiting reactors, preventive autotransformers
- Internal surge arrestors (Non-linear resistors / 'metrosils')
- Flux equalising windings on yokes or non-wound limbs
- Flux collectors and flux shunts
- Devices used for supporting, clamping or anchoring the internal assembly within the tank

Before making a mechanical assessment of a power transformer, it is necessary to know whether it is core-form or shell-form construction. The inherent high short circuit withstand capability of shell-form

construction together with its enveloping, rectangular core build (secured or clamped within the transformer tank structure) means that a mechanical assessment for shell-form is less complex than for core-form construction.

In comparison with core-form, shell-form experiences moderate short circuit stress because:

- i. the phase assembly has more coil groups and, being arranged side by side, they oppose each other under short circuit, keeping the phase assembly in compression,
- ii. the short circuit forces are distributed over the very large area of each pancake winding, and
- iii. the massive shell-form core and its integral fit with the tank structure, easily supports the phase assemblies

The mechanical assessment of shell-form transformers is normally confined to performing electrical testing to discern any disturbance to the geometry of the pancake windings and performing an internal inspection to check for looseness of wedges between each phase assembly and the core and observing mechanical damage to tapping leads and line lead exits of the phase assemblies.

Even with the mechanical assessment of core-form construction, it is helpful to know whether the winding design has multiple layer windings or disc windings or helical windings. Each winding type has special considerations for its short circuit withstand capability and each have particular failure modes. It is recommended to use transformer design reviews to learn these aspects whenever a new transformer design is procured. Later in the service life, these understandings assist in assessing the likely mechanical condition, especially when the transformer has been subjected to short circuit events of known severity and duration. There are other severe event conditions (such as synchronising errors) that could also be considered.

In order to make a mechanical assessment of core-form construction, the function and importance of clamping the transformer windings needs to be understood. Coil clamping is an important attribute of the mechanical strength of the active part. Applying the correct coil clamping pressure allows for:

- i. the complete internal assembly (core and coils) to be lifted by a crane,
- ii. the bracing of the transformer windings during transportation,
- iii. the reduction of the operating 'load noise' level of the windings and
- iv. most importantly, assists the transformer to survive short-circuit events.

Importance of coil clamping for the short circuit withstand of a transformer

During a short-circuit event on a transformer, the windings are effectively in unstable equilibrium. The coil clamping helps to maintain a near equilibrium (or minima of the short-circuit forces). Retaining the relative position of the windings, stabilizing their length and maintaining the residual clamping force is very important for short-circuit withstand of the transformer. Under a short-circuit fault, the windings repel and exert large axial and radial forces that effectively try to separate the windings. The size of the coil clamping force is designed to be commensurate with the expected magnitude of axial forces exerted during the short-circuit event. This keeps the windings in compression and therefore stable. However, any alteration to a winding's axial length, or its relative axial position through dislocation or collapse of the winding conductors, is likely to loosen the coil clamping. Furthermore, these changes also alter the winding ampere-turn distribution. Any change in the relative position of the ampere-turns will alter their fine positional balance and symmetry (between the winding groups) and that invariably results in substantial increases in the short-circuit forces. For these reasons, short-circuit withstand capability of a transformer is dependent on the status of its coil clamping and therefore coil clamping should not be compromised [48].

Relaxation of coil clamping

Coil clamping will be compromised if the residual clamping pressure has relaxed enough to then allow dislocation or mechanical deformation of the windings during a short-circuit event. Coil clamping may also be compromised either by the effects of repetitive short-circuit events or a single protracted short-circuit event. This can be due to the cumulative plastic deformation of the axial insulation components, or to a partial winding collapse or to conductor tilting. However, the most common way that the coil clamping relaxes is with shrinkage of transformer cellulosic insulation through ageing.

Paper shrinkage allows gradual relaxation of the residual clamping pressure. Despite some limited elasticity or recovery to expand again, the structures can still become loose. This can occur either during the vigorous oscillations of a short-circuit event or by eventually vibrating out after clamping pressure is lost. Not all of the insulation structure is necessarily keyed, glued, tied or dowelled to

prevent this dislocation. Furthermore, repetitive short-circuit events can cause cumulative bruising and dislocation of components in the insulation structure. Should an axial or radial spacer or support block be displaced due to these cumulative effects, or whether it simply vibrates loose (having lost its clamping pressure), then the void this creates provides room for winding dislocation during short-circuit events, resulting in even greater short-circuit forces to be exerted and further winding damage.

Transportation damage to the active part

Transport damage ought to be detected prior to transformer commissioning, however mechanical damage can sometimes go undetected. There are two aspects to the mechanical failure of the active part during transport. Firstly, the active part is typically locked, pinned or braced within the tank enclosure for securing it during transport. Of necessity, these points of contact with the tank employ solid insulation for electrical isolation, and these brittle materials are potentially subject to large mechanical forces during transport. Should the brittle insulation material shatter or split, then the entire active part could dislocate within the tank. Secondly, even if still secured by the fore-mentioned locating points, the core and coil assembly itself may distort, partially collapse or eject insulation blocks because of shocks, vibration or swaying motion during transport. Functional mechanical design of the active part usually focuses on withstand to short circuit through events and the ability to lift the completed assembly by crane. For that the phase assembly relies on its retention by axial compression, the radial support back to the core, and the hoop stress withstand of the conductors. Subjecting the active part to sudden shock, vibration or swaying may introduce very unusual mechanical stress to the structure. The core and coil assembly design may not properly address this transport stress, especially where the inertia of the heavy mass of the winding conductor may readily disturb the end insulation structure that is comprised of paper based collars, washers and block work insulation. Also, shipping transport in very heavy seas can conceivably establish inertial swaying of the massive core limbs because of the inherent ease for flexing of the laminations. Cumulative dislocation of the insulation, core limb and winding structure may then occur with the repetitive rolling motion during sea transport and this may not even register as a shock loading. Refer to CIGRE Technical Brochure 673 [49] for more information on transport damage.

Mechanical damage to windings and insulation structure

Short circuit events external to the transformer create severe overcurrent's that will interact with the magnetic leakage field permeating the winding structures, thus exerting electromagnetic forces on all those winding turns that are carrying the overcurrent.

In transformers of core-form construction with two concentric windings, the ampere turns are usually uniformly distributed and balanced, resulting in a magnetic leakage field that is predominantly axial over the greater part of the winding height. During external short circuit events, the axial leakage field tends to reduce the diameter of the inner winding (causing free buckling or forced buckling) by radial compression, and it tends to increase the diameter of the outer winding (causing hoop tension) through radial expansion. At the winding ends, the magnetic leakage field diverges (or fringes) creating a significant radial component in the leakage field that tends to compress the windings axially. Refer to Figure 8-1.

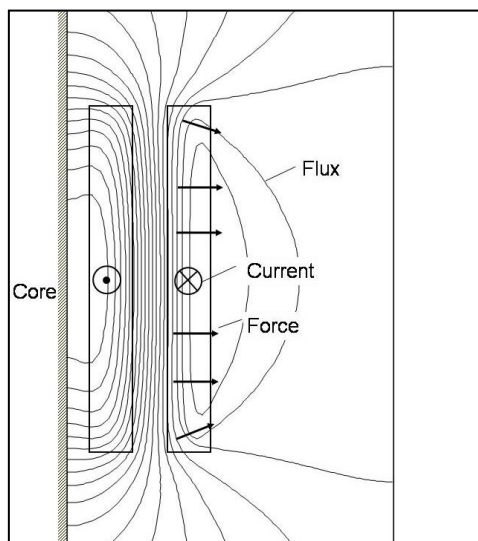


Figure 8-1 Electromagnetic force

The electromagnetic forces are oscillatory and act instantaneously. Because of their dependence on the square of the current, electromagnetic forces are unidirectional.

Because the windings are structurally elastic in the axial direction, they act as a tall stack of massive discs, each restrained by axially oriented damping springs (the insulation spacers, conductor paper insulation and the end insulation structural buffers). These springs contract when the peak axial force is attained and relax as the force cyclically drops to zero.

These endthrusts can severely damage the windings themselves and their end insulation structures. The greater the relative axial displacement, the greater is the magnitude of the end thrust because the radial leakage field is further concentrated at each winding end. Maintaining the symmetrical positioning of the winding groups is therefore imperative for minimising the end thrust forces [50]. Accordingly, the typical mechanical failure modes in windings caused by through faults can be classified as either radial failure modes or axial failure modes.

8.3.1 Radial failure modes

Hoop tension

Outward radial force causes stretching of outer conductors that produces excessive hoop tension in the conductor - this may break the conductor insulation or lead to axial collapse due to local bulging of conductors beyond the outer radial edge of their spacer supports.

Forced buckling

Inward radial force causes collapse of the inner winding due to beam failure which can break the conductor insulation or crush the axial duct insulation structure on the inside diameter of the winding. The inward radial force exerts compressive stress. If that stress exceeds the elastic limit of the conductor material, the conductors bend radially inwards between the axial strips supporting the inside diameter of the winding, producing a shape resembling a multi-pointed star Figure 8-2.

Free buckling

This is another collapse pattern caused by inward radial force being exerted on the inner winding. The inward radial force exerts circumferential compressive stress which causes the conductors to bulge inwards as well as outwards at one or a few specific points around the circumference Figure 8-3. This is essentially an unsupported buckling mode and the deformation extends along most of the winding length. The wavelength of the buckle does not correlate with the spacing of the axial sticks supporting the radial spacers. Free buckling occurs when the conductor strands possess very high rigidity such that they must buckle first before any appreciable force is exerted inwards onto structures that might support the inside diameter.

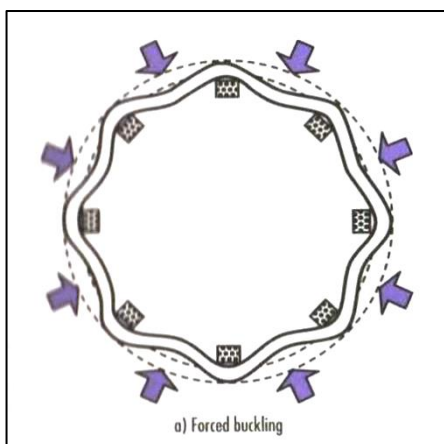


Figure 8-2 Forced buckling distortion of coil circle

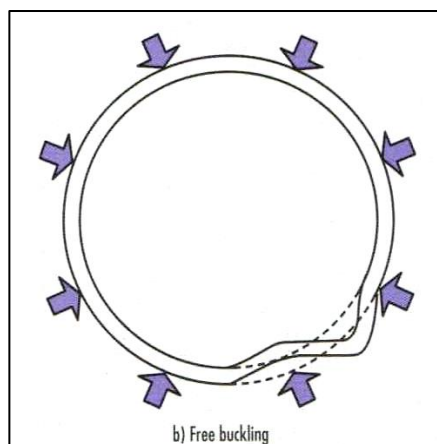


Figure 8-3 Free buckling distortion of coil circle

8.3.2 Axial failure modes

Tilting

Axial forces cause collapse of the winding due to conductor tilting. It is a failure mode mainly seen in disc windings where the whole set of conductors within the radial depth of a disc all tilt over in the same direction whilst the conductors in the adjacent disc (above or below) all tilt in the opposite direction. This results in the classic zig-zag pattern observed in this failure mode Figure 8-4. This

failure mode causes the conductor insulation to break, loss of clamping pressure, winding structure instability and even greater forces as a result of mmf imbalance and positional offsets when the conductors significantly dislocate. Thin conductors are more susceptible to tilting.

Beam failure

Axial forces exerted on each winding conductor act like a distributed load on a beam (the conductor) that is supported between the columns of winding spacers Figure 8-5. The conductor beams deflect and flex as in classic bending moment stress mechanics. Beam failure can break the conductor paper insulation.

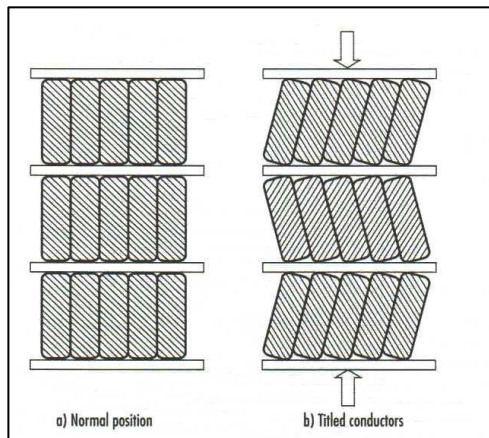


Figure 8-4 Conductor tilting

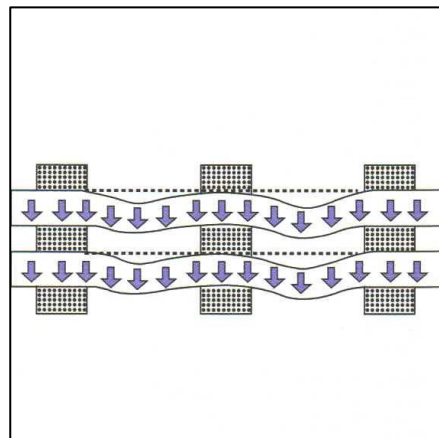


Figure 8-5 Beam failure of conductor span between winding spacers

Telescoping

A failure mode sometimes seen in narrow windings, especially winding radial groups used for layer windings, tertiary windings or tapping windings. This failure mode occurs when axial force drives some of the winding turns axially past adjacent turns. Telescoping of the winding is likely when narrow windings have radial looseness or inadequate axial support.

End insulation collapse

A failure mode where the hammering of a winding radial group into the top or bottom end insulation causes the winding end supports to collapse. The collapse occurs due to either over-stressed materials, misalignments or instability within the support structure (e.g. columns of spacers or block stacks), which can result in gross axial deformation of the winding, or magnified end thrust forces. The electrostatic shields located at the coil ends of HV windings can be impacted and damaged, especially the braid, strap or pigtail tie to the line lead. Lead exits, often with their complex insulation structures (e.g. snouts) will also be susceptible to mechanical damage by end thrust forces, let alone the electromagnetic forces on the line leads themselves complicating the situation. Relaxation of the residual axial clamping force may also allow excessive vibration, even under normal service conditions. Vibration, or alternatively the impacts of repetitive through fault events, will cause cumulative bruising of the axial support, gradually exacerbating the degree of collapse. Accordingly, a transformer subjected to a series of 'soft' repetitive through faults can ultimately fail without ever having seen an external short circuit event with full fault current (where the transformer impedance predominantly limits the current).

Fretting of conductor insulation

A failure mode where mechanical deterioration of conductor insulation may result from impact forces (hammering effect) or friction resulting from the relative movement between paper covered conductors and solid insulation or from deflection and bending of unsupported spans (causing permanent set deflection and plastic deformation of the conductors). The conductor insulation is sometimes reinforced by manufacturers where inter-disc crossovers or the transposition of paralleled conductors creates localised pressure points between conductors during external short circuits or repetitive through faults, yet these areas may remain susceptible. Misalignment of axial spacers (spacer lanes skewed or haphazard alignment) may also exert excessive stress on conductors and directed oil flow washers. This failure mode is extremely difficult to observe or detect and therefore usually impossible to assess due to the windings being inaccessible.

Spiralling of helical windings

A failure mode mostly seen in helical windings occurs when the end turns of the helical winding rotate away from their correct position. This failure mode occurs when axial force is applied on the end turns causing axial dislocation of the turns. Axial stretching of a helical turn tries to reduce its mean turn diameter, rather than elongate the conductor. However, the resultant effect on a helical winding having solid radial support will instead be to fore-shorten the turn, appearing as resultant rotational movement (or unscrewing). This failure mode may sometimes result in mechanical damage to top and bottom lead exits on helical windings due to this rotational dislocation and stretching of the lead connections. The greater the number of helical turns displaced axially, the greater is the cumulative rotational displacement.

Below are the main tests and transformer information to diagnose the mechanical risk of failure.

8.3.3 Tests and diagnostics

Frequency response analysis

Frequency response analysis (FRA) is an important test because it is sensitive to whole winding group dislocations, telescoping of windings, as well as some localized winding dislocations or movement, and is generally used to detect geometrical changes and electrical short-circuits in the windings caused by external short-circuit events. Provided sufficient movement of winding conductor occurs to alter the frequency response of the resistance, inductance and capacitance (RLC) network parameters of a winding, then FRA may be able to discern the presence of failure modes such as severe transport damage, buckling, hoop stress distension, telescoping, spiralling, and end insulation collapse caused by winding movement.

Interwinding capacitance

Interwinding capacitance can be sensitive to major distortion in the radial separation distance between a pair of coaxial cylindrical windings. Changes in the winding length or vertical position of one winding may also change the interwinding capacitance. Dimensional changes of a few percent are required to detect a recognizable change of interwinding capacitance, especially in three phase transformers where the capacitance of one phase contributes only a third of the total measurement. This test may detect the presence of severe forms of the buckling, hoop stress distension, telescoping and winding end thrust collapse.

Leakage reactance

Single phase leakage reactance can be sensitive to major displacement of the ampere turn positions of the windings. This is because the leakage field has been distorted and the ampere turn positional balance altered. The test is performed on each phase to facilitate comparisons between the phase assemblies.

Dimensional changes of a few percent are required to detect a recognizable change of the leakage reactance. This test may detect the presence of severe forms of transport damage, buckling, telescoping and end insulation collapse caused by winding movement.

Methods used to survey residual clamping pressure

An inspection of the internal assembly makes the following survey possible:

- i. Alignment of the phase assembly heights
- ii. Measurement of phase assembly height to a common datum
- iii. Checks for spacers, blocks and packing shims fallen out of position
- iv. Checks of alignment of spacers, block and shims
- v. Observation of relative winding movement or distortion of shape
- vi. Hand looseness of coil clamping devices
- vii. Hand looseness of any block work in the end insulation structure
- viii. Tonal differences when accessible block work is lightly tapped
- ix. Ease to dislocate accessible block work when again lightly tapped
- x. Measurement of residual clamping pressure (where this is practicable)

However, the extent of the coil clamping survey performed on a transformer may be limited by the available 'hands-on' access to the internal assembly. Access may be restricted by having only a few access covers, or by the limited reach of confined space access or the ability to remove the

transformer's bell cover or lid. For these reasons, such surveys are usually performed as a part of other major work on a transformer. Note that the required skills, resources and expertise increases in complexity down this list. This survey may detect the presence of transport damage, relaxation of clamping pressure, dislocated end insulation structures, hoop stress distension on an outside winding and severe forms of end insulation collapse caused by winding movement.

Measurement of the residual clamping pressure

The measurement of residual clamping pressure is not always practicable. However, any coil clamping arrangement that is adjustable, inherently had to have some way of measuring the force, effort, torque or applied pressure that was originally used to set it. There are two common methods.

- Torque wrench method

A common practice for pressure screw arrangements is to record the torque wrench loading at each pressure screw position and tabulate (or map) the load distribution on all the phase assemblies and compare them. Note that it is usual to measure the torque required to infinitesimally increase the existing load at each pressure screw (rather than the torque required to release it). The disadvantage with this method is that any adjustment at one single point may slightly influence the loading at the other pressure screw positions around the coil circle.

- Hydraulic jack method

A preferred and almost universal method to measure residual clamping pressure is to use a minimum of four hydraulic jacks (cylinders) to bear evenly down on the top pressure ring of the phase assembly. The main limitations are whether there is enough room to position the jacks and whether suitable pressure points can be found to extend each jack between the top yoke clamp and the winding pressure ring. By gradually and equally increasing hydraulic pressure on the four jacks, it is possible to record, one by one, how much force is required to overtake the existing clamping force at each clamping point. Hydraulic cylinders provide ease of applying force and a suitably slow rate of increasing the force. They also provide a means of quick release of force, if required for safety. Positioning the four hydraulic cylinders approximately 90° apart around the coil circle ensures that the application of force is simultaneous and evenly distributed around the coil circle [48].

Dissolved Gas Analysis

In coil clamping arrangements that employ pressure screws, it was common to use metal pads under each screw for distributing pressure onto the phase assembly. If the coil clamping relaxes enough to let loose the metal pads, then sparking of these floating metal objects will occur. This is due to the voltage they assume through capacitive coupling with the winding voltages below. The sparking will produce dissolved hydrogen and acetylene in the oil.

Below is a summary of the information and test results that would be recommended for a basic fleet assessment for the transformer active part mechanical risk of failure:

1. Frequency Response Analysis (FRA) Test. Ideally, a set of initial FRA signatures should be recorded at transformer commissioning for making later comparisons.
2. Single phase leakage reactance
3. Inter-winding capacitance (usually already obtained from dissipation/power factor tests)

Transformers of high capital value or ones that would pose severe consequences in the event of their failure may be considered worthwhile candidates for performing an internal inspection to ascertain a proper mechanical assessment of their active part. This would especially be considered following any severe external short circuit event.

8.3.4 Failure mode assessment

Mechanical condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

8.3.5 Scoring Matrix for Mechanical Assessment

The generic scoring matrix illustrated in 2.3 has been adapted to more easily allow the assessment of mechanical failure.

Table 8-2 Scoring matrix for mechanical assessment

	Scoring matrix for mechanical assessment
F	Severe mechanical damage in main tank, likely to lead to imminent failure. Electrical tests indicating obvious changes in leakage reactance, interwinding capacitance and FRA signatures. Indications of a dielectric failure may also appear
E	Indications of mechanical alteration in the windings because of some changes in leakage reactance, interwinding capacitance or FRA signatures. Visual evidence of dislodged blocks, and some winding movement may also be present. No coil clamping pressure on parts of a phase assembly
D	Not applicable to mechanical assessment
C	Not applicable to mechanical assessment
B	Indicators show small deviation from new, however follow normal degradation behaviour. Normal operation is expected for long term. Through fault(s) have occurred but no evidence of damage or change in electrical tests. Relaxation of coil clamping is evident
A	All test results normal: no trace of acetylene; normal levels of other gases and no indication of problems from electrical tests. No visible disturbance of internal assembly structures

9. Bushings and cable boxes

Bushings and cable boxes are critical components of a power transformer, as they are the principal interface item between the transformer winding and the rest of the substation or electrical system. Bushings are subjected to electrical, thermal and mechanical stresses, and are a leading cause of failure for transformers [51]. Therefore, knowledge of bushing condition is crucial in any comprehensive assessment. Bushings assessed to be in poor condition do not necessarily result in transformer replacement, as an economical assessment may show that bushing replacement is sufficient.

9.1 Transformer bushings

Note: The Technical Brochure (TB) by CIGRE Working Group A2.43, titled '*Transformer Bushing Reliability*' [52] to be published in 2019, provides a more detailed description of some of the topics covered by this section. Readers who require a more detailed understanding of bushings, should also refer to the WG A2.43 TB. As the specialists who developed the two TBs have different experiences and backgrounds, they may have slightly different views or opinions of some topics. Readers should consider the information in both TBs when determining how to best assess bushings.

The core insulation of bushing rated above 36 kV is generally composed of graded capacitance (a series of concentric layers made of conductive material) that allows a uniform distribution of the electrical field. In order to facilitate the bushing diagnostic, one layer (usually the last or the last but one layer) is accessible externally by a capacitive tap also called measuring tap or test tap. This capacitive tap will allow to measure exclusively the capacitance and power factor of the bushing even if the bushing is connected to the transformer winding that has usually a capacitance that is significantly higher than the capacitance of the bushing.

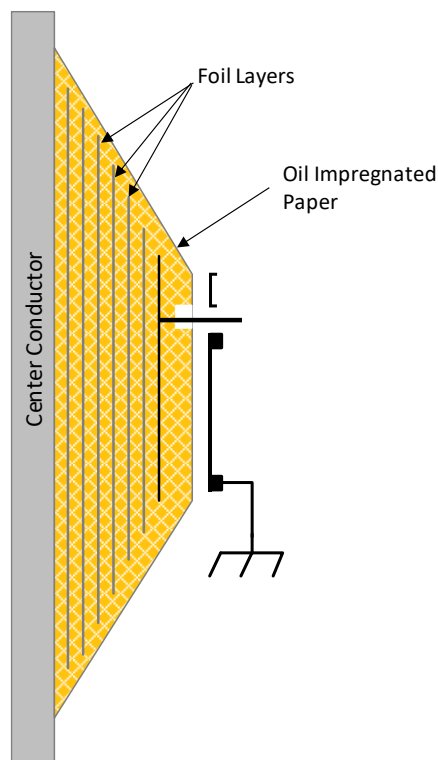


Figure 9-1 Core insulation of bushing

Table 9-1 notes some different bushing types, based on design and construction.

When the transformer is in service, the tap may be connected to a monitoring system for measuring of voltage, capacitance, dissipation factor, transient over-voltages and partial discharges. If no monitor is used, the tap is connected directly to ground. Any weakness of the connection loop to ground may cause an overvoltage stress and damage the bushing.

Some bushings don't have grading layers to even out electrical stress. Some older RBP bushings with grading layers don't have a measuring tap. In this case, the outermost layer is connected directly to the flange. Some of these bushings are mounted with an insulating plate between the bushing flange and the transformer tank to make the measurement of capacitance and dissipation/power factor possible. In service, both are connected by a short circuit cable or bar, but for measuring purposes these connections can be removed and the bushing flange is used as measuring electrode.

The core insulation may be made of the following material:

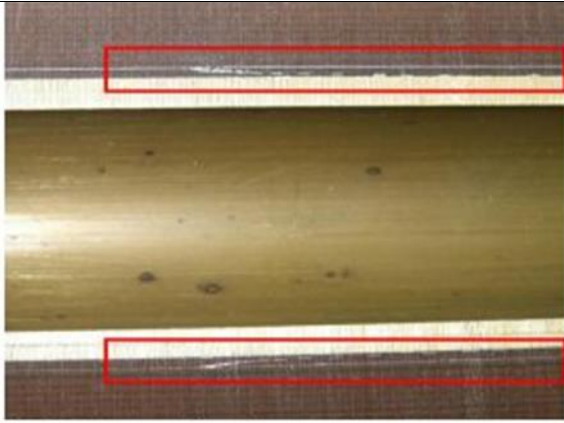
Table 9-1 Types of transformer bushings

Bushing core insulation	Definition
Oil impregnated paper (OIP)	They are made of paper impregnated by insulating oil and additional free oil between the core and the exterior housing.
Resin Bonded Paper (RBP)	This type of bushing core is made of paper coated with a uniform and thin layer of epoxy resin, with semi-conductive material such as graphite interspersed to form the concentric grading layers. Most manufacturers have discontinued the production of RBP bushings due to quality problems.
Resin Impregnated Paper (RIP)	This type of bushing insulation is made of paper impregnated with resin. A resin-impregnated paper bushing can be provided with an insulating envelope. The intervening space can be filled with an insulating liquid or another insulating medium. At lower voltages, the epoxy impregnated condenser core can have the silicone sheds directly moulded or extruded on its surface.
Resin Impregnated Synthetics (RIS)	This new type of bushing insulation is made of synthetic textile impregnated with resin. The resin can be filled with The insulating envelope is equal to the same as for RIP.
Other Bushing Types for Special or Unique Applications	For direct connection to gas insulated switchgear (GIS), gas insulated bushings are used. These bushings use SF ₆ gas as the main insulation medium in conjunction with concentric metallic screens to control the electric field. Also used for through-wall bushings in HVDC applications
Hybrid bushings	Using a combination of different insulating technologies are also used e.g. in HVDC applications.

Bushing envelope	Definition
Porcelain Insulator	Historically, porcelain has been the only material available to provide both the necessary mechanical strength and electrical insulator properties required for the outer shell of a bushing. In cases where the installed bushings are not suited to a heavy pollution situation, corrective measures include applying special grease on the porcelain or regularly washing the porcelain.
Composite Insulator	Composite insulators are made either with an internal tube of glass resin surrounded by a silicone shell, or with epoxy in the case of the full epoxy design. The condenser body would typically be of the RIP and RIS design but also OIP exist. Any space between the insulator and condenser core may be filled with oil, gel, solid foam or gas or, at lower voltages, with silicone directly moulded or extruded on the condenser body. The selection of silicone rubber quality is of outmost importance for the service life for the bushing

9.1.1 Transformer bushing failure modes

The below shows the typical defects and failure-modes of HV transformer bushings. Bushing failures often result in fire or explosions.



Contact problem between the outmost grading layer and the conductor tube.



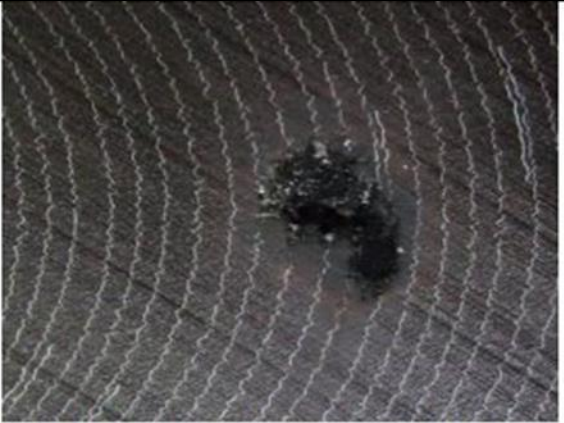
Ingress of moisture into the resin surface due to wrong storage



Contact failure at the measuring tap due to water ingress and corrosion



Contact failure between the inner measuring tap connection and the outmost grading layer due to a "cold" soldering joint.



Partial breakdown between grading layers or at the overlapping of a grading layer due to fast transient over-voltages.



Partial breakdown in a OIP insulation

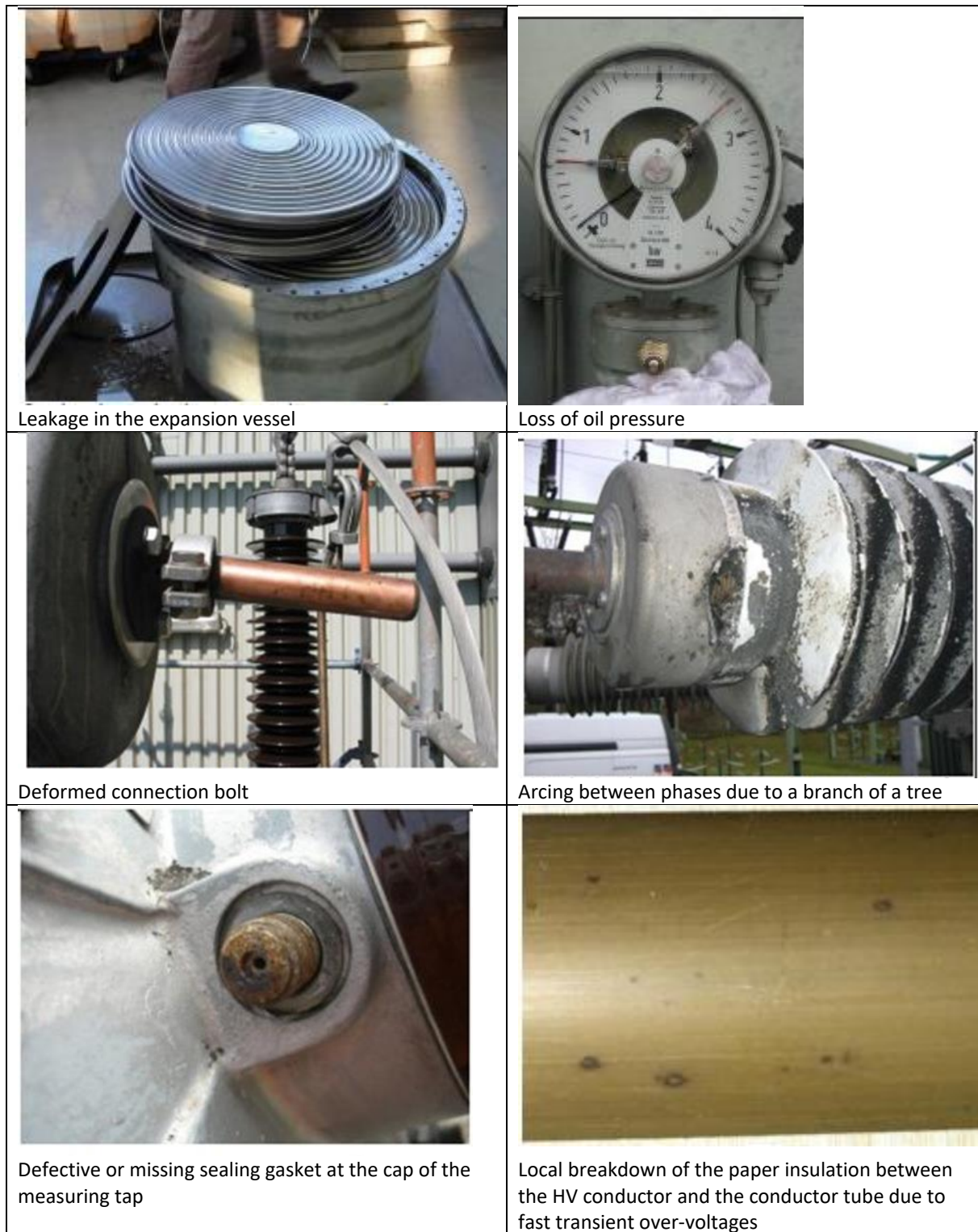


Figure 9-2 Bushing failure modes

The various bushing failure modes generally lead to a variation in the insulation – changing the leakage path to ground. No bushing is perfect, and there is a small leakage current to ground. As the insulation deteriorates the leakage current may rise and may degrade in terms of power factor (dissipation factor); partial discharge (PD) may be generated. Detecting these changes is performed through offline testing and inspection or through on-line monitoring.

9.2 Transformer bushing test and diagnostics

Table 9-2 Bushing assessment methods

	Type of fault							
	Moisture							
	Ageing							
	Contact problems and high impedance faults between layers							
	Shorted grading layers							
	Leakage							
	Partial discharges							
	Diagnostic techniques							
General	Visual inspection	OIP	5	1	5	5	5	5
		RBP	5	1	5	5	5	5
		RIP/RIS	5	1	5	5	5	5
	Thermography	OIP	4	5	4	3	4	5
		RBP	4	5	4	3	4	5
		RIP/RIS	4	5	4	3	4	5
Electrical tests	Capacitance	OIP	5	4*	1	3	4	4
		RBP	5	4*	1	3	4	4
		RIP/RIS	5	4*	1	3	4	4
	DF/PF	OIP	3	5	1	2	2	2
		RBP	3	5	1	2	2	2
		RIP/RIS	3	5	1	2	2	2
	Dielectric Response with FDS/PDC	OIP	3	5	1	1	1	1
		RBP	3	5	1	1	1	1
		RIP/RIS	3	5	1	1	1	1
	Partial discharge	OIP	1	5	3	1	5	4
		RBP	1	5	3	1	5	4
		RIP/RIS	1	5	3	1	5	4
Oil	Dissolved gas analysis	OIP	1	5	1	1	4	5
	Moisture in oil		5	5	4	5	3	1
	Colour of oil		5	2	5	1	1	4
1=very detectible, 2=moderately detectible, 3=detectible, 4=poor detectability, 5=not detectible *big leakages can be detected with hot collar test								

9.2.1 General Inspections

Visual inspection

A first basic way of assessing status of any kind of bushings and almost the only method to assess the so-called DIN bushings is visual inspection. Items to look out for are oil leaks, high or low oil level in the glass reservoir or sight glass, the colour of the oil in the reservoir, burn marks, insulator damages such as worn off flakes and micro cracks, indication of high stresses on the connection to the outer terminal etc.

Based on any findings during visual inspection a decision can be taken to undertake any further measurements.

Thermo-vision/ Thermography

Using an IR/sensitive camera, see picture below, hotspots on the bushing surface can be detected. Low oil level in an OIP bushing may also be detected using this method. At maximum rated current, the bushing's outer terminal reaches a temperature of about 35-40°C higher than ambient air. Significantly higher temperatures, especially at lower current loading, can indicate bad connections.

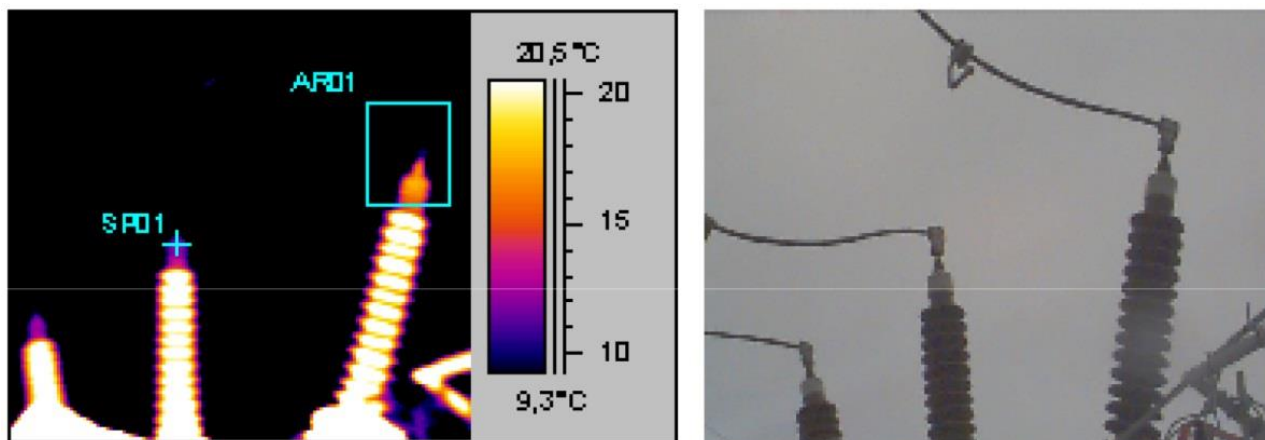


Figure 9-3 Overheating of the top terminal seen using an IR/sensitive camera.

9.2.2 Electrical measurements

Capacitance and dielectric dissipation factor (Tangent Delta) or Power Factor

A change in bushing capacitance or power factor (dissipation factor) can be an indicator of insulation deterioration, failure, ageing of insulation, partial discharges or partial breakdowns of capacitive bushing layers. It is important to understand causes of variation in test results – the Garton effect, for example shows that power factor (dissipation factor) results at higher voltages may be elevated due to contamination and moisture [53], but may also vary with temperature in ways unique to each bushing design.

The most common techniques for assessing a bushing's condition are dissipation/power factor and capacitance measurement. Test results are then compared to nameplate values or previous tests. Increases or decreases from reference values are usually an indication of contamination and/or deterioration of the insulation system. Limits for the maximum permissible change tend to be manufacturer and type specific; however, a doubling of the initial power factor value or an increase over 1% is often taken to indicate either more frequent monitoring/testing is required or replacement of the bushing. A 3% change in capacitance is also taken as an indication of a need for more frequent testing/monitoring or bushing replacement. The evaluation of a bushing is based on a comparison of tests on similar bushings on the transformer and the factory test report. Also, trend analysis and comparison to bushings of the same type are important – the limits for particular designs and vintages of bushings may vary.

If a change in capacitance and/or dissipation/power factor is indicated on an OIP bushing an analysis of the insulation oil is recommended.

Frequency Domain Spectroscopy (FDS) / Dielectric Frequency Response (DFR)

The change in these electrical measurements can be an indicator of moisture in the insulation, partial breakdowns of capacitive bushing layers, ageing and contact problems. As with all electrical tests, care must be taken to understand factors which may confound results, including oil contaminants and ageing by-products.

The principle of this measurement is the same as Capacitance and Dissipation/Power Factor described previously but differs by being applied at different frequencies, typically from 0.001 Hz to 1000 Hz. Interpretation of the result is based on frequency analysis of the dielectric parameters. In its application to high voltage bushings, the measurement of Dissipation/Power Factor at low frequencies enables the detection of moisture.

Polarisation Depolarisation Current (PDC) Method

The change in polarization/depolarization currents can be an indicator of moisture in the insulation, insulation failures, ageing of insulation and partial breakdowns of capacitive bushing layers.

The principle of this measurement is to apply a DC voltage across the insulation during a certain period of time and to measure the polarisation current. Then the insulation is short-circuited with the short circuit current measured (depolarisation current). Experience shows that moisture may strongly influence these measurements. The dielectric response of insulation can be recorded in the time domain (PDC) or the frequency domain (FDS). The PDC results can be transformed into the frequency domain and vice versa. They contain more or less the same information, except the information for higher frequencies above 0.1Hz which cannot be derived from the PDC measurement.

Care must be taken.

Partial discharge (PD) measurement

Partial discharge activity typically occurs within insulation voids, on ungrounded metal objects lying in an electric field, or as corona due to the intense electric stress on the insulation surrounding sharp edged/shaped electrodes. The long term effect of PD is destructive for insulation systems, by degrading their insulation qualities. A significant increase of the PD level or the rate of the PD can provide an early indication for evolving defects. The PD pulses generate electro-magnetic waves, acoustic signals, chemical reactions, local heating and optical signals. Different techniques can be used to detect these phenomena and thus infer the presence of PD.

For detection, different types of coupling are used such as bushing capacitance taps, high frequency current transformers or coupling capacitors. The test method outlined in IEC 60270 [40] is a useful tool, because it is a sensitive method and can be calibrated. However, electromagnetic noise external to the transformer can interfere with the measurement. For this reason, the method may be unsuitable for some on-site measurements. Multi-channel PD measuring systems which can record the PD simultaneously on several channels can sometimes be used to separate PD sources from each other and from external electromagnetic noise (interference). New Diagnostic Tools for High Voltage Bushings [54] [2] Frequency Dependent Dissipation Factor Measurements for the Assessment of High Voltage Bushings [55].

For PD measurements on new bushings limits are given in IEC 60137 [56] and IEEE C57.19.01 [57]. In case of OIP bushings oil samples may be taken and analysed when PD levels are deemed too high.

- Note: At the time of preparation of this document, there were no guidelines published regarding suitable or acceptable levels of PD in bushings in service. IEC 60270 does offer guidelines for new bushings during FAT (factory acceptance testing) but these values should not be used for bushings installed in transformers.

9.2.3 Oil

Dissolved Gas Analysis (DGA)

Oil sampling and connected-DGA analysis is a good method of assessing OIP bushings for insulation failures, ageing of insulation, partial discharges (e.g. bad contacts on capacitive layers of bushings), partial breakdowns of capacitive bushing layers. However, the practicality of taking oil samples may be difficult and there is a risk to the integrity of the bushing should a sample be taken poorly – introducing an increased risk of failure. It should be noted that oil sampling should be done by trained and certified labour as history has shown that failures of bushings have been started after improper conducted oil sampling. IEC Technical Report 61464 [58] gives a clear guidance how to take oil samples and what actions to take based on the results from the DGA.

The colour of the oil in the glass reservoir is also an indication of the degradation of the oil, and should be noted, together with the level of the oil in the reservoir.

9.3 Failure mode assessment

Bushing condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the assessment scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

The final assessment of the bushing condition is based on the weighing of several parameters such as tan delta/DGA/capacitance, application (GIS, shunt reactors), loading conditions, age and known weaknesses in the design. If possible and/or needed should the assessment be done together with the bushing manufacturer.

9.4 Transformer cable boxes

The purpose of a cable box is to protect and insulate the connection point between the cables and the transformer bushings. There are various types installed world-wide with a variety of insulating mediums including oil, various compounds such as pitch or tar, or air. The assessment techniques will vary depending on the construction type.

9.4.1 Cable box failure modes

Failure of the transformer cable box may require transformer cable box replacement, major work or corrective maintenance depending on the nature of the fault.

Liquid Filled Cable Boxes



Figure 9-4 Phase to Earth Fault in liquid filled 11kV Cable Box



Figure 9-5 Degradation and Crystallisation of insulating liquid in 11kV and 33kV Cable Boxes

Dry Type Cable Boxes



Figure 9-6 33kV Dry Type Cable Box Failure due to PD Activity

Failure modes include;

- Moisture ingress
- Loss of insulation medium
- Oil leakage
- Changes in oil quality
- Condensation
- Partial discharge activity
- Overheating
- Standoff supports / cable thermal expansions

9.4.2 Tests and information for assessment / Test and diagnostics

Proposed assessment methods for the two types

Air insulated cable boxes

1. Visual inspection, from outside and inside
 - a. Check for oil leaks from inside the box for example between the bushing flange and the transformer cover.
 - b. Look for signs of leakage current and PD-activities

Oil-filled cable boxes

2. Visual inspection from outside for oil leaks
 - a. Oil sampling and DGA analysis from the cable box. Evaluate the result as per 12 Insulating liquid.

Additionally, capacitance and dielectric dissipation factor (Tangent Delta) and Power Factor on the bushings in those cases where bushing test tap is accessible from outside. The assessment of the values as per table for failure modes for bushings. Note that this is not considered part of the cable box assessment but the assessment of the bushing component.

9.5 Failure mode assessment

Cable box condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

10. On load tap-changer (OLTC) / de-energized tap-changer (DETC)

On-load tap-changers (OLTC) can be divided into two groups according to the type of current limiting impedance used:

1. **Resistor type OLTCs** are suited to high voltage power transformers but require a high speed mechanism to limit time the resistors are inserted to typically 50 – 150 ms. These spring-operated mechanisms can be sophisticated and require more skills for maintenance. HV applications are more susceptible to dielectric failure. Therefore, oil quality is also a maintenance issue for most resistor type OLTCs.
2. **Reactor type OLTCs** are limited to lower voltage (less than 138 kV) power transformers but they can typically carry higher loads and have no limitation in transition time. The reactor type OLTCs all feature the use of a reactor that provides the OLTC with twice as many voltage positions when compared to a resistor type OLTC that uses the same number of steps in the tapping winding. High load applications make them more susceptible to coking, but most are equipped with slower motor-driven mechanisms that are easier to maintain. The reactor usually brings additional transformer cost, size and, occasionally, testing issues.

The OLTC incorporates one or more of the following components:

- Diverter Switches are always used in conjunction with a tap selector that comprises two sets of moving and stationary contacts that alternately carry the load. The operation of the Resistor Type OLTC involves two steps. First, the target tap is preselected with the unloaded moving contact. Then the Diverter Switch transfers the load to the pre-selected tap. This contrasts with the Reactor Type OLTC where the Diverter Switch has to first interrupt the load on one set of moving contacts before that set can be used to pre-select the next tap. Diverter Switches that are the Resistor Type must have high speed mechanisms (usually spring operated) in order to limit contact wear from arcing and to limit the temperature rise of their transition resistors.
- Selector Switches effectively carry out the two-step operation described above, but with only a single motion. This is achieved with an arrangement of auxiliary moving contacts that can bridge across the two taps for the necessary make-before-break transition. Selector Switches that are Resistor Type must also have high speed mechanisms for the same reasons outlined for the Diverter Switches. Selector Switches are usually spring-operated for high-speed action when used on Resistor Type OLTCs, and motor-driven with slow action on Reactor Types. These devices are usually limited to equipment voltages up to 138 kV and having less than 33 tap positions.
- Change-over Selectors (coarse or reversing) are designed to carry, but not to make or break, through-current. They are used in conjunction with the tap selector or selector switch to enable its contacts and the connected taps to be used more than once, thus increasing (doubling) the number of operation positions for the same number of taps.
- Vacuum type tap-changers can use all principles described above, but in most cases through the addition of a number steps in the sequence and by-pass or auxiliary transfer switches.
- Motor Drive Mechanism and External Drive Shafts are used for the operation and control of the OLTC.

For more details on the different switching technologies and applications, refer to IEC 60214 Parts 1 and Part 2 [59] [60].

Unlike the active part of the transformer, OLTCs could theoretically be kept in working conditions indefinitely by maintaining, repairing or replacing parts (including complete components). End of life decision is therefore not typically based on a simple condition criterion but rather occurs in the following situations [61]:

- Failure beyond economical repair
- Maintenance costs vs new technologies
- Sustainability i.e. availability of maintenance skills or parts

- Inadequacy of OLTC rating and specifications

Reliability surveys show that, depending on a number of different factors such as voltage, application and more importantly maintenance policies, OLTC can account for 10% and up to 50% of all transformer failures [62] [63] [16]. It is clear that the OLTC (and to a lesser extent DETC) will influence the overall reliability of the transformer and that it should be reflected by the transformer assessment index.

Also, transformer asset managers should be aware of possible sustainability issues, e.g. cases where the original OLTC/DETC manufacturer no longer offers parts or service. Problems related to this kind of situation: availability or high costs of spare parts, lack of technical support and upgrades, and scarcity of skilled service technicians will eventually impact the reliability of the transformer. Because it is not possible to factor this case into a transformer assessment index, based on the failure modes, asset managers should consider and handle these as special cases with an individualized action plan (ex: retrofit with a new OLTC technology). The same applies to OLTC vintages that have some known design related reliability issues that cannot be corrected by regular maintenance.

10.1 Failure modes

A number of failure root cause analysis are available through the literature. For example, IEEE C57.140 Guide for Evaluation and reconditioning of Liquid Immersed Power Transformers [6] has proposed a simplified failure mode structures for both OLTC and DETC shown in Figure 10-1 below. It clearly illustrates that DETC have only a subset of all the possible failures on an OLTC, and that these two components can therefore be treated altogether.

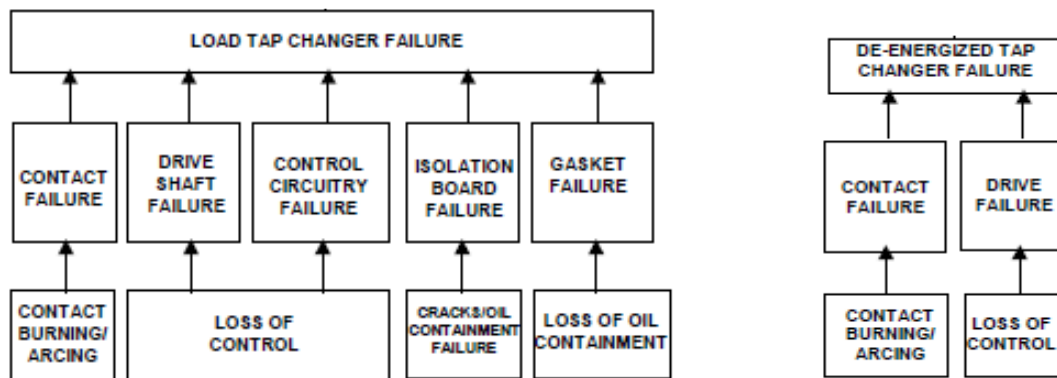


Figure 10-1 OLTC and DETC failure root cause analysis [6]

A more comprehensive failure model has been proposed by CIGRE in Technical Brochure 227 [64] in which failure modes have been grouped by components (Diverter, Selector and Drive Mechanism) and under three categories: Mechanical, Electrical (Thermal) and Dielectric. This model will be developed further in the following sections mainly by adding a new category (Chemical) to cover corrosive sulphur problems that have emerged more recently.

Indeed, surveys that have included maintenance issues as possible root causes show that these account for a significant part of the total OLTC failures. For instance, Figure 10-2 Major transformer failure due to OLTC Flashover - Root cause analysis [56] shows the OLTC failure cause breakdown statistics by a large North-American utility [62]. It can be observed that Insufficient or Incorrect maintenance work accounted for more than half of the major failures (flashover) originating from the OLTC.

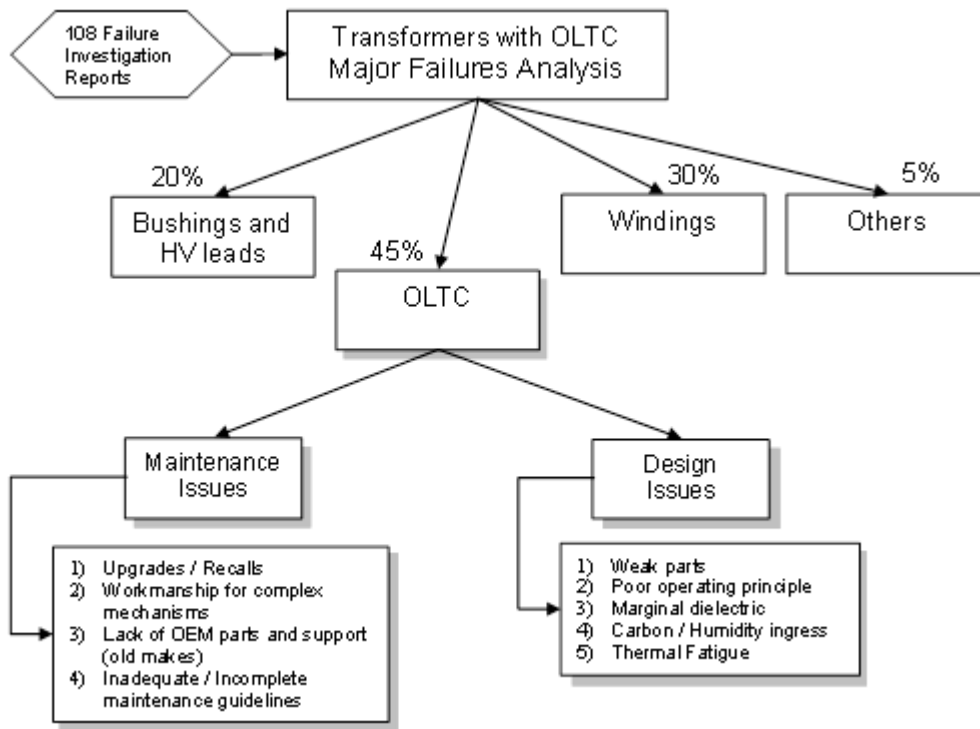


Figure 10-2 Major transformer failure due to OLTC Flashover - Root cause analysis [56]

10.1.1 Thermal

For this category, two types of thermal failure will be distinguished:

- **Overheating:** Overheating of the OLTC compartment would typically occur when transition resistors remain inserted due to an incomplete diverter switch operation. Depending on the switching principle, this may lead either to a quasi-instantaneous and disastrous failure of the resistor (load current through the resistors), or to a high heat release (up to a few kW) over possibly a long period of time (when resistors are inserted in parallel to load current path). The latter leads to significant temperature increase with a degradation of the oil and insulating materials which will eventually cause a flashover. Most likely failure causes include Incorrect or insufficient maintenance and weakened diverter springs.
- **Coking:** Coking can result either from a loose connection or the degradation of a contact surface. Reduced conductivity leads to a hot spot on which carbon particles will deposit to form a hard coating (coke) further aggravating the contact quality and heat dissipation. This whole sequence is unstable and eventually develops into a thermal runaway. Total heat dissipated typically range in the few hundreds of W, but as it is concentrated on a small surface it can be sufficient to cause severe local overheating, hence DGA techniques have been applied successfully. The overall temperature rise of the oil compartment over the main tank temperature will be in the range of 5 to 20 °C so this failure could also be detected at its latest stages by a thermograph scan.

10.1.2 Dielectric

In this category, only purely dielectric failures will be considered, in other words a component rupture that eventually results in a flashover will be treated as a mechanical failure. Other than external influences (over voltage, lighting, etc.) that leaves only two possible causes for a flashover or tracking: moisture ingress in insulation material and build-up of carbon deposits (or other contaminants/particles in oil) due to insufficient maintenance.

10.1.3 Chemical

Numerous failures of transformers have been recently related to the formation of copper and silver sulphide on metal surfaces and copper sulphide deposits in the insulating paper in the windings. CIGRE Technical Brochure 625 [37] offers an excellent summary of the current knowledge.

Concerning the possible sources elemental sulphur, there are a few possibilities according to recent findings:

1. High concentrations of elemental sulphur can be released by oil reclamation processes using reactivated clays due to the high temperatures used in the reactivation process. Elemental sulphur cannot be naturally present in the original oil. Many cases have been reported “*where the oil was believed to be non-corrosive beforehand or even tested and found to be non-corrosive and, sometime after the treatment, the silver contacts were found to be coated with silver sulphide*” [37]. In at least one reported case, it led to a dielectric failure just weeks after the treatment. This occurred even though investigations revealed that the reclamation processes could generally be considered appropriate. This situation would not generally cause a severe/immediate failure event if the OLTC didn't have silver plated contacts, since copper is less susceptible to a reaction with elemental sulphur.
2. Silver corrosion in OLTC was also observed in corrosive oils with DBDS or without DBDS, as well as in oils tested originally as non-corrosive, and on which no reclamation process had been performed. These are suspected to be result of oil becoming corrosive after it has reached and advanced stage of oxidation.
3. This situation would likely not cause any problem as long as the OLTC does not have silver plated contacts.

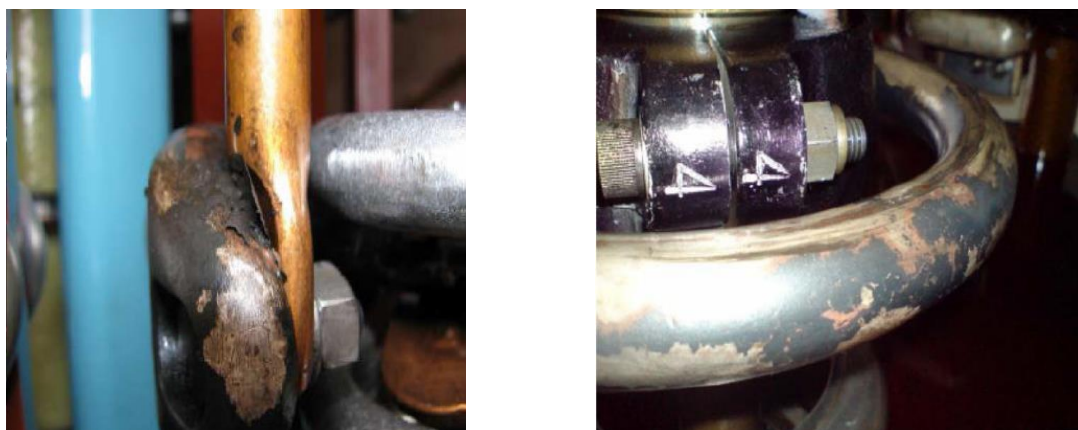


Figure 10-3 Examples of silver sulphide formation on OLTC contacts [37]

10.1.4 Mechanical

Mechanical failures can be divided into two groups: Motor drive failures and leaks.

Motor drive failures including:

1. Incomplete operation: Operation is interrupted before completion (ex: Motor trip)
2. Incorrect operation: OLTC does not reach the set position (ex: OLTC runs through)
3. Wrong tap indication: Actual position and relayed indication do not match (ex. Bad synchronism or erroneous position signal)
4. Drive linkage failure: Mechanical failure (ex: rupture, seizure) due to inadequate maintenance.

It will normally lead to minor material consequences, but still result in unplanned outages. Statistics about these minor outages show a strong correlation with age as shown in Figure 10-4. For instance, a 50 years old motor drive would have a yearly probability of an unplanned outage of more than 10%.

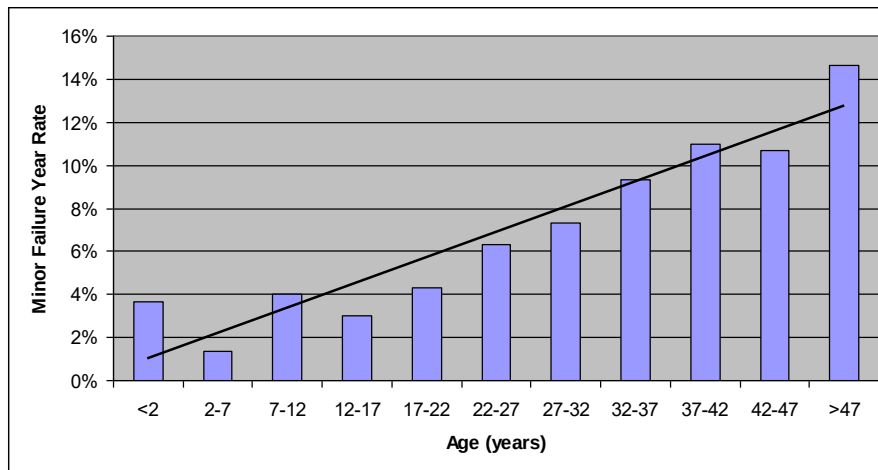


Figure 10-4 Probability of OLTC minor failures [62]

The probable cause of these minor failures by components has been established by the same utility as follows:

- 40% OLTC controls (relays, contactors, cams etc.)
- 14% Braking system
- 12% OLTC contacts (coking)
- 6% Motor
- 28% Other (lubrication, wiring etc.)

The relationship with age shown above, the nature of the of failures and field observations suggested that most problems are related more to slow degradation due to stressors (such as moisture, heat and dust) than to wear, as no evidence could be linked to the number of operations.

Definitely more serious consequences (including internal flashover) can result from a bad synchronism between the motor drive and the OLTC selector. Such event can occur on OLTCs that do not have a protection (weak point acting as a mechanical fuse in a location easily accessible for repair) and will be caused by the motor drive forcing through a mechanical stop at a limit position.

Leaks can be classified essentially as internal or external leaks. Internal leaks will mainly cause either a misleading interpretation for DGA monitoring of the active part when the oil flows from the OLTC compartment to the main tank or OLTC oil overflow when the oil is migrating in the opposite direction. The oil flow is essentially determined by the difference in conservator heights and many utilities prefer to have the oil leak from the main tank to OLTC to avoid transformer oil contamination with carbon and dissolved gases. The issue of controlling the overflow appears comparatively more manageable, typically a container collects the overflow of oil through the OLTC compartment drain pipe. However some leaks may develop to an extent that topping off the transformer oil level may be required on a regular basis.

External leaks are defined as direct leaks from the OLTC compartment to the environment, mainly through the gaskets and seals. These are less easily manageable as they can seep and drip from all around, therefore they can cause considerable cleaning and environmental costs.

Although leaks appear to have a marginal effect on transformer reliability, they still represent a major reason for transformer replacement as repairs can be quite difficult and very costly.

10.2 Tests

A number of on-line diagnostic tools have emerged recently for the condition assessment of OLTCs. Table 10-1 below gives an overview of the detection capabilities of the main diagnostic methods available.

Table 10-1 Failure detection capabilities of different OLTC tests [5]

DIAGNOSTIC METHOD		OLTC TYPE									
VIBRO-ACOUSTIC	Vacuum										
	Reactor										
	Resistor										
MOTOR TORQUE	All										
DISSOLVED GAS ANALYSIS	Vacuum										
	Reactor										
	Resistor										
IR THERMOGRAPHY	In-Tank										
	Compartment										
DYNAMIC RESISTANCE	All										
PROBLEMS											
	Linkage/Gears							G	G	G	G
	Timing/Sequence	G							E	E	E
	Control/Relays							E			
	Motor							G			
	Brake							E			
	Lubrication							E			
	Contacts alignment							G	G	G	G
	Arcing				M	G	E		E	E	E
	Overheating/Coking		G	L	M	E	E		L	M	L
	Contact wear	M							E	M	L
	Transition	E							E	E	E

L-Low
M-Moderate
G-Good
E-Excellent

2. Review maintenance records and policies including:

- Last date of service and observations from the inspection
- List of critical modifications and upgrades performed
- List and type of alarms/trip during service
- Qualification of the service technician
- Maintenance practices and more particularly deviations from the OEM's recommendations (time, operations, replacement/repair criteria)

3. Accumulated number of operations and operating range

The number of operations is normally not a relevant indicator of the expected reliability provided that the OEM maintenance recommendations are applied and that the accumulated number stays within the maximum lifetime number rated by the OEMs. This evaluates the overall wear of the OLTC and gives a reference criterion above which the unit's reliability starts to be affected and its condition is considered to be beyond cost-effective repair.

For modern products, typical ranges of values issued by OEMs are:

- Diverter switch : 800,000 to 1,200,000 operations
- Selector switch: 1,000,000 to 4,000,000 operations
- Vacuum interrupters: 600,000 to 1,000,000 operations
- Motor drive unit: 1,000,000 to 4,000,000 operations

For older OLTC vintages, different OEM limits (typically lower) may apply. For a normal network transformer with approximately 5,000 yearly operations, it is clear that this criterion will more likely not impact the transformer assessment index significantly.

4. Load history

Load deviations from the rated load can affect ageing and wear mechanisms. Main components that can be affected are insulating material degradation and contact wear. For this purpose, the average load factor and general characteristics of overload occurrences (peak and yearly frequency) should normally be available.

10.4 Scoring Matrix for tap Changers

The generic scoring matrix illustrated in 2.3 has been adapted to more easily allow the assessment of tap-changers.

	Scoring matrix for tap-changers
F	Indications of extreme degradation or ageing, or excessive maintenance delays have occurred. A failure is likely to occur in the short term.
E	Indications of severe degradation or aging, or critical maintenance operations are overdue. Reliable operation is impaired.
D	Indications of advanced degradation or aging, important delay in required maintenance. Reliable operation is compromised.
C	Indications of significant aging or deterioration, important delay in required maintenance. Reliability may be affected
B	Indications of slight aging or deterioration but still within the manufacturer's limit. Reliability is not affected
A	All tests results are well within normal limits and maintenance is up to date

11. Cooling system, transformer tank and ancillary components

This section focuses on the cooling system, transformer tank and ancillary components. Although these are not part of the main power circuit it does have a major influence on the capability and reliability of the power transformer. It is thus important to monitor the condition of these components closely and resolve any problems as soon as they develop as it can affect the long term operational performance of the power transformer.

11.1 Cooling System

The cooling system plays an integral part in the design and operating performance of a power transformer. A description of each type of cooling system can be found in TB 445 [5], IEC 60076-2 [24] and IEEE C57.12.00-2015 [65]. It is very important that the energy from the losses within the power transformer is efficiently transferred out into the surrounding environment. Working group A2.18 have identified that one of the defective/faulty conditions typically attributed to the general overheating or abnormal rise of the oil temperature may be due to cooling deficiency [64].

Failure modes of cooling system

The critical wearing out of cooling system components affect the two basic functions of the cooling system i.e. oil circulation and heat exchange. Inefficient oil circulation is affected by the convective ability of the oil (ON) and pumping forced oil/water flow (OFAF, ODAF, OFWF, ODWF). The heat exchange (loss dissipation) efficiency is affected by loss of oil forced flow, oil flow blockage, pump deficiency (opposite direction of rotation, bearing wear, pump motor failure, improper oil flow rate, fan deficiency/failure, coolers/ pipe contamination and air flow blockage due to contamination).

Fan failure

The cooling fans form an important part of the cooling system, especially for large transformers. Usually with more than one cooling fan failure the transformer load must either be decreased or the transformer taken out of service due to the significant drop in air flow which reduces the rate of heat transfer from the transformer oil to the environment. This decreases the efficiency of the cooling system with heat starting to build up within the transformer and if the transformer was running at full load, it will result in higher oil/winding temperatures until it is alarmed / tripped.

Wrong direction of fan

When maintenance work is done on the coolers there may be a chance of the fans being connected in the wrong direction resulting in inefficient cooling due to recirculation of warm air. This will result in elevated oil temperature.

Pump failure

The correct flow and circulation of oil in the transformer tank, windings and piping is important to allow for the effective heat transfer from the windings to the external environment. If a pump fails, the oil flow rate decreases and decreases the efficiency of the cooler. This will result in elevated oil temperature.

Wrong direction of pump

When maintenance work is done on the coolers there may be a chance of the pumps being connected in the wrong direction. This has the effect of circulating oil in the wrong direction through the coolers and reducing the efficiency of the coolers. This will result in elevated oil temperature. This failure mode can affect the general cooling of the transformer as it not only reduces the cooling efficiency of the system but also hinder the natural flow of oil from bottom of tank to top of tank and the result may be disastrous as it will work against of the basic principle of cooling by convection.

Control circuit failure

The control circuit of the cooling system plays an important role in ensuring that the correct amount of fans and pumps from the coolers are activated for the relevant loading. If there is a problem with the control circuitry or components this will impact the operation of the cooling system and efficiency of cooling. This will result in elevated oil temperature.

Blocked ducts/pipes

Usually with old/aged oil there is a high level of particles and sludge formation which may build up in the cooling ducts, piping and radiators. The effect is a layer of material on the surface of these components insulating the metal surfaces and reducing the efficiency of heat transfer. Another mechanism is the blockage of the ducts, cooling channels and pipes due to sludge formation or foreign debris thus impeding oil flow and reducing heat transfer. Oil-water heat exchangers can also be blocked on the water side due to deposits or corrosion causing decreased cooling efficiency. See Figure 11-1.



Figure 11-1: Particle build-up in water cooler

Closed radiator valves

Usually after maintenance some of the radiator valves can be left closed. This prevents oil from circulating within the radiators and thus decreases the efficiency of cooling.

High oil viscosity

The viscosity of oil affects the oil movement in the convection process and the through the winding cooling ducts. High oil viscosity reduces the cooling efficiency. The viscosity of oil is affected by dissolved particles and oil ageing. The viscosity is sharply dependant on the oil temperature. The correct oil type must be selected considering the expected minimum ambient temperature and precautions must be taken at transformer cold-load pickup [3].

Dirty radiators and fins

Radiators, pipes and headers may experience a build-up of rust, corrosion, dust deposits and peel off paints which could reduce the efficiency of heat transfer to the environment.

Leaks

The cooling circuit may develop oil leaks usually found at the interface points to the tank and ancillary components. Corrosion due to galvanic action of dissimilar metals in the lower portion of the cooler at the juncture of the tubes and the tube sheets can cause oil leaks. Thermal expansion and contraction can cause leaks by means of deforming the relatively soft tubes especially at the joints. Leaks cause loss of oil, which must be environmentally controlled. Moisture ingress also occurs at the point of leaks.

Oil temperatures below zero Degrees Celsius of WF coolers can lead to ice creation in water pipes. Ice may destroy the integrity of WF coolers and may cause water penetration to the oil side of the cooler. Water pipes integrity can be affected also by coolers ageing, repetitive cycles of pipes thermal extension/contraction and sealing wear. Water in oil can drastically reduce its dielectric strength and cause accelerated insulation aging.

Elevated air ambient temperatures

Although ambient temperature is an external factor and not controllable in open air environments; it may affect the heat transfer from the transformer to the environment. The effect is significant for transformers located in enclosed areas (buildings). This usually occurs if the transformer was not properly designed for the environmental conditions and will have to be managed accordingly. The higher the ambient temperature the slower the heat transfer and the equilibrium temperature is higher. This causes the internal temperature of the transformer to increase.

Elevated external cooling water inlet temperature

For water coolers it is important that the inlet cooling water temperature is low enough or as per designed specifications to allow for efficient heat transfer. During summer months the cooling water temperature is usually higher and may decrease the cooling efficiency. This usually occurs if the

transformer was not properly designed for the environmental conditions and will have to be managed accordingly.

11.1.1 Test and diagnosis

Working group A2.18 have described diagnostic methods for the cooling system consisting of forced oil flow rate/pressure, pump vibration, pump motor current, metal-in oil analysis, difference in inlet-outlet cooler temperature, temperature distribution across the transformer tank and oil pipes, temperature distribution across the cooler, radiator air flow rate /direction, fan motor current and bearing monitoring system indicating wear [64].

Infrared scanning

Infrared scanning provides a relative difference in surface temperature. On a transformer infrared scanning can locate areas of elevated temperature and assist in identifying for example a radiator valve being closed. One of the advantages of infrared scanning is that it can be performed without an outage. Figure 11-2 indicates how IR scanning can be used to identify blocked radiators.

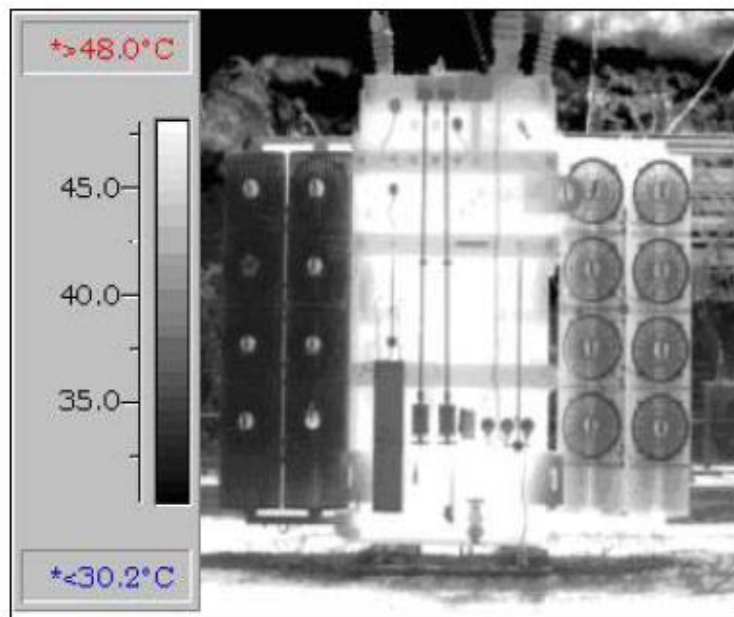


Figure 11-2: IR Image showing blocked radiators [66]

Temperature monitoring

Temperature monitoring is an important parameter for identifying an increase in energy within the transformer and problems with the cooling system. The top oil temperature is usually used as an indication of the inlet temperature of the coolers. Monitoring the bottom oil temperature either at the bottom main tank or at the outlet of the cooler allows for a differential temperature calculation to be made. This value can then be compared between cooler circuits to identify low differential values which may be an indication of blocked piping, poor radiator surface condition, etc.

Working group A2.27 Recommendations for Condition Monitoring Facilities recommended that the following temperatures should be available for condition monitoring [67]:

- **Top oil** – at a measurement point representative of the temperature of the oil at the top of the tank and as close as possible to where it was measured during the factory temperature rise test.
- **Bottom oil** – at a measurement point representative of the oil entering the bottom of the windings which is normally the oil temperature at the cooler outlet close to the transformer tank.
- **Cooler inlet oil** – measurement taken from the cooler inlet oil and may be used to monitor cooler efficiency. In some transformer designs the top oil measurement can be used.
- **Cooler outlet oil** – measurement taken from the cooler outlet oil and can be used to monitor cooler efficiency. In some transformer designs the bottom oil measurement can be used.
- **Cooling medium at inlet to coolers** – a measurement representative of the temperature of the cooling medium (normally air or water) at the inlet to the coolers. In the case of an air temperature

measurement, the sensor should be mounted in the shade. Air ambient temperature can be used if this measurement is not available. For water cooling medium a sensor or thermometer pocket should be included at both the cooler inlet and outlet.

- **Cooler Performance Index: Δ Temp (Inlet – Outlet)** - This cooler performance index is derived by finding the difference between the Inlet and Outlet temperatures of the cooler and comparing between coolers. This can be done easily if there is temperature monitoring on the inlet and outlet of the cooler. If there are none available, portable temperature scans can be taken externally to get these values on a routine basis. This delta is then compared with the designed values for the cooling circuit. Temperature deltas more than 30% of between coolers should be investigated further.
- **Ambient temperature** – Monitor the ambient temperature if the transformer is located in an enclosed area (building). This temperature must be alarmed for immediate investigation when the alarm value is triggered.

Oil flow monitoring

Oil flow is a very important parameter, especially for forced oil (OF, OD) cooling systems. The designed flow rates must be maintained to achieve the associated cooling efficiency. Cooling circuits usually have oil flow indicators which confirm that there is oil flow. It is recommended to install analogue oil flow meters so that the flow rate can be measured and trended per cooler. Any flow impediment can then be easily and quickly identified.

- **Cooler Flow Index: Δ Flow (Each cooler circuit)** - Particular cooling system designs have oil flow meters installed to display oil flow in cooling system branch or in each cooler. Where there are no such meters, periodic measurements can be performed with a portable flow meter at load or as close as possible to transformer rating at a given cooling stage. Such measurements can be done on the transformer online, without the interruption of the customers power supply. Where oil flow is defined by a working pump, the measured value should be compared with about 80% of indicated on oil pump nameplate or in pump manual (considering 20% oil flow reduction due to oil path hydraulic resistance). Each cooler flow can also be compared with the average of all the cooling system coolers/branches to identify any anomalies between coolers.

Visual inspection

Visual inspection of the transformer oil-cooling loop components should be performed as regularly as prudent but should not exceed 12 month frequency.

Fans should be manually energized to ensure proper operation. Any significant noises (e.g. grinding, rubbing, scraping) should be noted, investigated further, and corrected. Fan blade guards must meet government safety standards. Obvious airflow obstruction through coolers via debris should be noted and cleaned. Periodic infrared imaging of the coolers/radiators should be taken to ensure proper oil flow and cooler/radiator heat transfer. All areas including piping, valves, and surrounding ground area must not show evidence of oil leakage. Typical leakage points will include all pipe flange connections, valve stems, pump electrical connections, radiator/cooler headers, etc.

Any missing, failed, or defective fans should be repaired or replaced. If the transformer's future expected service includes loading beyond rated power, a possible solution could be increasing the number of fans as part of the reconditioning of the transformer. However, any cooling revisions and or uprating of the transformers should be developed and designed by OEMs/Experts with specific knowledge of the transformer's cooling design, and original designed parameters of the transformer. These may include other methods most suitable for the design.

- **Cooler Condition Index: Visual Inspection** - Cooler Condition Index forms part of the routine inspections of the transformer. This requires a visual assessment of the general condition of the cooling system. Any deviations identified must be rectified as soon as possible or at the next available maintenance schedule.

Sludge

Sludge usually starts building up in oil and start getting deposited on tank surfaces, windings and core and then start blocking in the cooling ducts provided in windings and in header piping's as the condition becomes excessive. This is a slow process and indication of this starts with the gradual increase in temperature under normal loading and ambient conditions. Although the formation of sludge can usually be detected by a visual inspection, IEC 60422 [42] suggests tests such as

colour/appearance, acidity, dielectric dissipation factor (DDF), resistivity, acidity and interfacial tension (IFT), which can provide indications of sludge components before visible sludging actually occurs.

11.1.2 Failure mode assessment

The cooling system condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the assessment scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

11.2 Transformer tank

Power transformer tanks are generally manufactured of hot-rolled, unalloyed steel sheets. In case of low ambient temperature at the site of installation of the transformer, fine granular steel is used to manufacture the transformer tank. The areas of the tank where strong eddy currents can be generated due to high currents are made of non-magnetic (austenite) steel. The transformer tank is an integral part of transformer where the main isolating and coolant (oil) of the active part is stored. Thus, its mechanical strength is of utmost importance to hold the pressure of the oil. The tank must also be capable of handling vacuum pressure (during oil filling) and a sudden pressure increase which might be experienced during 'through faults.' e.g. during faults in overhead lines connected to transformer bushings.

11.2.1 Failure modes of transformer tank

Transformer tanks are designed depending upon the site requirements and it plays an important part in the successful operation and utilisation of the asset throughout the life cycle of the transformer. Generally, the tank is a very robust part of the transformer and does not hinder the operation of transformer, but it can be a problematic area if there were any issues during transport, installation at site, due to climatic conditions or proper maintenance not being done. The transformer tank can have following failure modes where proper action plans for mitigation and control is required.

Oil leakage from main tank gasket joint and other welding joints

There are many joints in a transformer tank and interconnecting pipes, accessories and instruments where any leakages must be attended to as a priority. The main tank may be of Top Cover or Bell Tank type (Bottom Joint) and the joint may be provided with gasket, chord or O ring. In all situations, leakage from these joints are critical to the reliability of the transformer especially due to moisture ingress and the slow drop in the oil level from main tank can trigger the low oil alarm and can further operate the Buchholz relay alarm and trip contacts forcing the transformer out of service. Depending on the design of the tank, it may be possible for water to enter and collect in hollow steel components, e.g. U-channel stiffeners. In cold climates this can freeze, effectively destroying the tank.

Corrosion and improper painting

Transformer tanks are painted with a primer and external paint coatings as per site condition specifications, but it is important that proper maintenance is carried to check for any corrosion, rust and paint feel off. These conditions do not have an immediate effect but the corroded/rusted areas will eventually have an effect on the mechanical strength of the tank which can be critical when experiencing excessive pressures developed in main tank due to abnormal conditions.

Another problem is the excessive or improper thick coatings of paint applied to the transformer over time. This includes the effects of using the wrong type of paints (such as metallic based) which significantly affects the heat transfer. The surface of tank forms an important heat transferring medium and if it is coated with too much paint, or the wrong type of paint, the radiating capacity of tank can be reduce resulting in the transformer overheating under normal load conditions.

11.2.2 Test and diagnosis

The transformer tank must be routinely inspected visually for defects. While doing visual inspection, the notable problem shall be identified, and necessary additional tests may be conducted. For leakage issues, the replacement of gaskets and repair of the tank must be planned as early as possible. For problems associated to welding cracks/poor surface paintings/peel off the root cause must be ascertained and corrective or preventive action must be taken.

For an improperly painted transformer tank with bad surface conditions, the paint thickness must be measured, and it shall be within permissible limits. During major maintenance, overpainted/high thickness areas can be repainted as per OEM specifications.

11.2.3 Failure mode assessment

The transformer tank condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the assessment scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer, but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

The findings from the inspection of one transformer of the fleet must be recorded and any defects identified must be verified for possible recurrence on similar transformer designs in the fleet.

11.3 Ancillary components

A power transformer makes use of important ancillary equipment which could also affect the performance when these components fail. The most important or generally used components are discussed in this section.

11.3.1 Failure modes of components

Buchholz relay

The Buchholz relay provides an important protection for the transformer detecting gas build up and rapid movement of oil due to a fault. Failure or malfunction of the Buchholz will either result in the unnecessary tripping of the transformer or catastrophic failure to the transformer if an incipient fault is not picked up with the build-up of gas. The Buchholz relay must be routinely maintained and calibrated to ensure optimum performance.

Pressure Relief Device (PRD)

The pressure relief device is used when there is an over pressure condition within the transformer. Failure of the PRD can either result in false operation with related draining of oil or failure to operate causing the pressure not to be released and thus rupturing of the tank. The PRD must be routinely inspected for leaks around joints, diaphragm cracking and tested to ensure optimum performance.

Preservation system

Liquid immersed transformers usually have a liquid preservation system as specified in IEC 60076-1 [47]. The most common types of liquid preservation systems are free breathing system or conservator system, diaphragm or bladder type, inert gas pressure system, sealed-tank system with gas cushion or sealed system.

Breather system

Refrigerating and dehydrating breathers with silica gel are normally used to prevent moisture from entering the transformer tank. Failure of the breather usually results in moisture ingress into the transformer which promotes the aging processes of the insulation.

Temperature devices

The oil and winding temperature devices provide information on the temperature of the top oil and the simulated temperature of the windings respectively. These component settings could drift over time and thus must be routinely calibrated.

Oil level Indicator

The oil level indicator is usually a dial type gauge which provides an indication of the oil level within the transformer. Failure of this device usually results in the wrong oil level and if the oil level is low an alarm is generally received. This device must also be routinely tested and calibrated.

Fire suppression system

The fire suppression system is used to extinguish or contain a fire on a power transformer. The fire system does not affect the performance of the transformer, but it is required when a severe fault condition is present. The failure modes are mainly related to corrosion, leakage or blockage.

Lightning arresters

Usually power transformers connected to overhead lines utilise lightning arresters to protect the transformer from insulation failures arising from lightning and other overvoltage conditions such as switching surges. Failure of arresters is usually due to damaged, contamination (moisture, dirt and corrosion), lightning strokes, long-duration surges due to switching and misapplication. Lightning arresters should be inspected annually for looseness, broken parts, dirt, and other deposits.

Instrument Transformers

Instrument transformers like current and voltage transformers play an important role by linking the primary and secondary system of the power transformer to enable for protection and measurement circuits to be connected. The insulation system must be capable of withstanding high voltages and currents and maintain the isolation of the systems. Other key characteristics of instrument transformers are the accuracy, thermal and burden considerations.

11.3.2 Test and diagnosis

The ancillary components can be inspected, routinely tested and calibrated to ensure the reliability. The following tests discussed are relevant to surge arresters. IEC 60099-4 [68] and 60099-5 [69] provides a detailed guide on the testing of surge arresters. The following are basic tests that can be utilised to either assess the condition of surge arresters or to make the replacement decisions.

Visual inspection

Visual inspection of the transformer components should be performed as regularly as possible. Inspection frequency should not exceed 12 months. Inspection activities should include as a minimum:

- Inspect the pressure relief device on a routine basis for the general condition and check if the yellow (oil) or blue (silicon) rod/button is visible
- For sealed or pressurised breathing systems, check the pressure gauge (against weekly data sheets) for seasonal temperature changes. If there is no change it is an alert that the device may be defective. Monitor the pressure and if below acceptable levels investigate possible leaks and top up gas to correct pressure
- For sealed systems with bladders, on a medium-term routine (5 yearly), inspect visually the conservator internally through the inspection flange
- On desiccant breathers, confirm that colour change takes place from the bottom of the breather. If the colour change should take place from the top, check the seals, flanges, union joints and pipes for leaks.
- On instrument transformers [IEC60044-1], verification of terminal markings must be carried out especially after any work has been done in the circuit.
- Inspection of surge arresters to identify dirt, particle or salt deposits on the arrester housing. These must be routinely cleaned. Inspections must also identify cracks and damage to the porcelain housing. These must be either repaired or replaced depending on the severity of the damage.

Calibration

Calibration testing must be performed on components that provide indications of plant condition, such as temperature, pressure, flow or level. These can be done:

- On oil and winding temperature devices. This is usually performed by simulating the heat by means of a hot oil bath setup.
- For oil level indicators with magnetically coupled float mechanism, remove the outside mechanism for testing without lowering the transformer oil. A magnet can be used to confirm function by rotating the magnet and confirming operation on the dial.
- Calibrate pressure monitoring devices on the sealed or pressurised breathing systems.

Functional, alarms and trip verification

Routine verification must be performed to check correct functioning, alarming and tripping of the auxiliary components. These can be done;

- Manually on the Buchholz relay on a routine a basis by pumping a small amount of air into the top chamber as per OEM testing procedure. Observe correct float operation and associated alarm or trip contacts activating.

- Use a volt-ohmmeter to check the integrity of the alarm and trip switches / contacts on the oil and winding temperature devices, pressure relief device, Buchholz and oil level indicator.

Routine testing

All piping should be flushed to clear debris and to verify the spray pattern is still effective.

Infrared scanning

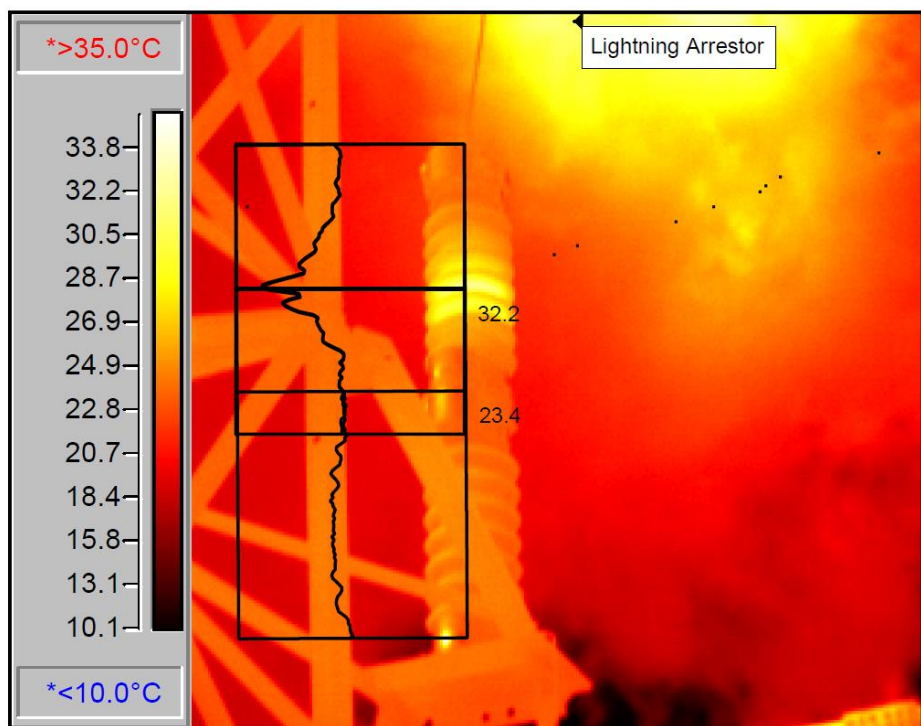


Figure 11-3: IR image showing hot spot [66]

Infrared scanning provides a relative difference in temperature which can locate “hot spots” on arresters. One of the advantages of infrared scanning is that it can be performed without an outage. Figure 11-3 provides an example of a localised hot spot. Thermal scans can be compared between arresters of the same transformer or from sister units to pick up any anomalies.

Leakage current

Leakage current provides a good measure of the condition of the surge arrester, especially the metal oxide type. The total leakage current consists of a capacitive and resistive component with the capacitive component being the most significant. However, any change, degradation or aging in the arrester causes a significant increase to the resistive component of the leakage current with little change to the capacitive component. Thus, by measuring the arrester’s resistive leakage current, a diagnosis of its condition can be obtained. An increase of the leakage current usually leads to an increase in the element temperature which in turn leads to a further increase in leakage current. This then results in thermal runaway and failure of the arrester.

A sudden increase in the resistive leakage current is usually an indication of water contamination which is the major cause of arrester failures.

Insulation resistance testing

Insulation resistance test maybe conducted to assess the condition of the surge arrester. Compare results to the previous test results.

Power factor testing

Power Factor tests are effective in detecting problems in lightning and surge arresters which are evaluated on the basis of dielectric loss (milliwatts or watts) of comparable units or previous year’s benchmark test data. Abnormal dielectric losses can be divided into higher than normal and lower than normal and which are indicative of the problems as listed below:

Type	Higher than normal losses	Lower than normal losses
Silicon Carbide	Contamination by moisture, dirt or dust. Corroded gaps	Broken shunting resistors. Poor contact and open circuits between elements
Metal Oxide	Contamination by moisture, dirt or dust. Corroded gaps in early designs (newer designs are gapless)	Discontinuities in internal electrical configuration

11.3.3 Failure mode assessment

The ancillary components condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the assessment scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

12. Insulating liquid

12.1 Introduction

Transformers **may use** one of three classes of insulating liquid for their non-solid insulation and as a cooling medium; dielectric liquids, pressurised gas, or atmospheric air.

The various dielectric liquids, often loosely referred to as oils, **are almost universally used** in high voltage power transformers and **are** the sole consideration for this chapter. Refer to CIGRE TB445 [5] for discussion on condition assessment of the pressurised gas used in gas insulated transformers and for more details regarding some of the mentioned tests.

The dielectric liquid **is used** for heat transfer and as an insulation medium, as well as for impregnating paper-based, solid insulation, and for diagnostics purposes, being responsible for providing information for assessment of normal / abnormal behaviours:

- Dielectric liquid **transfers** heat through its circulation. It **takes** up heat at transformer heat sources (including the core, windings and stray loss heating in tank structures) and **transfers** heat to the transformer heat coolers (involving either heat dissipation or heat exchange);
- Dielectric liquid **acts** as an insulation medium to provide insulation between the live parts and the live parts to ground. To be a functional electrical insulator in power transformers, dielectric liquid **must attain and retain** certain dielectric, chemical and physical properties. Changes in these properties **are** where the condition assessment of the dielectric liquid **is mainly focused**;
- Dielectric liquid **impregnates** paper-based, solid insulation, establishing a compatible composite insulation system. This insulation system **establishes** both a solid and liquid medium having optimal resistivity and permittivity in zones subject to high electrical stress. This insulation system **is ideally suited** to the complex winding geometry, connections and electrode shapes found in power transformers. The liquid, in principle, **eliminates** gas voids within the paper. Otherwise gas voids **would initiate** partial discharge activity;
- Dielectric liquid **is** responsible for transmitting information of any abnormal processes which **may be undergoing** inside of the transformer. Through dielectric liquid analysis, understood hereby as the assessment of key physical and chemical parameters of the liquid and, especially, tracking its variation along time, and the dissolved gas analysis, it **must be** possible to identify potentially harmful processes of transformer active part and internal components.

A possible division of dielectric liquids **may include** four groups – mineral oil, natural esters, synthetic esters and silicone. Despite one **can identify** more types of dielectric liquids, this brochure **will only cover** the listed ones. Mineral insulating oil **is** the most common and **is produced** from refining crude oil. Refer to CIGRE Technical Brochure 436 [70] “Experiences in Service with New Insulating Liquids” for a comparative study of these four transformer dielectric liquid types.

Regardless of the dielectric liquid, it **deteriorates** in service. It **degrades** and **is contaminated**. This **is recognised** by there being international standards that **address** either new or used dielectric liquid. The principal international standards currently being used for mineral oils, natural ester liquids, synthetic ester liquids and silicones **can be found** in Annex D.

These standards **are intended** to set the essential requirements for transformer dielectric liquids. Most power companies, network operators and transformer manufacturers **have further developed** their own transformer dielectric liquid specifications, guidelines and supervision methods. Accordingly, there **is** a wide variety of procedures and criteria for monitoring dielectric liquids.

Monitoring and maintaining dielectric liquid quality **is** essential to ensure the reliable operation and service life of a transformer. Like most materials, dielectric liquid deteriorates. It **diminishes** both in quality and its ability to withstand electrical stress and transfer heat from the core and coils to the environment. Invariably, dielectric liquid condition **deteriorates** though contamination or degradation (ageing), and, depending on the type of dielectric liquid, forming acids, and other polar compounds, such as metallic sulphides. Dielectric liquid degradation **can be premature**. Rather than ‘fail’, dielectric liquid **breaks** down if subjected to arcing, partial discharges or overheating. It then **forms** gases of low molecular weight (the gases monitored in DGA) and sometimes carbon particles or coke. The behaviour of each of the dielectric liquid types **will differ** in the quantity of each by-product formed, especially the coking tendency.

This chapter will compare the value and significance of various standardised dielectric liquid tests. It will provide a broad understanding of degradation and contamination of dielectric liquid. It will explain why they are detrimental to a transformer and discuss the available methods to slow, prevent or correct degradation and contamination of dielectric liquid. It will also illustrate how to apply criteria for the evaluation of test results and test data so that informed assessments can be made about the condition of the dielectric liquid in a transformer.

12.2 Degradation and contamination

Dielectric liquid degradation occurs throughout its use in the transformer, the rate of degradation, assessed by the regular dielectric liquid analysis, is key for the differentiation between a normal or an abnormal behaviour of the transformer. Dielectric liquid condition is also degraded by the presence of physical and chemical contaminants. Furthermore, some contaminants can increase the rate of dielectric liquid degradation. Dielectric liquid left in poor condition leads to increased risk of transformer failure. In extreme cases, it can lead to impairment of the essential cooling of the transformer. This happens through deposition of decay products (in the case of mineral oil, for example), or as a result of the increase in the viscosity of the dielectric liquid (in the case of the esters, for example).

The physical contaminants will be considered first because they are the contaminants usually encountered in the dielectric liquid of power transformers. They include water, particles and fibres.

Degradation and contamination of the dielectric liquid which can result in the failure of the transformer may require corrective maintenance prior to having detrimental effect on the active part or result in a transformer failure. The decision between treating/reclaiming or replacing the dielectric liquid make take into considering the level of degradation and the related costs.

12.2.1 Physical contamination

In general, there is a correlation between breakdown voltage of the dielectric liquid and its water content relative to saturation, which is the relative humidity of the dielectric liquid (defined by the ratio of absolute water content to the maximum water content which can be dissolved in the dielectric liquid at certain temperature). There is also a correlation between breakdown voltage of the dielectric liquid and its particle content, even if non-conductive particles, as the interfaces between them and the dielectric liquid may create preferential paths for the discharges. High moisture, together with high particle or high fibre content (concentration) in the dielectric liquid, may significantly lower the breakdown voltage. Usually, dielectric liquid reconditioning processes, essentially formed by filtering, drying and degassing, are used to remove moisture, gases, fibres and other particles from the dielectric liquid, thereby restoring the breakdown voltage of the dielectric liquid. More details can be found in the Technical Brochure 349 [35] regarding the moisture contamination and in [71] regarding particles.

Water

The majority of the water in a transformer is actually held within the paper insulation (> 99% for a mineral oil filled transformer and about 95% for an ester oil filled transformer). One unavoidable source of moisture is its formation as a by-product of paper degradation. Under the influence of heat, insulation paper will decompose forming carbon oxides, water and furans. When the moisture remains impregnated in the paper it has a catalytic effect for paper degradation, almost as a chain reaction (self-induced).

Also, moisture may enter the unit as a contaminant from the atmosphere due to transformer leaks, cracked seals and from poorly maintained breather systems. The high moisture content in the insulation paper may increase the risk of bubble formation or cause electrical faults such as partial discharge.

For minimizing the moisture ingress, free breathing transformers should be fitted with a desiccant breather that has an oil cup seal (hydraulic seal) or, preferentially, fitted with a mechanical barrier between the dielectric liquid and the ambient, such as rubber bags, diaphragms or air cells. Gaskets and oil seals should be replaced before wet oil leaks develop. Check also dielectric liquid seals and breathers in any other dielectric liquid compartments of the transformer (on-load tap-changer, cable boxes, passage terminals and external wiring).

In natural ester filled transformers the moisture present in the paper (including the moisture generated by paper degradation) may migrate to the dielectric liquid, due to its higher saturation point. Hydrolysis of the dielectric liquid (chemical reaction between the dielectric liquid and water) then results in

consumption of that water, having as by products free fat acids. This is claimed to keep both the dielectric liquid and paper dry, increasing the neutralization index (acidity), which, in the case of natural esters, is due to the increase of concentration of long chain organic acids, which are not harmful for the transformer.

Particles and fibres

Transformers unavoidably retain some quantity of residual paper insulation fibres, rust, particles and swarf. Despite metallic particles (conductive) are most critical, the non-conductive ones, especially if associated with high moisture content, are also of concern, with variations according to the type of dielectric liquid. Metal fragments are always created during transformer manufacture through component fabrication, core cutting, coil winding, phase assembly, brazing, crimping and tanking activities and, despite the several stages of decontamination and cleaning, residual content may eventually be found. Dielectric liquid flowing through transformer cooling systems and dielectric liquid transfer operations may introduce fibre and particle contamination. Carbonaceous particles can also be introduced into the dielectric liquid through the pyrolysis of paper. Severely overheated dielectric liquid or sparking in dielectric liquid usually causes localised carbon coking. Carbon particles can also be suspended in dielectric liquid as a result of electrical arcing.

Accordingly, the particles and fibres present in dielectric liquid are an amalgamation of different types and sizes and accumulate from different sources. Limiting the velocity of dielectric liquid flow during the design of the cooling equipment may reduce the formation of particles. Other actions to minimize the contamination with particles are: the use of dedicated handling equipment, always suitably clean, and flushed before use; pre-clean dielectric liquid valves and dielectric liquid hose connection devices; regularly maintain tap-changer diverter switch; take extreme care during internal inspections and on-site openings of the tank.

Dielectric liquid properties affected by physical contaminants

The following properties of the dielectric liquid are tested to check whether physical contaminants have affected the dielectric condition of the dielectric liquid:

Moisture content must be kept low. It is important to use relative moisture content of dielectric liquid, rather than absolute moisture content, especially when dielectric liquid immersed paper insulation is present. In assessing the significance of moisture content in dielectric liquid, it must be known that the moisture content in dielectric liquid changes with the temperature of the transformer. Therefore, when obtaining the samples of the fluid, the temperature of the sample taken must be recorded. This is because there is a dynamic moisture exchange between dielectric liquid and paper that is governed by their temperatures. Sampling the dielectric liquid at different loading conditions and temperatures is an effective way to evaluate the total content of water inside the tank, both in solid and liquid insulation. An accurate estimation of the moisture content in solid insulation based on the moisture content in the dielectric liquid is possible for transformers operating at stable conditions over a period of six months to one year, having dielectric liquid temperature above 40C, as the equilibrium between the components would only be reached after a relatively lengthy period of stable temperature. Otherwise it may be an inaccurate method. If the relative moisture content is high, then the electrical strength of the dielectric liquid will be impaired and moisture will be absorbed into paper insulation. This reduces insulation life and increases the risk of dielectric breakdown. For all dielectric liquid types, the breakdown voltage is proportional to the relative moisture saturation value (that is moisture content divided by moisture saturation at same temperature).

Breakdown voltage needs to be of sufficiently high value to ensure the dielectric liquid does not breakdown under electrical stress. The uneven distribution of the moisture inside the tank may also affect partial discharges.

Particle count must be kept low. Dielectric liquid is filtered to reduce the count of the larger particles and fibres. Low particle and fibre content are especially important in extra high voltage transformers and HVDC converter transformers because the presence of such contaminants, especially in the presence of moisture, can considerably reduce the electrical breakdown strength of the dielectric liquid.

12.2.2 Chemical contamination

Chemical contaminants are undesirable chemicals that have been introduced to the dielectric liquid by one of the following mechanisms:

- a. intentionally introducing an additive that later proves to have a deleterious effect on the transformer
- b. cross contamination (mixing with an already contaminated dielectric liquid, for example with a product like PCB or silicone)
- c. contact with an incompatible solid material during dielectric liquid storage, or handling, or within the transformer that introduces a chemical contaminant, or an undesirable chemical effect (materials such as plasticisers, vulcanised rubber, some plastics, some paints, or the effect of stray gassing)
- d. inadvertently producing undesirable by-product (during dielectric liquid refining, transfers, storage, reconditioning or reclamation)
- e. accidentally adding a foreign liquid when topping up the dielectric liquid level (such as diesel)
- f. accidentally adding an incompatible dielectric liquid type (even if the dielectric liquids are miscible and compatible, concern may arise because of the unintended additive dilution or introduction of an additive)
- g. contamination due to contact with ambient air
- h. Furthermore, if a chemical contaminant does not impair the actual condition of the dielectric liquid, it may still cause concern. For example, in the following situations;
 - i. if it is PCB, the transformer must now be managed according to local applicable regulations;
 - j. if it causes stray gassing, it complicates interpretation of dissolved gas analysis;
 - k. if it has a deleterious effect on other components of the transformer (examples include corrosive sulphur affecting copper and silver, accelerated ageing of paper insulation, dilution of metal passivator, lost compatibility with some solid materials, and increased wear of the on-load tap-changer)

Dielectric liquid properties affected by chemical contaminants

In order to assess whether chemical contaminants have affected the condition of the dielectric liquid, the following properties of the dielectric liquid are checked. This is to ensure the overall quality of the dielectric liquid is maintained.

- Dissipation factor (IEC “ $\tan \delta$ ”) or power factor (IEEE “ $\cos \phi$ ”) both parameters are directly related with presence of polar components in the dielectric liquid. They have to be sufficiently low, depending on the dielectric liquid type, and must not show an excessive increase over time. Despite IEC using dissipation factor and IEEE using power factor, the difference in their numerical value is small in the value range of concern. Both measurements will respond to dielectric liquid deterioration (oxidation by-products) or to chemical contamination, being essential to the dielectric liquid quality assessment, however, there is no direct correlation between them and the breakdown voltage. Dissipation factor is less sensitive to high water content, unlike breakdown voltage, whereas dissipation factor is more sensitive to track polar ageing products than breakdown voltage.
- DC resistivity has to be sufficiently high, showing variations according to the insulation system, and it must not show an excessive decrease over time.
- Breakdown voltage needs to be of sufficiently high value to ensure the dielectric liquid does not breakdown under electrical stress.
- Interfacial tension (IFT) has to be sufficiently high, in the case of mineral oil, to ensure that chemical contamination or oil oxidation have not introduced polar compounds into the oil. Despite having no direct relation with any of the dielectric liquid functions, the IFT is a sensitive measurement for polar contaminants. In dielectric liquids having a slightly polar characteristic, such as esters in general, this meaning of IFT is lost, since the value shows very little to no variation when new dielectric liquid is compared with highly oxidized dielectric liquid.
- The neutralization index / total acid number (TAN) must be sufficiently low. Since the parameter indicates only the total content, not the types, the assessment of the reactivity of the acids compounds is important. Short-chain acids are more corrosive than long chain ones. High operation temperature may result in increasing the reactivity. In general, the acids identified in mineral oil filled transformers are short chain, very reactive in all temperatures. In synthetic esters the acids are of intermediary chain length (C7~C9), presenting limited reactivity at low temperatures, but becoming a concern for temperatures above 120°C~130°C. In natural esters

the acids are long chain (C18), presenting very low reactivity at all temperatures (mostly limited to 160°C~180°C). Reactive acids have catalytic effect to paper degradation and may be critical for other bare surfaces and components. When associated with reduction of interfacial tension a high value of neutralization index is strong indication of sludge generation.

- Determination of dielectric liquid corrosiveness to copper and silver; deposition of copper sulphide on the copper conductor and insulating paper, and deposition of silver sulphide on OLTC contacts. Sulphur corrosion may be introduced into the mineral oil due to contamination by adding additives based on disulphides, by residual disulphides left in the oil after refining, or by introduction of undesirable by-products into the mineral oil during oil reclamation (elemental sulphur after improper oil reclamation); but also may be present as consequence of mineral oil degradation, as observed in oxidized mineral oils in service (usually uninhibited oils of lower degree of refining).

Such tests only indicate that the dielectric liquid has been affected in a detrimental way. By themselves, most tests cannot determine the type of chemical (or physical) contamination involved. To ascertain the actual contaminant present, further investigation of the contamination event and specialised laboratory testing of dielectric liquid samples may be required. These tests can include sulphur speciation, liquid chromatography, specific additive content in the dielectric liquid, scanning electron microscopy and chemical compatibility tests for materials known to be present in the transformer. Chemical compatibility is an evaluation of the interaction of the dielectric liquid with other materials but does not assess long term performance of the material, despite being an important parameter. For the chemical compatibility the control parameters are the dielectric liquid properties, which may be affected by any reaction of the dielectric liquid with, or even some extraction of substances from, the tested material.

12.2.3 Dielectric liquid degradation

Despite all the usually performed treatments, dielectric liquid will always contain some dissolved oxygen, leading to oxidation. Oxidation is the major process of thermal degradation or ageing of the dielectric liquid under normal operating conditions. It is the rate at which the dielectric liquid oxidises, that condition assessments are aimed to monitor. As the refining process removes most of natural protection from dielectric liquids, highly refined liquids require the use of oxidation inhibitor. In free-breathing equipment, where dielectric liquid remains directly in contact with air, oxidation is relatively rapid. This is not a recommended use condition for natural ester liquids, as, despite the use of oxidation inhibitor, they are less stable than mineral oil. Being a thermal degradation process, oxidation is accelerated by higher temperatures. The presence of metals, especially bare copper, organo-metallic compounds, or both, may act as a catalyst for oxidation.

For temperatures above recommended limits, oxidation process becomes intensive. In general, the maximum temperatures indicated at the standards are:

- Mineral oil: continuous service temperature according to IEC 60076-2 [24], which considers usual the free breathing condition, is defined as 100°C. For IEEE C57.12.00, considering essentially oxygen free applications, the limit temperature is set to 105°C. For emergency conditions both standards limit the temperature to 115°C.
- Ester: both IEEE C57.154 [25] and IEC 60076-14 [26] limits the continuous temperature to 130°C and, at emergency condition, 140°C.
 - In the case of synthetic ester liquid, which is more stable than mineral oil regarding oxidation, this limitation is mainly caused by the reactivity of the acids.
 - For natural ester liquids, as the long chain acids are much less reactive, for an essentially oxygen-free condition, producer of natural ester liquids accepts a continuous operation temperature as high as 160°C and a limit for emergency condition of 180°C.

As thermal degradation takes place, dielectric liquid changes colour. It forms acidic compounds, and at an advanced stage of oxidation, precipitation of sludge may occur. Precipitation of sludge may cause the dielectric and, in extreme circumstances, the thermal properties of the transformer to be impaired.

The ageing mechanisms of mineral oil are complicated. For reference, verify CIGRE TB 526. Generally, oxygen reacts with certain hydrocarbons by a free radical process. This is a chain reaction mechanism that generates hydroperoxides. Hydroperoxides are unstable and quickly decompose to form ketones and water. Ketones oxidise to form carboxylic acids or cleave to make aldehydes. The presence of hydroxyl groups will result in the production of alcohols and phenols. The final products of mineral oil oxidation are acids, water and sludge. Most oxidation products will have a negative effect

on the electrical properties of mineral oil. The carboxylic acids that are produced will either dissolve in the oil or volatilise into the headspace. Dissolved acids may cause damage to the paper and copper windings, while volatile acids may corrode the top of the unit.

The oxidation mechanism in natural esters process is very similar to mineral oil. The lower stability of natural esters is attributed to the presence of unsaturated links (double bonds) in the hydrogen-carbon chain. Accordingly, the use of natural esters usually excludes the use on free breathing transformers, when the dielectric liquid is in continuous contact with ambient air. Eventual contact or contact with confined air is acceptable, as the availability of oxygen is limited. Constructions such as corrugated tank or expandable radiators, gas headspace (which may be nitrogen or ambient air, which will be confined) or dielectric liquid conservator having a rubber bag or bladder are the most frequent alternatives suiting for the application of natural ester liquids. Another difference is the behaviour of the free radicals, which will recombine with the ester molecule, increasing the length of the hydrocarbon chains. This process is called *oligomerization*. This results in the long term, in a potential increase in the viscosity of the dielectric liquid, but, positively, prevents most other side effects of oxidation (much less concern with the acids, almost zero formation of sludge, no degradation of breakdown voltage), making an increase in viscosity the primary condition assessment parameter for the natural esters liquids. The transformer operating temperature is much less critical for natural esters. This allows the safe operation of natural ester filled transformers at much higher temperatures, provided the availability of oxygen is limited. The operating temperature, according to international standards, is limited to 130°C, despite special designs have been developed using natural esters for temperatures up to 160°C. The good stability at low oxygen conditions and the low reactivity of the long chain free fat acids of the natural esters are key for this application. Natural ester has negligible sludge formation.

Synthetic esters, like the natural esters, have a dielectric liquid molecule comprising of an ester group (two oxygen atoms in each of its four chains) with the majority of the dielectric liquid molecule being hydrocarbon. Synthetic esters, unlike natural esters, have saturated chains (no double bonds), leading to their oxidation stability being higher than that of mineral oil. The synthetic esters also remain stable at higher temperatures than mineral oil. Again, good control of oxidation is critical, especially in applications where operating temperatures exceed 130°C, when the behaviour of the acids, whose molecular weights are higher than mineral oil acids, but much smaller than natural esters, must be carefully evaluated. Synthetic ester has negligible sludge formation.

Degradation of dielectric liquid, by oxidation, can be controlled by oxidation inhibitors. Mineral oils can be used with and without inhibitors. In recent years, application of inhibited mineral oils is increasing. Natural and synthetic esters are also used with inhibitors, being especially important for the natural ester based liquids.

Oxidation inhibitors became important additives for highly refined mineral oils. Natural esters and synthetic esters also have oxidation inhibitors in their typical formulation. In contrast, silicone liquid does not. Advances in refining techniques of mineral oils have resulted in higher extraction of their natural oxidation inhibitors, so synthetic inhibitors have to be added. Whether natural, or added after refining, these inhibitors added to mineral oils and esters retard oxidation mechanisms by destroying free radicals and initiators, or by reacting with peroxides form more benign products. Free radical reactions will take place more quickly in the presence of catalysts such as iron, copper and aluminium. The effect of these catalysts can be neutralised by metal passivators. Natural inhibitors such as phenanthroline and some organosulphur compounds act as radical scavengers as well as deactivating metal catalysts. Some of the phenols, which are produced by oxidation of aromatic compounds, also act as antioxidants, by donating a hydrogen atom to a free radical, stabilising it and terminating the free-radical reaction. There are certain phenols which are particularly effective as antioxidants and these are usually the synthetic additives used in inhibited oil.

The effectiveness of the inhibitors (whether natural or additive) is measured by performing the oxidation stability test. This test is intended to evaluate the dielectric liquid's response to pre-defined accelerated ageing conditions and is not intended to represent actual transformer operation. The dielectric liquid is subjected to a set amount of excess oxygen (typically by bubbling oxygen or air), a defined temperature and a measured amount of metal catalyst (typically copper). Although different standards exist to measure the oxidation stability of the dielectric liquid, importantly they all measure the secondary effects of the oxidation. The test measures the quantity of by-products formed by the dielectric liquid degradation. These include the quantity of volatile and soluble acids and their effect on the electrical properties of the dielectric liquid. In the case of mineral oil the quantity of sludge and sediment precipitated is also considered, while in natural ester this is replaced by the variation of viscosity. Different dielectric liquid types may produce different proportions of the by-products. This

needs to be taken into account in making an assessment. Comparing test result values with acceptance limits given in the standards, only provides information on the relative condition of that dielectric liquid type. The test does not allow comparisons between dielectric liquid types. There is no direct correlation between the results and the service life of the dielectric liquid.

There are more sophisticated tests that do allow comparisons between the different dielectric liquid types. These require specific equipment to allow measurement of the primary effects of the oxidation reaction and the energy released by the oxidation reaction (exothermic reaction). One such test is ASTM WK21616, which uses Differential Scanning Calorimetry (DSC) to evaluate the Oxidation Induction Time (OIT). This is the only test effectively allowing an accurate comparison of different liquids, as it measures a primary effect of the oxidation, presuming that all insulating liquids are inhibited with similar amount of inhibitor. Most of other applied tests can just be used as “approved / reproved” statement.

The oxidation inhibitor content depletion should be monitored and maintained as required.

12.3 Dielectric liquid Testing and Dissolved Gas Analysis

The use of dielectric liquid in a transformer provides a valuable way of assessing changes in the internal condition of the transformer. Removing an dielectric liquid sample for dielectric liquid testing and dissolved gas analysis works similarly to a human “blood test” for the transformers. The detection of abnormal patterns of behavior of the dielectric liquid properties offers information about the dielectric liquid itself, as well as transformer components immersed in that dielectric liquid. These dielectric liquid tests are essential for the condition assessment, for the predictive maintenance and for preventing unexpected outages, but they are much more an investigation than just a test.

This chapter however, remains focussed on detecting and assessing the condition of the transformer dielectric liquid. Other chapters will deal with the use of dielectric liquid testing and dissolved gas analysis to detect and assess the condition of other transformer components immersed in dielectric liquid.

In most cases of dielectric liquid testing, it is important to consider not only the measured absolute values, but also the trending. Also relevant to an assessment is any correlation between rates of change, or variations, with how the transformer is operated or maintained. Some dielectric liquid properties have defined continuous service limiting values, which are strongly recommended to be followed. However, when a relevant variation of the properties is identified, shorter sampling intervals and even some early investigations may be needed to intercept serious consequences or make timely interventions. More details may be found in the references IEEE C57.104 [10] and IEEE C57.155 [30], respectively for mineral oil and ester oils.

The integrity of dielectric liquid test results and dissolved gas analysis is dependent upon appropriate dielectric liquid sampling procedures and operator competency. Rigid criteria, extremely careful procedures, use of quality hardware are key aspects of proper dielectric liquid sampling. Dissolved gas analysis measures gas content in “parts per million – ppm” or even “parts per billion – ppb”, so it may be affected by the tiniest error in the sampling procedure.

12.3.1 Dielectric liquid tests

The suggested grouping of tests differs from Table 1 IEC 60422 [42], reflecting the recommendations of this working group. This technical brochure is informative, not normative. The minimum suggested surveillance checks for degradation of transformer dielectric liquid are highlighted in Group 1 in the following table.

GROUP 1 Minimum set to monitor the dielectric liquid for continued service	GROUP 2 Specialised dielectric liquid quality Evaluate dielectric liquid for continued use	GROUP 3 Suitability for type of transformer Environmental/operational compliance
Acidity	Oxidation Stability	PCB Content
Colour	Sediment & Sludge	Corrosive Sulphur
Water Content	Particle Count	Aromatic Content
Breakdown Voltage	Flash Point	Biodegradability
Dissipation Factor	Pour Point	Toxicity
DC Resistivity	Density	
Interfacial Tension*	Viscosity*	
	Inhibitor Content	
* For natural ester oils, interfacial tension does not offer relevant information regarding oil degradation, being typically replaced by viscosity measurement. Interfacial tension may be replaced by viscosity in the Group 1 tests.		

Group 1 Dielectric liquid Tests are commonly performed on the dielectric liquid of transformers, for condition assessment, throughout their service life. All the tests listed in this group provide useful information on the dielectric liquid condition and represent the minimum set of tests for condition assessment.

Group 2 Dielectric liquid Tests are infrequently performed on in-service dielectric liquids. Inhibitor content is used to monitor inhibited dielectric liquid or reclaimed dielectric liquid. The oxidation stability and sediment/sludge tests are useful to check dielectric liquid following mixing, treatment or suspected contamination. The particle count test is more often performed for EHV and HVDC converter transformers. The verification of the oxidation inhibitor content in natural esters may be included in Group 1.

Group 3 Dielectric liquid Tests are performed on in-service dielectric liquids by exception. They are used to prove environmental compliance, or to detect whether the dielectric liquid's constituent base, or contamination, is harmful to the transformer or harmful to personnel. They are more often performed before commissioning, or during dielectric liquid handling or intrusive maintenance, or to evaluate the side effects of a dielectric liquid treatment. The corrosive sulphur test is an example of a test needed to evaluate a dielectric liquid's risk to the transformer.

For more information on these tests, see the Annex D and the bibliographical references [5] [42] [41] [71] [72] [73] [74] [75] [76].

The results of dielectric liquid tests can also correlate for certain dielectric liquid problems. The matrix that follows serves to illustrate how any, or all, of the dielectric liquid test groups and dissolved gas analyses may be used to detect particular dielectric liquid problems. This matrix applies only to the behaviour of mineral oil. It should be noted that these dielectric liquid problems may also be consequential to a failure, degradation or maloperation of another component of the transformer. Nevertheless, any detrimental effect on transformer dielectric liquid should be detected as early as practicably possible.

Type of Dielectric Liquid Problem											
Corrosive Sulphur											
Two liquids Incompatible (top up or mixed)											
Over-processing (by liquid reconditioning)											
Oxidation											
Contamination by foreign liquid											
Contamination by fine particles or fibres											
Contamination by gross water ingress											
Incompatible Paint, Solids or Additive											
Excessive Gas Content (N ₂ , O ₂)											
Diagnostic Dielectric liquid Test											
DIELECTRIC LIQUID TEST GROUP	1	Acidity			•			•			
		Colour and Appearance		•	•	•	•	•			
		Water Content (and Relative Saturation)			•			•			
		Breakdown Voltage	•	•	•	•	•				
		Dissipation/Power Factor		•	•	•		•			
		DC Resistivity		•	•	•	•	•			
		Interfacial Tension		•	•		•	•			
	2	Oxidation Stability					•		•	•	
		Particle Count				•					
		Inhibitor Content						•		•	
		Flash Point		•			•			•	
		Sediment & Sludge		•		•		•		•	
	3	Corrosive Sulphur		•			•				•
		Aromatic Content							•	•	
		Passivator Content								•	
	Dissolved Gas Analysis		•	•			•	•			

12.4 Failure mode assessment

The assimilation of dielectric liquid test data, into an overall condition assessment of the dielectric liquid component, will differ in criteria depending on which of the four dielectric liquid types were used. To establish a consistent approach to the assessment for each dielectric liquid type, and for that matter any other component of the transformer, five levels of dielectric liquid condition are defined. They are defined in such a way that each level of condition should influence the overall transformer condition to the same degree, irrespective of which component was being considered. Together, the five levels provide a transition or stepping down in dielectric liquid condition. The transition starts with dielectric liquid in 'as new condition' and finishes with dielectric liquid degraded to 'unserviceable, or end of life, condition'. When dielectric liquid exhibits the first perceptible signs of degradation, it becomes level two. As dielectric liquid degradation advances, its condition presents a flag for optimal condition based intervention. If left untreated, its condition then affects other transformer components.

Dielectric liquid condition can be assessed using the tests listed above, supplemented by family history, results of inspections and inductive reasoning as appropriate and in conjunction with the assessment scoring matrix. The tables in Annex A can also be used as a starting point when assessing a transformer but users should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in their scoring matrix. User should refer to Chapters 1 to 7 before assessing their transformers.

12.5 Scoring Matrix for insulating liquids

The generic scoring matrix illustrated in 2.3 has been adapted to more easily allow the assessment of insulating liquids.

	Scoring matrix for dielectric liquids
F	De-energize as soon as possible. Don't return to service until problem is repaired.
E	Very poor condition – high likelihood of failure. May expected damages on the active part or other components due to the dielectric liquid's poor condition. May also lead to a transformer failure. The causes for the dielectric liquid degradation should be repaired e.g. leakage. Reconditioning/reclaiming or replacing the dielectric liquid as soon as possible is recommended.
D	Poor condition. The causes for the dielectric liquid degradation should be repaired e.g. leakage. Reconditioning/reclaiming or replacing the dielectric liquid should be considered within the short term. Detrimental effects on the active part may take place e.g. acceleration of degradation rate of insulating paper due to acids or accumulation of sludge.
C	Acceptable condition with significant signs of degradation. Triggering an investigation into the causes for degradation. After repairing the issue consider reconditioning/reclaiming or replacing the dielectric liquid.
B	Good condition. Some signs of degradation are evident. Reduce the sampling interval to monitor the degradation rate.
A	Minimal signs of degradation. As new condition

Annex A Transformer Condition Assessment Tables

This Annex provides tables of diagnostic information that is formatted to allow assessment using the methods described in this technical brochure. The information is from IEC & IEEE guides, CIGRE publications, CIGRE experts, and other industry experts. Users can use these values as a starting point when assessing a transformer but should consider if the values are suitable for their fleet of transformers, operating conditions, maintenance practices, and the time scales used in the scoring matrix.

Users must note that although the tables provide six columns for assessment information (labelled A-F) in accordance with the scoring matrix used throughout the technical brochure, information is not provided in every column for every assessment method covered in the table. In some cases, it is not appropriate to provide six levels of information e.g. where the test or diagnostic method normally provides a 'binary result' (yes/no or good/bad). In other cases, the working group did not have adequate information to provide suitable values or considered that appropriate assessment values would vary with the age or design of the equipment or the operating environment and conditions. Users should consider their experience with their own equipment to determine suitable values where required for an assessment.

Table A-1: Dielectric Condition Assessment

Component			Degradation / Failure mode	Condition Indicator	Test	Unit	Applicable to which insulation liquids? ¹	F	E	D	C	B	A	Reference
Core	Insulating Liquid	Major	Core Ground	DC Resistance	Insulation resistance	MOhms	MO, NE, SE	< 1	1 - 10	11 - 25	26 - 50	51 - 500	> 500	C57.152 Z1
			Arcing	Acetylene	DGA	ppm	MO, NE	> 50	36 - 50	11 - 35	1 - 10	< 1	Non-detectable	C57.104 IEC 60599 C57.155
			Aging, Moisture, contamination	Dissipation factor	DF @25 C	%	MO, NE, SE	See insulation liquids tables						C57.106 IEC 60422 C57.147 IEC 62975
			Partial Discharge	Hydrogen	DGA	ppm	MO, NE, SE	> 3000	1800 - 3000	700 - 1799	100 - 699	30 - 99	< 30	C57.104 IEC 60599
			Moisture Content of Liquid	Moisture Content	Karl Fisher Moisture Content	ppm	MO, NE, SE	See insulation liquids tables						C57.106 IEC 60422 C57.147 IEC 62975
	Winding Insulation	Major	Elevated Moisture, aging, contamination	CHL, CH, CL of transformer	PF/DF	%	MO	> 2.5	2.1 - 2.5	1.6 - 2.0	1.1 - 1.5	0.5 - 1.0	< 0.5	C57.152
			Elevated Moisture, aging, contamination	CHL, CH, CL of transformer	PF/DF	%	NE	> 4	3.0 - 4.0	2.0 - 3.0	1.5 - 2.0	1.0 - 1.5	< 1	
			Displaced winding caused by aging	CHL, CH, CL of transformer	Capacitance	% Variance from factory	MO, NE, SE	> 8	7 - 8	6 - 6.9	5 - 5.9	0.5 - 4.9	< 0.4	C57.152
			Elevated Moisture, aging, contamination	Estimated Moisture	DFR	% wt / wt	MO, NE, SE	> 5	3.1 - 5.0	2.1 - 3.0	1.5 - 2.0	1.0 - 1.5	< 1	CIGRE TR 414
			Partial Discharge	Apparent Charge	Induced/ Applied Voltage Test	pC	MO, NE, SE	> 5000	2001 - 5000	751 - 2000	501 - 750	250 - 500	< 250	C57.152 IEC 60076-3
			Insulation degradation by moisture/contamination	Polarization Index	Polarization Index (PI)	ratio of resistance at 10 min to 1 min	MO	< 1.0	1.0 - 1.1	1.11 - 1.25	1.26 - 1.5	1.51 - 2.0	> 2	C57.152 Z2
		Minor	Winding turn strand to strand short circuit or open circuit	Frequency response of stray loss	Leakage Reactance vs Frequency	Plot of Leakage reactance vs frequency	MO, NE, SE	Slope of leakage reactance to frequency should be same for all phases						

Condition assessment of power transformers

Component			Degradation / Failure mode	Condition Indicator	Test	Unit	Applicable to which insulation liquids? ¹	F	E	D	C	B	A	Reference
			Winding turn to turn short circuit	Excitation current	Excitation Current	Amps	MO, NE, SE	Pattern - Depending on core configuration there will be a high-low-high or a low-high-low pattern.						C57.152 Z4
			Winding turn to turn short circuit	Ratio	TTR	% Variance from nameplate	MO, NE, SE	> 1.5	1.26 - 1.5	1.1 - 1.25	0.51 - 1.0	0.4 - 0.5	< 0.4	C57.152
			Thermal aging of solid insulation	DP Value	DP Test	number	MO, NE, SE		< 200	200 - 300	301 - 400	401 - 500	≥ 500	
			Z1 *put comments in this field relating to pattern/consistency											
			z2 Ratio of resistance at 10 minutes to 1 minute											
			z3 Slope of leakage reactance to frequency should be same for all phases											
			z4 Depending on core configuration there will be a high-low-high or a low-high-low pattern.											
			Z5Generally inferred from Furanic compound analysis, loading history, temperature, age, accessible paper samples. For inference from Furans, refer IEEE C57.140-2017.											
			Note 1 – MO = Mineral Oil, NE = Natural Ester liquid, SE = Synthetic Ester liquid											

Table A-2: Thermal (Solid Insulation) Condition Assessment

Failure Mode				Condition Indicator	Test / Criteria	Unit	Applicable to which insulation liquids? ¹	F	E	D	C	B	A	Reference
Accelerated Cellulose Aging	Hot Metal Fault Covered by Paper			Carbon Monoxide / Cellulose Gassing + Hot Metal Gassing	CO	ppm	MO, NE		< 5000	< 2500	571 -1400	351 - 570	< 350	IEEE C57.104-2008 IEEE C57.155
				Carbon Dioxide / Cellulose Gassing + Hot Metal Gassing	CO ₂	ppm	MO, NE		< 35000	< 17500	4001 - 10000	2500 - 4000	< 2500	IEEE C57.104-2008 IEEE C57.155
				Cellulose Aging	CO ₂ /CO Ratio		MO, NE			< 3			> 3	IEEE C57.104-2008 IEEE C57.155
				Degree of Polymerization	Inferred from Furans or Methanol and service history		MO, NE	< 100	> 100	> 300	> 500	> 700	> 800	CIGRE TB 323
				Winding Hot Spot Temperature	Design Calculation	°C	MO	> 160C	< 160C	< 140C	< 130C	< 120	< 110C	IEEE C57.91.2011 CIGRE TB 659
				Moisture in Oil			MO, NE, SE	See insulation liquids tables						IEEE C57.152
				Interfacial Tension			MO, NE, SE	See insulation liquids tables						
				Acidity			MO, NE, SE	See insulation liquids tables						
				Moisture in Cellulose	Oil Equilibrium curves	%	MO, NE, SE	> 3.0%	< 3.0%	< 2.0%	< 1.5%	< 1.0%	< 0.5 %	
					Dew Point									
				Note* The limits given apply for cellulose accelerated aging - not for increased dielectric risk	DFR or FDS Test									
				Winding Insulation Power Factor		%	MO	> 2%	> 1%	< 1%	< 0.7%	< 0.5%	< 0.3%	IEEE C57.152
				Thermal aging of solid insulation	DP Value		MO, NE, SE		< 200	200 - 300	301 - 400	401 - 500	≥ 500	
				Hot Metal Gassing (if indicated by Duval Triangle or IEC Ratio Scheme)	Ethylene	ppm	MO, NE	> 1000	< 1000	< 500	< 200	< 100	< 50	IEEE C57.104-2008 IEEE C57.155
				Winding DC Resistance	% increase over factory tested values		All	10%	5%	4%	3%	2%	1%	IEEE C57.152
				Core Overheating	Design Calculation	°C	MO, NE, SE			< 170	< 150	< 130	< 100	X1

Condition assessment of power transformers

Failure Mode				Condition Indicator	Test / Criteria	Unit	Applicable to which insulation liquids? ¹	F	E	D	C	B	A	Reference
	Hot Metal Fault Covered by Paper			Cellulose Gassing + Hot Metal Gassing	CH ₄	ppm	MO, NE	> 2000	> 1500	< 1200	401 – 1000	121-400	< 120	IEEE C57.104-2008 IEEE C57.155
				Cellulose Gassing + Hot Metal Gassing	C ₂ H ₄	ppm	MO, NE	> 700	> 400	< 250	101 – 200	51 – 100	< 50	X2
				Cellulose Gassing + Hot Metal Gassing	C ₂ H ₆	ppm	MO	> 800	> 500	< 200	101-150	66-100	< 65	Duval Triangle 5 Type 'C', 'T2' and 'T3'
		Bubbling		Moisture in cellulose versus Winding Hot Spot temperature curves.	Calculated Bubble Evolution Temperature minus Calculated Winding Hot Spot Temperature for Defined Overload Condition	°C	MO, NE, SE	< 0	< 10					IEEE C57.91.2011 + CIGRE TB 349
X1 - IEEE Transactions on Power Delivery, Volume 17, Issue 4, Oct. 2002 Page(s)991 – 995 X2 - Duval, Michel; Calculation of DGA Limits, DEIS Volume. 24, No.5 September/October 2008 Note 1 – MO = Mineral Oil, NE = Natural Ester liquid, SE = Synthetic Ester liquid														

Table A-3: Mechanical Condition Assessment

Failure Mode	Degradation Mechanism	Condition Indicator	Test	Unit	Applicable to which insulation liquids?	F	E	D	C	B	A	Reference
Mechanical	Loose Clamping	Clamping pressure	Residual Pressure Survey		All			Unequal phase pressures			Good agreement in pressures	
	Winding Displacement	Impedance	Single phase reactance	% variation between phases	All			> 2% change from FAT	1% - 2% change from FAT		≤ 1% change from FAT or SAT	TB 445
		Attenuation versus frequency	FRA		All			Signature differences to FAT or SAT signatures in the interpretation range			Good agreement	IEC 60076-18 TB 342
		Between windings	Visual		All			Poor alignment			Good alignment	
		Within windings			All			Poor alignment			Good alignment	
	Loose Clamping	Condition of blocks, spacers, sticks within winding assembly	Visual/manual		All			Loose or displaced blocks, spaces or sticks within winding assembly			No loose or displaced blocks, spacers, or sticks	
		Coil clamping devices, platforms or blocks next to top and bottom frames that support or compress the phase assembly	Visual/manual		All			Loose or displaced clamping devices, platforms or support blocks			No loose or displaced blocking	
		Change in tone	Tonal Checks		All			Tonal difference identified			No tonal difference identified	

Table A-4: Bushing Condition Assessment

Failure Mode	Bushing type	Affected Parameter/ Investigative Test	Criteria	Unit	F	E	D	C	B	A	Reference
Oil leakage	OIP	Visual inspection	Correct Oil level	%		Oil level not visible	10% full	20% full	30% full	50% full	IEEE C57.152
Colour of oil in sight glass (by Visual Inspection)	OIP	Visual inspection	Colour change of oil in sight glass		Black/ Dark Black	Dark and/ or turbid *	Light Brown/ Brown	Yellow/ Dark Yellow	Light Yellow	Clear & without visible contamination	IEC 60422, Table-5
Colour of oil in sight glass (by lab test)	OIP	Visual inspection	Colour change of oil in sight glass		Dark Brown (5.5 to 8.5)	Amber (4.0 to 5.5)	Bright yellow (2.5 to 4.0)	Yellow (1.0-2.5)	Pale Yellow (0.5-1.0)	Clear (0.0 to 0.5)	IEEE C57.152
DGA Oil	OIP	DGA in Bushing Oil	Hydrogen (H2)	ul gas / l oil		>140				≤140	IEC 61464
			Methane (CH4)	ul gas / l oil		>40				≤40	IEC 61464
			Ethylene (C2H4)	ul gas / l oil		>30				≤30	IEC 61464
			Ethane (C2H6)	ul gas / l oil		>70				≤70	IEC 61464
			Acetylene (C2H2)	ul gas / l oil		>2				≤2	IEC 61464
			Carbon monoxide (CO)	ul gas / l oil		>1000				≤1000	IEC 61464
			Carbon dioxide (CO2)	ul gas / l oil		>3400				≤3400	IEC 61464
Overheating in bushing top	OIP/RIP/ RIS/RBP	Thermal scanning	Temperature increase above ambient	°C	>40	31 - 40	21 - 30	11 - 20		1 - 10	IEEE C57.152
Overheating in bushing top	OIP/RIP/ RIS/RBP	Thermal scanning	Temperature difference of similar and comparable bushings	Δ°C	≥35	10 ≤ t < 35		5 ≤ t < 10		<5	CIGRE WG A2.43
pD activities in condenser core	OIP	DGA Ratio	H2/CH4 (pD)			>13				≤13	IEC 61464
Overheating in the condenser core			C2H4/C2H6 (Thermal fault)			>1				≤1	IEC 61464
pD activities in condenser core			C2H2/C2/H4 (pD)			>1				≤1	IEC 61464
Overheating in the condenser core			CO2/CO (Thermal fault in paper)			>20 or <1				1 – 20	IEC 61464
Contamination/ layers shorting/excess ive moisture	OIP/RIP/ RIS	DF/PF/Dielectric frequency response	tan delta increase	%		Initial value *3	Initial value *2	Initial value *1.5		Initial test (rating plate value)	IEEE C57.152, CIGRE TB445
	RBP	DF/PF/Dielectric frequency response	tan delta increase	%		>0.6		0 - 0.6		Initial test (rating plate value)	CIGRE TB 445

Condition assessment of power transformers

Failure Mode	Bushing type	Affected Parameter/ Investigative Test	Criteria	Unit	F	E	D	C	B	A	Reference
	OIP/RIP/ RIS	C1 measurement	Change of Capacitance (C1)	%	>10	5.1 - 10	2.1 – 5	0 – 2		Initial test (rating plate value)	IEEE C57.152, CIGRE TB445 Y1
Damaged sheds on insulator	OIP/RIP/ RIS/RBP	Visual inspection	Defined by supplier			Major	Minor			No damage	
Corrosion due to poor installation of test tap cover	OIP/RIP/ RIS/RBP	Visual inspection voltage tap	Burning marks		Many	Few				Clean	
Y1 – Capacitance of RPB bushings is not stable in service due to oil penetration into the condenser body. This condition measure is much less significant for RBP condition assessment compared with other bushing types.											

Table A-5: Cable Box Condition Assessment

Failure Mode	Description	Affected Parameter/ Investigative Test	Type of Cable Box	Unit	F	E	D	C	B	A
Oil leakage	Correct oil level	External Visual inspection	Oil filled				Traces of main oil leaking through gland plate or covered openings on box			Correct oil level in cable box, and/or no oil leaks
	Evidence of main oil leaking through bushings	Internal Visual Inspection	All types				Evidence of main oil leaking through bushings			
Changes in oil quality	Please see chapter 12 and Table A8 for oil assessment	Oil analysis	Oil filled							
Moisture ingress	Moisture in Oil	Moisture in oil test DGA	Oil filled	% Saturation						
		Breather condition Inspection	Air filled				Discolouration desiccant			Correct colour of desiccant
	Dry / Corrosion	Internal Visual Inspection	All types							
	Dry / Corrosion	Internal Visual Inspection	All types							
Condensation	Dry / Corrosion	Internal Visual Inspection	Air filled							
		Internal Visual Inspection	Pitch filled							
		Internal Visual Inspection	Gel filled							
		Internal Visual Inspection	Compound filled							
Partial discharge	Tracking	External Visual inspection	All types							
		Internal Visual Inspection	Internal Visual Inspection				corona powder deposits on cable termination insulation,			
	pD-activity	RFI	RFI							
		Acoustic measurement	Acoustic measurement							
	Hydrogen and methane gases present	DGA	Oil filled	ppm						
Overheating	Wall Temperature	Thermal scanning	All types	°C			Wall Temperature increase above ambient			
	Overheated components	Internal inspection (if inspection practicable)	All types	°C			Overheated components			
	Heating gases present	DGA	Oil filled	ppm						
Miscellaneous	Shrinkage Cracks	Visual inspection	Pitch filled		Cracks in Pitch					Pitch normal
	Composition	Visual inspection, sample for physical tests	Gel filled							Homogenous, natural colour and composition
		Visual inspection, sample for physical tests	Cable compound filled							Homogenous, natural colour and composition

Table A-6: OLTC Condition Assessment

Failure Mode	Sub-component(s)	Degradation Mechanism	Condition Indicator	Test	Units	F	E	D	C	B	A	Reference
Thermal: Overheating	Diverter switch or Load Selector switch - Resistor type	Diverter blocked in-between positions -> Transition resistors inserted	Workmanship	Training/Qualification				OLTC serviced by a technician without the OEM certification			OLTC serviced by a technician with the OEM certification	
			OEM Specs	Visual	Time or operation count	100% late	80% late	60% late	40% late	20% late	Maintenance according to schedule	
			Transition time	Vibro-Acoustic or DRM	ms	Outside OEM Specs					Within OEM Specs	
			Temperature differential	Thermography							OLTC tank temperature lower than Main tank temperature	
Thermal: Coking	Main current carrying contacts (OLTC or DETC)	Contact degradation -> Carbon deposits -> Thermal runaway	Temperature differential	Thermography (bolt-on OLTC only)	Relative	$T_{oltc} >> T_{main}$		$T_{oltc} \geq T_{main}$				
			Gas Pattern Trending	DGA	Duval Triangle or other Ratios	Significant deviation in gassing pattern over a few weeks		Significant deviation in gassing pattern over a few months		Stable gas pattern - Slight differences with the benchmark population	Stable gas pattern - In agreement with benchmark LTC population	IEEE C57.139
			Time	Operating history - Time elapsed since last operation.	Years		> 30	> 10	> 3	> 1	< 1	
Dielectric: Tracking and Flashover	Insulation parts in Diverter compartment	Carbon deposit and moisture ingress in insulation material	OEM Specs	Visual	Time or operation count	100% late	80% late	60% late	40% late	20% late	Maintenance according to schedule	
			Water content	Karl Fischer	ppm		> 50	< 50	< 40	< 30	< 20	IEC 60814
	Oil	Insufficient dielectric properties	Breakdown Voltage		kv	< 20	> 20	> 25	> 30	> 40	> 50	IEC60156 2.5mm
Chemical: Corrosive	Fixed and mobile contacts in the main transformer	Flashover in main tank	Corrosivity test				Positive and presence of silver contacts in main tank				Negative	DIN51353

Condition assessment of power transformers

Failure Mode	Sub-component(s)	Degradation Mechanism	Condition Indicator	Test	Units	F	E	D	C	B	A	Reference
	tank (OLTC or DETC)	Hotspot on tap selector switch	Corrosivity test					Positive			Negative	ASTM 1275-15 Copper strip
Mechanical: Incomplete operation	Motor drive and linkage	Motor trip	Torque	Motor current	A			Deviation from benchmark or over OEM specification			Agreement with benchmark or within OEM specification	
	Tap selector (OLTC or DETC)											
Mechanical: Incorrect operation	Motor drive	Wrong direction, run through, intermittent stops, etc.	Age of the Motor Drive		Years			> 50	<=50	<30	< 20	
Mechanical: Wrong tap indication	Motor drive and linkage	Bad synchronism motor drive to LTC	Synchronism	Vibro- Acoustic or Manual	OEM Specs		Outside OEM Specs				Within OEM Specs	
		Wear of linkages	OEM Specs	Visual	Time or Operation Count	100% late	80% late	60% late	40% late	20% late	Maintenance according to schedule	
		Erroneous position signal	Age of the Motor Drive		Years			> 50	< 50	< 30	< 20	
Mechanical: Drive shaft failure	Motor drive and linkage	Linkage seizure	OEM Specs	Visual	Time or operation count	100% late	80% late	60% late	40% late	20% late	Maintenance according to schedule	
		Incorrect coupling -> Motor drive forcing through a mechanical stop	Workmanship	Training/ Qualification				OLTC serviced by a technician without an OEM certification			OLTC serviced by a technician with an OEM certification	
Mechanical: Leaks	LTC Oil compartment	Internal leaks: Migration of gasses and carbon particles into main tank or low/high levels of oil in	Material	Design	Type			Diverter compartment or barrier board made of cellulosic material (Bakelite, Formica, etc...)			Diverter compartment or barrier board made of fiberglass or other insulating composite material	

Condition assessment of power transformers

Failure Mode	Sub-component(s)	Degradation Mechanism	Condition Indicator	Test	Units	F	E	D	C	B	A	Reference
		the LTC compartment	Oil level alarms	Occurrences within 5 years	Count		> 5	> 3	> 1	1	No occurrences	
			Presence of LTC typical dissolved gasses, in particular C ₂ H ₂	DGA	ppm		Typical LTC gas pattern + C ₂ H ₂ > 1000	Typical LTC gas pattern + C ₂ H ₂ > 500	Typical LTC gas pattern + C ₂ H ₂ > 300	Typical LTC gas pattern + 0 < C ₂ H ₂ < 100	No detectable LTC typical gasses	
		External leak	Rate	Visual or Maintenance record			Major leaks visible down to ground - Events of low level alarm - Topping off the oil level required	Significant leaks visible, some may reach ground	Minor leaks - Do not reach ground		No visible leaks	

Table A-7: Cooler/Radiator Condition Assessment

Component	Failure Mode	Degradation Mechanism	Condition Indicator	Condition Assessment techniques	Units	F	E	D	C	B	A
Coolers/Radiators	Fan or control failure / fan in wrong direction	Inadequate air flow (elevated oil temperature)	Elevated temperatures	Visual inspection		Cooling Circuit complete failure	More than one Fan Failure /wrong direction		One fan Failure / Wrong Direction		All fans working
	Failure of pump / pump in wrong direction / valve closed incorrectly / blockage	Inadequate oil flow (elevated oil temperature)	Elevated temperatures	Measurement difference of inlet and outlet oil temperature. Compare between coolers	°C	$\Delta T < 30\%$	$\Delta T < 50\%$	$\Delta T < 60\%$	$\Delta T < 80\%$	$\Delta T < 90\%$	$\Delta T \geq 90\%$
			Oil flow relay, no pump current	Oil flow indication / measurement		No Oil Flow		Restricted Oil Flow			Normal Oil Flow
	Dirty radiators and cooling fans	Insufficient heat transfer (elevated oil temperature)	Blocked cooling ducts	Visual inspection				Poor air flow		Restricted air flow	Normal air flow
	Oil leaks	Low oil level (transformer alarm/trip)	Oil level gauge/alarm	Visual inspection				Oil level alarm			No oil level alarm
		Moisture and oxygen ingress	Elevated levels of moisture and oxygen	DGA etc.	ppm					H ₂ O > 20/400ppm (oil/ester) O ₂ > 5000 ppm	H ₂ O < 20/400ppm (oil/ester) O ₂ < 5000 ppm
	Elevated ambient air/cooling water inlet temperatures	Higher oil and winding temperatures (elevated oil temperature)	Unexpected OTI/WTI alarms	Visual inspection				Unexpected WTI/OTI alarms			No WTI/OTI alarms
Tank	Oil leaks	Defective or deteriorated gaskets and seals, incorrect tightening	Oil on ground or wet spots on tank	Visual inspection			Oil level alarm	Oil level drop, contamination issues, etc.	Moderate leaks	Minor leaks	No leaks

Condition assessment of power transformers

Component	Failure Mode	Degradation Mechanism	Condition Indicator	Condition Assessment techniques	Units	F	E	D	C	B	A
	Corrosion and improper painting	Heavy polluted site conditions / improper handling during installation or maintenance	Rust, corrosion, stains etc.	Visual inspection				Oil leaks and weakening of the tank,		Minor rust	No rust
Components	Incorrect operation of components	Failure of components / Drifting of alarm or trip settings	Failure / Out of calibration	Routine checks			Failed or mis-operating components	Drifting in alarm or trip settings			No defects
	Lightning Arresters	Damage / Contamination/ Lightning strikes/ Misapplication	Elevated temperatures, power factor or leakage current	Routine checks			Cracks / high temperature difference	Dirty with major particle build up / moderate temperature difference	Moderately dirty, with some particle build-up		No defects

Table A-8: Mineral Oil Condition Assessment

Failure Mode			Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
Physical Contamination			Corrosive Sulphur	All		Corrosive					Less than the detection limit.	IEC 60422
			Flash Point	All	° C		<122		130 to 135		>135	
	Chemical Contamination		Particle Count (ISO Class)	All			>18/15	16/13 to 17/14	11/8 to 15/12	9/6 to 10/7	Up to 8/5	IEC-60422 Annex B Table B.1
			Particle Count IEC 60970 above 5mm	All			>130 000 / 100ml	130 000 / 100ml	32 000 / 100ml	1 000 / 100ml	250 / 100ml	From table B1 of IEC 60422:2013
			Moisture in Oil	<72.5	ppm		>40		30 - 40	<30	<20	IEC 60422
		72.5-170				>30	20 to 30		<20	<10		
		>170				>20	15 to 20		<15	<10		
		BDV (IEC 60156 2.5mm)	<72.5	kV		<30		30 to 40	>40	>55	IEC 60422	
			72.5-170			<40		40 to 50	>50	>60		
			>170			<50		50 to 60	>60			
		DDF (Tan δ) @90°C (IEC 60247)	<72.5			>0.05		0.1 to 0.5	<0.1	≤ 0.015	IEC 60422	
			72.5-170			>0.2		0.1 to 0.2		≤ 0.010		
			>170									
		Resistivity	<72.5	GΩm		<0.2		3 to 0.2	>3	> 60	IEC 60422	
			72.5-170			<3		3 - 10		>10		
			>170					0.15 - 0.30		<0.15		
		Acidity	<72.5	mg KOH/g		>0.30		0.10 - 0.20	<0.10	≤ 0.03	IEEE C57.106, IEC 60422	
			72.5-170			>0.20		0.10 - 0.15				
			>170			>0.15						
		IFT	All (Inhibited)	mN/m		<22		22 - 28	>28	>35		
			All (Un-inhibited)	mN/m		<20		20 - 25	>25			
			Colour (by Lab testing)	All		Dark Brown (5.5 - 8.5)	Amber (4.0 - 5.5)	Bright Yellow (2.5 - 4.0)	Yellow (1.0 - 2.5)	Pale Yellow (0.5 - 1.0)	Clear (0.0 - 0.5)	IEEE C57.152
			Colour (by Visual Inspection)	All		Black/Dark Black	Dark and/or turbid *	Light Brown/Brown	Yellow/Dark Yellow	Light Yellow	Clear & without visible contamination *	IEC 60422, Table-5
			Inhibitor	All	%		<40% of original value		40% to 60% of original value	>60% of original value	>90% of original value	IEC 60422
			Sediment and Sludge	All	%		>0.02% by mass				<0.02% by mass	IEC 60422

Condition assessment of power transformers

Failure Mode	Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
Environmental Hazard	PCB	All	mg/kg			>2			<2	IEC 60422
Note:										
1. * Values referred from										
2. ** Values referred from IEC 60422, Table-3										

Table A-9: Natural Ester Condition Assessment

Failure Mode			Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
Physical Contamination			Corrosive Sulphur	All	°C	Slightly Corrosive					Non-corrosive	IEC 62535 or ASTM D1275B
			Fire Point	<72.5	°C		<300				>300	(ASTM D92 or ISO 2592)
				72.5-170								
				>170								
	Chemical Contamination		Particle Count (ISO Class)	All			>18/15	16/13- 17/14	11/8-15/12	9/6-10/7	Up to 8/5	IEC-60422 Annex B Table B.1
			Particle Count IEC 60970 above 5mm	All			>130 000 / 100ml	130 000 / 100ml	32 000 / 100ml	1 000 / 100ml	250 / 100ml	From table B1 of IEC 60422:2013
			Moisture in Oil (ASTM D1533, IEC60814)	<72.5	ppm		>550	450 - 550	385 - 450	300 - 385	<300	ASTM D1533 or IEC 60814
				72.5-170			>450	350 - 450	240 - 350	150 - 240	<150	
				>170			>400	300 - 400	200 - 300	100 - 200	<100	
		Degradation	BDV	<72.5	kV		<30		30 - 40	>40	>50	(IEC 60156 2.5mm)=
				72.5-170			<40		40 - 50	>50	>60	
				>170			<50		50 - 60	>60	>60	
			DDF (Tan δ) @25°C (ASTM D924)	All	%		> 3%		1 – 3	< 1	≤0.5	IEEE C57.147
			Acidity	All	mg KOH/g		>1	0.3 - 1	0.15 - 0.30	0.06 - 0.15	≤0.06	(ASTM D974, IEC 62021-3)

Condition assessment of power transformers

Failure Mode			Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
			Colour (by lab testing)	All			>1.5			≤1.0	≤0.5	
			Colour (by Visual Inspection)	All		Black/Dark Black	Dark and/or turbid	Light Brown / Brown	Yellow / Dark Yellow	Light Green / Light Yellow	Clear & without visible contamination	
			Viscosity (ASTM D445, ISO3104)	All		>50% of original value	>30% of original value	>20% of original value	>10% of original value	>5% of original value	<50 cSt	IEEE C57.147
			Inhibitor	All	%		<40% of original value		<60% of original value	<90% of original value	>90% of original value	Contact the manufacturer
			Sediment and Sludge	All	%	>0.02% by mass	<0.02% by mass				Less than the detection limit.	
Environmental Hazard			PCB	All	mg/kg	> 2					< 2	

Table A-10: Synthetic Ester Condition Assessment

Failure Mode			Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
Physical Contamination			Corrosive Sulphur	All	°C	Slightly Corrosive					Non-Corrosive	IEC 62535 or ASTM D1275B
			Fire Point	<72.5 72.5-170 >170	°C		<300				>300	ASTM D92 or ISO 2592
	Chemical Contamination		Particle Count (ISO Class)	All			> 18/15	16/13 - 17/14	11/8 - 15/12	9/6 - 10/7	Up to 8/5	IEC-60422 Annex B Table B1
			Particle Count IEC 60970 above 5mm	All			>130 000 / 100 ml	130 000 / 100 ml	32 000 / 100 ml	1 000 / 100 ml	250 / 100 ml	Table B1 of IEC 60422:2013
			Moisture in Oil (ASTM D1533, IEC60814)	<72.5	ppm		>800	600 - 800	400 - 600	<400	<200	
				72.5-170			>600	400 - 600	300 - 400	<300		
				>170			>400	300 - 400	200 - 300	<200		
		Degradation	BDV (IEC 60156 2.5mm)	<72.5	kV		<30		30 - 40	>40	>50	
				72.5-170			<40		40 - 50	>50	>60	
				>170			<50		50 - 60		>60	
			DDF (Tan δ) @25°C	All	%					<0.01	≤0.005	ASTM D924
			Acidity	All	mg KOH/g		>3.0	2.0 - 3.0	0.5 - 2.0	<0.5	≤0.1	ASTM D974 IEC 62021-3
			Colour (by lab testing)	All			>1.5			≤1.0	≤0.5	ISO 2049

Condition assessment of power transformers

Failure Mode			Affected Parameter/ Investigative Test	Voltage Class of the equipment	Unit	F	E	D	C	B	A	Reference
			Colour (by Visual Inspection)	All		Black/Dark Black	Dark Brown	Light Brown/Brown	Yellow/Dark Yellow	Light Yellow	Clear & without visible contamination	
			Inhibitor	All	%		<40% of original value		40-60% of original value	>60% of original value	>90% of original value	
			Sediment and Sludge	All	%	>0.02% by mass	<0.02% by mass				Less than the detection limit.	
Environmental Hazard			PCB	All	mg/kg			>2			<2	

Annex B How to Develop a TAI

B.1. Transformer Assessment and Development of TAIs

Condition assessment is a process by which a transformer is assessed considering all aspects that could affect future performance. This Technical Brochure has provided some suggestions on how to go about building a tool to understand the condition of a transformer and its intrinsic components. This section illustrates how these tools can be used in practice.

After completing a condition assessment, if the assessment or the score indicates a problem, users could also refer to Figure 1 found in CIGRE TB 445, Guide for Transformer Maintenance, which may assist in determining the appropriate course of action. For convenience, a small section of TB 445 is included at the end of this Appendix. Note however, that the terminology used in TB 445 does not directly align with the terminology in this Technical Brochure.

Section 2.3 of this Technical Brochure indicates that there are five steps required to develop a Transformer Assessment Index (TAI) for a fleet of transformers:

Step 1 – Determine the purpose of the Transformer Assessment Score and Index

Step 2 – Identify the Failure Modes & Mechanisms to be included in the TAI

Step 3 – Determine how each failure mode will be assessed

Step 4 – Design a calibrated system for categorising failure modes (Scoring Matrix)

Step 5 – Calculate a TAI score for each transformer

The following examples illustrate how these five steps can be applied in practice using the techniques explained in this Technical Brochure. **Note that the data is fictitious and has been generated only to illustrate the techniques.**

Step 1 - Determine the purpose of the Transformer Assessment Score and Index

Three separate indices are considered in this Annex:

- **Replacement Index:** to identify the transformers in the fleet which should be considered for replacement.
- **Repair Index:** to identify the transformers that should be considered for repairs or major maintenance activities. Note that this index **is not** designed to identify transformers that are due for routine maintenance e.g. maintenance of the On-Load Tap Changer. Routine maintenance should generally be carried out in accordance with the organisation's maintenance plan.
- **Refurbishment Index:** To identify transformer that should be considered for major refurbishment work i.e. work that is more complex and costly than the repair work that might be carried out to the transformers highlighted by the repair index.

Note that the Repair index has not been calculated in the examples, but TABLE B1 shows the failures mechanisms that could be considered for a Repair index. The repair index could be constructed similarly to the Replacement index but using the appropriate failure mechanisms.

Step 2 & Step 3 - Identify the failure modes and mechanisms to be included in the TAI and assessment techniques

TABLE B1 indicates the failure modes and mechanisms, and the related assessment techniques that **could** be used to develop the three simple TAIs, using information that many users would have available for a mineral oil filled transformer with Oil Impregnated Paper (OIP) bushings and an oil filled OLTC. Note that table B1 is intended only as an example to assist in understanding the methods described in the is brochure. Users should determine the failure modes and mechanisms that are appropriate for the index that they wish to develop. As the indices are being developed for different purposes, not all failure modes are included in each of the indices.

The table also indicates which failure mechanisms and diagnostic technique the user has chosen to include in each of the indices being developed for these examples. The diagnostic techniques chosen are only for the purpose of the example. Users should refer to Appendix A for details of other diagnostic techniques that are applicable to the failure modes and mechanisms that they wish to include in the index.

The TAI is also influenced by the user's experience, level of expertise, spare parts and replacement transformers available, the way in which the fleet is generally operated, etc. Therefore, it is expected that **users designing a TAI will consider their own circumstance and will likely choose different failure mechanisms and test techniques to include in their indices.**

Note also that users with more assessment data would likely develop more complex TAIs. ***The examples have been kept simple to better illustrate the methodology.*** The examples **are not** intended to be "Industry Best Practice" TAIs or to be CIGRE's preferred TAIs.

For the example, the 'user' has determined:

- High moisture content is not a driver for replacement, as the moisture can generally be removed as part of a refurbishment. However, a transformer will not be considered for refurbishment if the paper is already significantly degraded (High Furans). Similarly, transformers with high levels of Partial Discharge or Arcing will not be considered for refurbishment as it is unlikely that these problems can be easily corrected during the refurbishment process.
- Bushings can be replaced as part of either the Repair or Refurbishment process. However, as replacing bushings can be expensive if identical bushings are not available, defective bushings are also one of the drivers for replacement.
- OLTCs are maintained in accordance with original equipment manufacturer's instructions by suitably trained technicians. It is therefore assumed that any problems with OLTCs cannot be easily corrected by further maintenance or a simple repair. OLTC problems could be addressed during a Refurbishment and are also considered a driver for replacement.
- Most problems with the cooling system, transformer tank and ancillary components can be addressed by a refurbishment or repair and are not considered drivers for replacement.
- Mineral oil can be easily replaced and is not considered a driver for replacement. However, if oil tests indicate that the oil has sulphur compounds that could potentially react with the winding material to create copper corrosion, corrosive sulphur may have already been deposited in the winding. The user has therefore decided not to undertake major refurbishment work on transformers that have been filled with this type of oil. Although this type of oil is not considered a driver for replacement, it is used in the Repair Index as filter to prevent transformers that have been filled with corrosive oil from being refurbished.
- Diagnostic techniques have been chosen for the selected failure modes based on the techniques currently in use in the user's organisation, and data readily available to the user. Even though this may not give the most accurate assessment of condition, it ensures that data is available for most transformers being assessed.

Step 4 – Design a calibrated system for categorising failure modes (Scoring Matrix)

The generic Scoring Matrix in section 2.3.4 can be used to allow test results and transformer assessment data related to each of the failure mechanisms to be converted into one of five levels. (Green, Yellow, Orange, Pink or Red) in a consistent way. Note that variations of this scoring matrix have been used elsewhere in this brochure to better allow assessment of the components of the transformer.

Users should also consider and define the time scales used in the scoring matrix to ensure consistency during assessments i.e. the terms 'as soon as possible' and 'short term' should be replaced with times that suit the user. The times chosen may align with the times required to arrange maintenance or obtain spare components for transformers, or with longer term replacement or repair budgets.

Note that users can develop their own scoring matrix, with any number of levels; provided that a clear description of each level is assigned to ensure that scoring of all components and sub-components of the transformer are consistent.

Step 5 – Calculate a TAI score for each transformer

Multiple methods of calculating scores will be used in these examples to illustrate and compare different scoring techniques. Refer to Table 2-2 Methods of calculating a Transformer Assessment Score for other methods used to calculate a final score.

B.2. Examples

B.2.1. Worst Case Component Scores

For these examples, scores have been allocated to the following sub-components using a “Worst Case approach” scoring system. “Worst Case” scores are allocated to the following sub-components:

- Core and Windings – Repairable
- Core and Windings – Non - Repairable
- Bushings
- OLTC
- OLTC Insulating Oil
- Cooling System
- Transformer Tank
- Ancillary Components
- Corrosive Sulphur
- Main tank insulating Oil

Scores for each failure mechanism have been allocated for a transformer (Transformer 1) by the transformer engineer using the scoring matrix and the data available for the transformer. For each sub-component, one or more failure mechanisms has been considered and the worst score for all the considered failure mechanisms is allocated to the sub-component. Note that the test data and other information shows the scores allocated for the selected failure modes/degradation mechanisms and shows how the worst case scoring method has been used to allocate scores to the sub-components.

Table B1 Worst Case Sub-Component Scores

Core and Windings (Main Tank)								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
Thermal aging of solid insulation Kraft paper	Furanic compound analysis	oil sample	Yes	No	Filter	Pink	Core and Windings - Non-repairable	Pink
Partial Discharge	Hydrogen	DGA	Yes	No	Filter	Orange		
Arcing	Acetylene	DGA	Yes	No	Filter	Yellow		
Core Ground	DC Resistance	Insulation resistance	No	No	Yes	Green	Core and Windings - Repairable	Pink
Moisture Content of Liquid	Moisture Content	Karl Fisher Moisture Content	No	No	Yes	Pink		
OIP Bushings								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
	DF/PF or C1 measurement	tan delta or PF increase, or change of Capacitance (C1)	Yes	Yes	Yes	Orange	Bushings	Orange
Shed Damage, Oil leakage, or oil degradation, Issues with Voltage Tap	Check oil in sight glass and sheds of insulator, check voltage tap for burn marks	Visual inspection	Yes	Yes	Yes	Yellow		
Overheating in bushing top	Thermal scanning	Temperature increase above ambient	Yes	Yes	Yes	Green		

OLTC								
Failure / Degradation Mechanism	Cause of Degradation	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
Main Current Carrying ContactsContact degradation -> Carbon deposits -> Thermal runaway	Loss of conductivity: pressure or surface degradation	Temperature differential	Yes	No	Yes	Green	OLTC	Orange
		Thermography (bolt on OLTC only)	Yes	No	Yes	Green		
	No operation over a long period (Change-over switch or DETC)	Gas Pattern Trending - DGA	Yes	No	Yes	Orange		
	Moisture ingress	Water content - Karl Fischer	No	Yes	Yes	Pink	OLTC Insulating Oil	Pink
OLTC Oil - Insufficient dielectric properties	Presence of contaminants or particles	Breakdown Voltage - IEC60156	No	Yes	Yes	Orange		
Cooling System								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
Failure of fan / Pump Cooling control circuit failure Wrong direction of fan / Pump	Inadequate air or oil flow volume (overheating)	Routine Inspection and Testing, Visual Inspection	No	Yes	Yes	Green	Cooling System	Pink
Dirty radiators and cooling fins	Inefficient heat transfer (overheating)	Routine inspection	No	Yes	Yes	Orange		
Leaks caused by: galvanic corrosion thermal expansion and contraction	Lower oil level (transformer trip)		No	Yes	Yes	Pink		
	Moisture ingress Oxygen Ingress (Premature aging of insulation)		No	Yes	Yes	Pink		
Transformer Tank								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
Oil leakage from main tank gasket joint	Gasket defective/damaged Improper tightening Loose bolts	Look for oil leaks and check the gasket and bolts	No	Yes	Yes	Orange	Transformer Tank	Pink
Leakages from other joints, welding joint/welding crack of transformer	Gasket defective/damaged Improper tightening Loose bolts Cracks	Visual inspection	No	Yes	Yes	Pink		
Corrosion, rusting, dents and paint peel off	Due to poor maintenance Heavy polluted site conditions Improper handling during maintenance/ Installation	Visual inspection	No	Yes	Yes	Orange		
Overpainting / improper painting	Poor maintenance	Visual inspection	No	Yes	Yes	Orange		

Ancillary Components								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
Incorrect operation of components • Dehydrating Breather • Temperature Devices	• Failure of components • Components out of setting	Routine Inspection and testing / calibration	No	Yes	Yes	Green	Ancillary Components	Green
Mineral Oil								
Failure / Degradation Mechanism	Criteria / Condition Indicator	Diagnostic Method	Replacement Index	Repair Index	Refurbishment Index	Transformer 1	Sub-Component	Sub-Component Score
	Corrosive Sulphur	Test of Oil Sample	No	Yes	Filter	Green	Corrosive Sulphur	Green
Degradation	BDV	Test of Oil Sample	No	Yes	Yes	Orange	Main tank Insulating Oil	Red
	Tan Delta	Test of Oil Sample	No	Yes	Yes	Pink		
	Resistivity	Test of Oil Sample	No	Yes	Yes	Orange		
	Acidity	Test of Oil Sample	No	Yes	Yes	Red		
	IFT	Test of Oil Sample	No	Yes	Yes	Orange		
	Colour (by Visual Inspection)	Test of Oil Sample	No	Yes	Yes	Orange		
	Inhibitor					Pink		
	Sediment and Sludge					Orange		

Refer to section 2.4.2 for details of how the above information can be used to calculate an assessment score for the transformer

B.2.2. Assessed sub-component scores for the fleet

Sub-Component scores, which have been generated from data from 10 fictitious transformers using the scoring matrix and the worst case scoring method (to derive sub-component scores), are provided below, and will be used to generate TAI scores for each of the transformers. (Note that the raw data has not been provided).

Table B2 Assessed Sub-Component scores for the fleet

Sub-Component	Main Tank - Repairable	Main Tank - Non Repairable	Bushings	OLTC	OLTC Insulating Fluid	Cooling System	Tx Tank	Ancillary Comp.	Corrosive Sulphur	Mineral Oil
TX 1 Score	Pink	Pink	Orange	Orange	Pink	Pink	Pink	Green	Green	Red
TX 2 Score	Green	Green	Red	Green	Yellow	Yellow	Orange	Yellow	Green	Orange
TX 3 Score	Red	Orange	Red	Yellow	Yellow	Pink	Pink	Orange	Green	Red
TX 4 Score	Yellow	Orange	Green	Yellow	Green	Yellow	Orange	Green	Red	Yellow
TX 5 Score	Red	Red	Orange	Orange	Red	Pink	Pink	Orange	Green	Red
TX 6 Score	Yellow	Yellow	Green	Green	Yellow	Yellow	Yellow	Green	Green	Yellow
TX 7 Score	Yellow	Green	Green	Green	Yellow	Yellow	Yellow	Green	Green	Green
TX 8 Score	Pink	Red	Red	Orange	Pink	Pink	Pink	Yellow	Red	Pink
TX 9 Score	Yellow	Yellow	Green	Green	Green	Yellow	Yellow	Green	Green	Orange
TX 10 Score	Red	Orange	Orange	Yellow	Orange	Red	Red	Red	Green	Orange

B.2.3. Replacement Index

The sub-components applicable to the Replacement Index and the scores for the 10 transformers are detailed in the following table.

A number of different ways of generating a total score for the transformer are shown. The scores can assist the Asset Manager to determine which transformers should most urgently **be considered** for replacement.

Note however, that transformers that have only bushings or OLTCs scored as Red, may also be suitable for repair or refurbishment.

Table B3 Calculation of Replacement Scores

	Main Tank - Non Repairable	Bushings	OLTC	Simple Score	Red	Pink	Orange	Yellow	Green	Simple Numeric Score	Hybrid Score	Non Linear Score
TX 1 Score	Pink	Orange	Orange	1 Pink	0	1	2	0	0	7	7 (Pink)	45
TX 2 Score	Green	Red	Green	1 Red	1	0	0	0	2	4	4 (Red)	83
TX 3 Score	Orange	Red	Yellow	1 Red	1	0	1	1	0	7	7 (Red)	93
TX 4 Score	Orange	Green	Yellow	1 Orange	0	0	1	1	1	3	3 (Orange)	13
TX 5 Score	Red	Orange	Orange	1 Red	1	0	2	0	0	8	8 (Red)	99
TX 6 Score	Yellow	Green	Green	1 Yellow	0	0	0	1	2	1	1 (Yellow)	5
TX 7 Score	Green	Green	Green	3 Green	0	0	0	0	3	0	0 (Green)	3
TX 8 Score	Red	Red	Orange	2 Red	2	0	1	0	0	10	10 (Red)	171
TX 9 Score	Yellow	Green	Green	1 Yellow	0	0	0	1	2	1	1 (Yellow)	5
TX 10 Score	Orange	Orange	Yellow	2 Orange	0	0	2	1	0	5	5 (Orange)	21

The Asset Manager will have to further consider the appropriate action for the transformers being considered for replacement.

The **Simple Score** shown in the table is a worst-case score – but the number of sub-components which have that score is also indicated. Tx 2, Tx 3, Tx 5 and Tx 8 all need urgent attention. However, it is obvious from the score Tx 8 has 2 sub-components that need urgent attention.

The **Count per Category** approach provides more information about the overall condition of the transformer. Although displaying the result in this way is not as simple as a single number, the 5 columns of numbers can be easily ordered, so that the transformers that most urgently need attention are at the top of the list.

The **Simple Numeric Score** is calculated in this example, by allocating marks as follows. **Green = 0, Yellow = 1, Orange = 2, Pink = 3 and Red = 4**. Although this type of scoring provides a simple numeric score, the urgency, or the time scale for action, is lost. Tx 2 has a score of only 4, even though one sub-component needs urgent attention! Tx 10 has a worse score, even though the worst sub-component score is Orange, and it does not need urgent attention.

The **Hybrid Score** combines the simple numeric score with the **Worst-Case** sub-component score. This gives an indication of the time scale for action (from the worst case score), along with an indication of the overall condition of the transformer (from the simple numeric score). Tx 2 has a score of 4 (Red). This indicates that the overall condition of the transformer is good, but a sub-component needs urgent attention.

The **Non-linear Score** is calculated using the formula:

$$TAI = \sum_{n=0}^{k-1} x_n i^n$$

Where:

- i is the number base / radix, which is equal to or greater than the number of failure modes included in the TAI
- x_n is the number of failure modes per category
- k is the number of categories included in the failure mode assessment

Although it is not immediately obvious, the single numeric score per transformer indicates the timescale for action. A score above 81 can only be achieved if at least 1 sub-component needs urgent attention (i.e. is scored as Red). Similarly a score above 27 can only be achieved with at least 1 sub-component scored as Pink.

B.2.4. Use of Weighting Factors

The designer of the TAI has decided that although bushings and OLTCs are drivers for transformer replacement (it may be more economical to replace a transformer if multiple sub-components need attention), they can also be addressed during a repair or refurbishment process. Weighting factors have been added to the scores for bushings and OLTCs to reduce the scores of those transformers whose bushings and OLTCs need attention. (Due to the simplicity of the example, this leaves only 1 sub-component without a weighting factor.) In the following example, bushings have been weighted by a factor of 0.3, and OLTCs by a factor of 0.5. These weightings have been chosen only to illustrate the effect on the resultant scores. Users wishing to use weighting factors should determine appropriate weighting factors to suit their TAI and their needs.

Table B4 Scores with Weighting Factors

	Main Tank - Non Repairable	Bushings (0.3 Weighting)	OLTC (0.5 Weighting)	Simple Score	Red	Pink	Orange	Yellow	Green	Simple Numeric Score
TX 1 Score	Pink	Orange	Orange	1 Pink	0	1	0.8	0	0	4.6
TX 2 Score	Green	Red	Green	0.3 Red	0.3	0	0	0	1.5	1.2
TX 3 Score	Orange	Red	Yellow	0.3 Red	0.3	0	1	0.5	0	3.7
TX 4 Score	Orange	Green	Yellow	1 Orange	0	0	1	0.5	0.3	2.5
TX 5 Score	Red	Orange	Orange	1 Red	1	0	0.8	0	0	5.6
TX 6 Score	Yellow	Green	Green	1 Yellow	0	0	0	1	0.8	1
TX 7 Score	Green	Green	Green	3 Green	0	0	0	0	1.8	0
TX 8 Score	Red	Red	Orange	1.3 Red	1.3	0	0.5	0	0	6.2
TX 9 Score	Yellow	Green	Green	1 Yellow	0	0	0	1	0.8	1
TX 10 Score	Orange	Orange	Yellow	1.3 Orange	0	0	1.3	0.5	0	3.1

The dangers of using a Simple Numeric Scoring system with weighting factors can be clearly seen from this example! Tx 2 now has a score of only 1.2. If the transformers are ordered based on their simple numeric score, Tx 2 would be number 8 on the list of 10. This transformer will not get the attention of an Asset Manager, despite the fact that it has a failure mode that needs to be addressed urgently! Note however that other scoring systems still show the timescale for required action.

B.2.5. Refurbishment Index

The refurbishment index uses the Main Tank – Non Repairable and the Corrosive Sulphur scores as filters. If either of these scores are Pink or Red, then the transformer is not considered for refurbishment, as it may have limited remaining life. On this basis, Tx 1, Tx 4, Tx 5 and Tx 8 have not been scored or considered for refurbishment. Scores for 8 other sub components have been used to determine which transformers are most in need of refurbishment.

Table B5 Filtering to develop a Refurbishment Index

Main Tank - Non Repairable	Corrosive Sulphur	Sub-Component	Main Tank - Repairable	Bushings	OLTC	OLTC Insulating Fluid	Cooling System	Tx Tank	Ancillary Comp.	Mineral Oil
Pink	Green	TX 1 Score	-	-	-	-	-	-	-	-
Green	Green	TX 2 Score	Green	Red	Green	Yellow	Yellow	Orange	Yellow	Orange
Orange	Green	TX 3 Score	Red	Red	Yellow	Yellow	Pink	Pink	Orange	Red
Orange	Red	TX 4 Score	-	-	-	-	-	-	-	-
Red	Green	TX 5 Score	-	-	-	-	-	-	-	-
Yellow	Green	TX 6 Score	Yellow	Green	Green	Yellow	Yellow	Yellow	Green	Yellow
Green	Green	TX 7 Score	Yellow	Green	Green	Yellow	Yellow	Yellow	Green	Green
Red	Red	TX 8 Score	-	-	-	-	-	-	-	-
Yellow	Green	TX 9 Score	Yellow	Green	Green	Green	Yellow	Yellow	Green	Orange
Orange	Green	TX 10 Score	Red	Orange	Yellow	Orange	Red	Red	Red	Orange

Refurbishment scores are show in the table below

Table B6 Refurbishment Scores

	<i>Red</i>	<i>Pink</i>	<i>Orange</i>	<i>Yellow</i>	<i>Green</i>	Simple Numeric Score	Hybrid Score	Non Linear Score
TX 1 Score	-	-	-	-	-	-	-	-
TX 2 Score	1	0	2	3	2	11	11 (Red)	4250
TX 3 Score	3	2	1	2	0	22	22 (Red)	13392
TX 4 Score	-	-	-	-	-	-	-	-
TX 5 Score	-	-	-	-	-	-	-	-
TX 6 Score	0	0	0	5	3	5	5 (Yellow)	43
TX 7 Score	0	0	0	4	4	4	4 (Yellow)	36
TX 8 Score	-	-	-	-	-	-	-	-
TX 9 Score	0	0	1	1	4	3	3 (Orange)	76
TX 10 Score	4	0	3	1	0	23	23 (Red)	16584

As there are now 8 sub components being scored into the five categories, the non-linear score gives much larger numbers. Red scores are multiplied by 8⁴ as per the formula.

B.3. Extract from TB 445

If the assessment or assessment score indicates a problem, or multiple problems of different severity, then choices need to be made with respect to actions, and determination are they MAJOR or MINOR work.

MINOR WORK, could be classified as “Corrective Maintenance” which is an operation carried out to restore any part of the transformer, which has “failed” or “degraded” to the point where it needs corrective action to avoid loss of performance or a major failure. Examples might include oil processing, cooling fan replacement, repair of leaks, possibility bushing replacement, and OLTC minor

repairs. It implies that the unit, is out of service during this work for a short period of time, BUT is not removed from the site to affect these repairs or component replacements.

MAJOR WORK, could be classified under a few definitions, all of which should be passed through a Technical and Economic Evaluation to arrive at the decision to either Refurbish or Repair, or scrap and replace.

Technical & Economic Evaluation

The cycle of transformer operation and routine corrective maintenance is not perpetual. When an asset suffers severe damage or when its reliability is no longer satisfactory, a technical and economic evaluation must be made to decide the best option between scrap and replace, repair or refurbish and, if the work is to be performed on-site or in a workshop. CIGRE TB 248 Guide on Economics of Transformer Management, provides an excellent normative approach to this process.

Major Repair, ranges from replacement or refurbishment of major components, such as ALL the bushings (at one time interval), OLTC overhaul, or complete removal and replacement of the cooling system. This still implies work which may be done at site, but over a longer outage duration.

In most cases ANY movement of a unit from the site to a repair facility is considered MAJOR work.

REFURBISHMENT, is an option, that may have been decided based on an economic evaluation that it is worthwhile, such as complete replacement of major insulation, or complete new windings. Any work that involved complete removal, and disposal of the dielectric fluid can be defined as major work.

Scrap and Replace, (or END of Life) The service life of a transformer should end when its condition is such that it cannot be kept in service, nor be put back into service, primarily because a technical and economic evaluation has determined that its return to a serviceable condition is not economical, nor potentially unsafe.

Annex C Literature overview

Introduction

Many other groups and individuals have developed and published information about Transformer Assessment Indices; although they have generally referred to their indices as a “Health Index.” For the benefit of readers, this Appendix aims to provide an overview of some of the current documents that have been published about this topic, and the methodologies that these documents describe. Many other documents have also been written about this topic, but it was not possible to identify, review and reference all documents.

It must be noted that concepts, methodologies and terminology detailed in these publications, will in many cases vary from what is presented in the Chapters of this Technical Brochure. The references to these documents are however provided to allow readers to consider the concepts and methods proposed by others. It should also be noted that terminology e.g. “Health Index,” used in this Appendix is the terminology generally used by the authors of the reference documents and may vary from the terms, or the definitions of these terms, used in the Chapters of this brochure.

This document does not aim to provide a detailed description or to make recommendations or comments regarding the published methodologies. The interested reader should refer to the content of the references for further information. As this is a relatively new topic, research is still being undertaken and no consensus has been reached in the literature on a preferred method.

Overview

Generally, the process encompasses several steps [1], as shown in Figure 1, and this document presents an overview of this process with references to published literature.

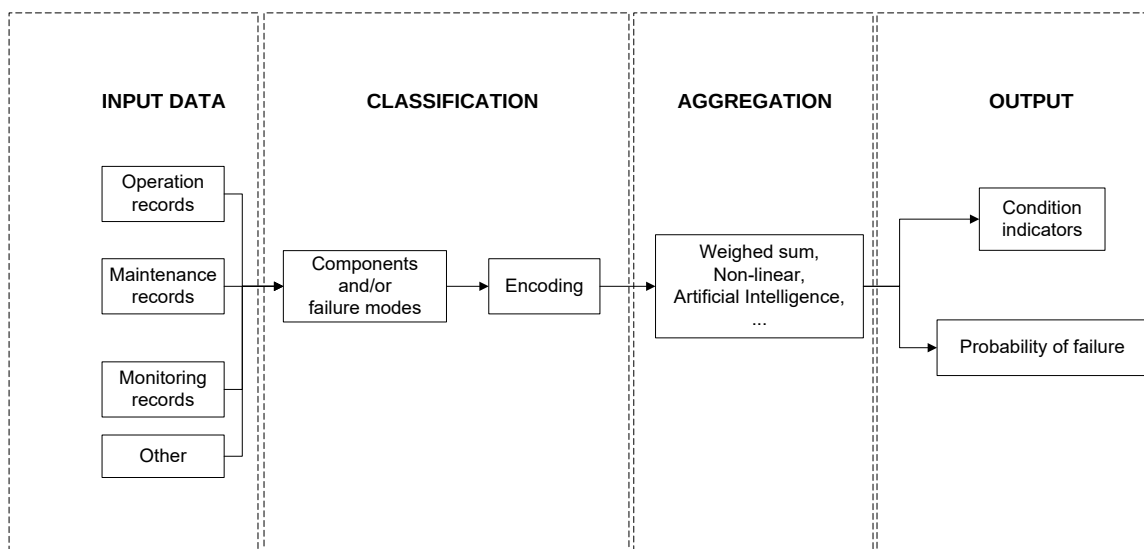


Figure C.1 – Process for the calculation and uses of the health index

Input data

- Origin of data

The pragmatic approach, as presented in [2], is to collect the available data about the transformers' condition that will later be fed into the aggregation algorithm. Gathering these pieces of data is a crucial task since the value of the health index, the output of its analysis and its potential applications directly depend on the quality of the original data. Maintenance guide and practices including typical routinely performed diagnostic tests can be found in [3-5].

Besides results from diagnostic tests, other general information such as a transformer's age, load history, service conditions and previous maintenance actions are often recorded by utilities and may also be fed into the health index [6-8]. Authors in [9] include economic parameters in their assessment to use the health index result to prioritize maintenance activities.

Often, data is not usable due to values being aberrant or unavailable. In these cases, different choices can be made. In [10, 11], authors chose not to include transformers for which data is missing. In [12],

the authors do not take into account nonexistent data for the calculation [10]. Authors have also assumed a good condition but with a lower confidence in the assessment. In [13], each data input is associated with a reliability index.

When very little data is available, statistical curve-fitting techniques [14, 15] may also be used to try to determine a trend within the few results present. These curves are then compared against known values. This protocol may help to assess the validity of the data. When possible, experts also try to deduce data from other parameters that are already known [15].

- Encoding of data

Input parameters generally consist of direct values taken from the aforementioned tests and service condition records. The number of input parameters varies widely from one paper to another as it ranges from less than 5 to more than 20 [1]. When a lot of parameters are taken into account, two different aggregation approaches are considered. Some authors keep all their parameters as individual inputs [1, 8, 9, 11, 13, 16-21] whereas others group some parameters into sub-categories or sub-indices [1, 7, 14, 22-32] related to one component (e. g.: the active part or the oil). The available data can also be classified and associated to the failure modes of the components to be assessed [33, 34].

Both approaches result in the clustering of data of different natures, for instance a number obtained by an objective measurement and a word obtained by a subjective (visual) evaluation. To place all data on the same plan, authors in [13, 24, 35] use standardized dimensionless numbers between 0 and 1. Similarly, it is also possible to establish a scoring system. In this case, results from maintenance or monitoring tests are all assigned a score according to their value. To harmonize all the values before aggregating the data in a health index, scores can be put on the same scale (e. g. from 1 to 5) [1, 9, 11, 14, 20, 21, 23, 26, 28, 30, 31, 34, 36-39]. Scales can be determined by classifying values and applying thresholds. These scoring scales were questioned [16, 34, 40] as most papers use linear scoring scales and this linearity may give too much or not enough weight to some values. It could also suggest a proportionality factor within the results that may not actually exist.

Aggregation methods

- Weighted sum

In this approach, the input values are all given weights depending on their relative importance for the condition assessment of the transformer. Weights may also be determined considering the final use of the health index [7].

Authors from [41] perform a common factor analysis to identify correlations between results from the oil tests and to determine which weights should be assigned to the different values of the oil tests.

A weighted sum is then calculated, and the output value is a number. Various scales can be used for the output: a digit ranked on scale starting from 0 or 1 [1, 10, 11, 27, 39, 42] or a scale ranging from 0 to 1, such that the number can be represented as a percentage [1, 7, 13, 14, 20, 21, 23-26, 29, 35, 38, 41, 43].

In [44], failure statistics and the score related to each failure mode are associated to calculate the individual failure rate of the transformer.

Authors from [32] choose to calculate a quadratic mean of all the parameters.

However, according to [16, 34, 40], the use of this procedure leads to some failure modes being hidden or averaged by others.

- Sum of non-linear scores

Authors in [37] use a non-linear scoring scale and in [34], authors use a logarithmic scale for the standardization of the input parameters. The goal is to make the process capable of pointing out a specific component's poor condition regardless of other components in good condition.

- Worst case

Statistics can be another approach for the calculation of the health index. Several models are presented in the literature. In [33, 40, 45], authors analyse the different failure modes using the Failure Mode and Effect Analysis (FMEA).

In [33, 40, 45], the input data is combined into assessment functions that are not specifically detailed in the papers. Authors from [33, 40] combine the input data for each failure mode to determine the

transformer's probability of failure. They also use comparative statistics to determine trends in data and to identify sudden changes that could account for an unexpected degradation of the apparatus.

In [45], results from the FMEA provide information about the worst case scenario for a transformer's failure. The combination of the assessment functions and this worst case approach results in the health index.

- Statistical regression

In [19] and [28], authors aggregate the data by performing a statistical regression. For instance, authors in [19] use binary logistic regression. Input parameters are sorted into only two categories: healthy or unhealthy. Weights that should be assigned to each input are determined by a maximum likelihood criterion. The health index is calculated using a non-linear equation. A logistic regression is then performed on the entire fleet's results to assess their validity and, if need be, it can be used to refine the values of the weighing coefficients.

- Artificial Intelligence

The development of computer science and data processing over the last years has resulted in the creation of new algorithms such as pattern recognition algorithms [31], neural networks [9, 16, 18], wavelet networks [17], fuzzy logic [30, 46], optimization methods [36, 47], all based on machine learning techniques. These new tools are capable of analysing large datasets [9], for instance results from monitoring records. However, the accuracy of the algorithm mostly relies on the model chosen by the user and the amount of data that the algorithm is fed with [9]. As a consequence, an increase in the amount of input data leads to an improvement of the algorithm's performance.

The result of the calculation is a number, typically between 0 and 1 [9, 16-18, 30, 47]. Output values are compared against those obtained via the weighted sum but they could be equally applied to any non-linear approaches.

In [12], authors develop a computer package based on Boolean logic to aggregate all the input data. The package provides the user with maintenance recommendations.

- Summary

As shown previously, the methods used to aggregate the data into a health index are various and research is still being made on this topic. Table 1 summarizes the link between the aggregation methods and the published papers.

Table 1 – Summary of the aggregation methods

Aggregation methods	Papers
Weighted sum	[1, 7, 10, 11, 13, 14, 20, 21, 23-27, 29, 32, 35, 38, 39, 41-44]
Sum of non-linear scores	[34, 37]
Worst case	[33, 40, 45]
Statistical regression	[19, 28]
Artificial Intelligence	[9, 12, 16-18, 30, 31, 36, 46, 47]

Outputs

- Condition indicators

As stated in [14], the health index represents the asset condition in an easy way (a colour, a number or a text string) to be used by the Asset Manager. The classification of the transformers may contain three [18, 39, 46, 47], four [8, 16, 22, 27, 35, 38] or five classes [2, 9, 20, 21, 23, 25-27, 30], the condition of the transformer being ranged from excellent/very good to mediocre/very bad. A colour gradient from green to red may also be used [7, 8, 15, 25, 35, 45, 48] to easily bring to light the state of the transformer. This gradient may also provide information about the quality of the original data. In [45], the health index is displayed on a 3D graph. One axis represents the condition of the unit with a colour gradient from green to red; a second axis represents the quality of the original data by fading the colour as the confidence-level on the data decreases. A third axis gives information about the number of transformers per category. Further analyses of the input data may then provide information for fault diagnosis. For instance, authors in [12] propose a diagnosis of the transformer by combining input data using logic gates.

- Probability of failure

As for the Health Index, there is no consensus in the literature regarding a methodology to assess the probability of failure.

In the literature related to the health index of transformers, the probability of failure is determined using statistical distributions [10, 13, 15, 23, 24, 26, 28, 42-44], with specific failure analysis techniques such as FMEA or probabilistic fault trees [33, 40], directly via the health index output value [2, 9, 11, 22, 26], or indirectly via the apparent age concept. The concept of apparent age – or effective age – translates the condition assessment information in a corrected age that may be used to estimate the present and future probability of failure based on statistical analyses [10, 26, 42, 43].

Authors in [25, 26] use the probability of failure as a basis for estimating the remnant life of the transformer. In [15], the remnant life is estimated directly using the input data from maintenance and monitoring records. A flowchart is then used to determine the need for maintenance of the transformer.

References used in the Literature review

- [1] A. Azmi, J. Jasni, N. Azis, and M. Z. A. A. Kadir, "Evolution of transformer health index in the form of mathematical equation," *Renewable and Sustainable Energy Reviews*, vol. 76, pp. 687-700, 2017.
- [2] D. T. Hughes, "The use of health indices to determine end of life and estimate remnant life for distribution assets," *Proceeding of 17th International Conference on Electricity Distribution*, 2003.
- [3] CIGRE Working Group A2.18, "Life management techniques for power transformer," Brochure 227, 2003.
- [4] CIGRE Working Group A2.34, "Guide for transformer maintenance," Brochure 445, 2011.
- [5] M. M. Islam, G. Lee, and S. N. Hettiwatte, "A review of condition monitoring techniques and diagnostic tests for lifetime estimation of power transformers," *Electrical Engineering*, pp. 1-25, 2017.
- [6] IEC White Paper, "Strategic asset management of power networks," 2016.
- [7] A. Babizki, M. Foata, K. Lindl, D. Beerboom, E. Pawlowski, M. Zdrallek, et al., "Practical approach for fleet management of transformers considering different stakeholders' perspectives," 4th International Colloquium "Transformer Research and Asset Management", 2017.
- [8] N. Dominelli, M. Lau, D. Olan, and J. Newell, "Equipment health rating of power transformers," in *Conference Record of IEEE International Symposium on Electrical Insulation*, 2004, pp. 163-168.
- [9] H. Zeinoddini-Meymand and B. Vahidi, "Health index calculation for power transformers using technical and economical parameters," *IET Science, Measurement and Technology*, vol. 10, pp. 823-830, 2016.
- [10] C. Rajotte and P. Picher, "The use of health index for condition assessment and replacement prioritizing," *TechCon North America*, 2017.
- [11] A. Singh and A. G. Swanson, "Analysis and improvement of the plant health index used by ESKOM to assess the condition of the power transformers in its distribution fleet," *Proceeding of the 19th International Symposium on High Voltage Engineering*, 2015.
- [12] C. Roncero-Clemente and E. Roanes-Lozano, "A multi-criteria computer package for power transformer fault detection and diagnosis," *Applied Mathematics and Computation*, 2016.
- [13] B. Nemeth, C. Voros, and G. Csepes, "Health index as one of the best practices for condition assessment of transformers and substation equipments - Hungarian experience," *CIGRE*, 2014.
- [14] CIGRE Working Group C1.16, "Transmission asset risk management," Brochure 422, 2010.
- [15] M. Vermeer and J. Wetzter, "Transformer health and risk indexing, including data quality management," *CIGRE*, 2014.
- [16] M. M. Islam, G. Lee, and S. N. Hettiwatte, "Application of a general regression neural network for health index calculation of power transformers," *International Journal of Electrical Power and Energy Systems*, vol. 93, pp. 308-315, 2017.
- [17] M. Ahmed, M. Elkhatib, M. Salama, and K. B. Shaban, "Transformer health index estimation using orthogonal wavelet network," in *2015 IEEE Electrical Power and Energy Conference: Smarter Resilient Power Systems, EPEC 2015*, 2016, pp. 120-124.

- [18] A. E. B. Abu-Elanien, M. M. A. Salama, and M. Ibrahim, "Determination of transformer health condition using artificial neural networks," in INISTA 2011 - 2011 International Symposium on INnovations in Intelligent SysTems and Applications, 2011.
- [19] W. Zuo, H. Yuan, Y. Shang, Y. Liu, and T. Chen, "Calculation of a health index of oil-paper transformers insulation with binary logistic regression," *Mathematical Problems in Engineering*, vol. 2016, 2016.
- [20] M. A. Martins, "Condition and risk assessment of power transformers: A general approach to calculate a Health Index," *Ciencia e Tecnologia dos Materiais*, vol. 26, pp. 9-16, 2014.
- [21] W. Wattakapaiboon and N. Pattanadech, "The new developed Health Index for transformer condition assessment," in CMD 2016 - International Conference on Condition Monitoring and Diagnosis, 2016, pp. 32-35.
- [22] Y. Sukrislismo and A. Kartika, "Case study on asset health indices implementation on distribution transformer asset at PLN Bali Indonesia," in IET Conference Publications, 2015.
- [23] I. G. N. S. Hernanda, A. C. Mulyana, D. A. Asfani, I. M. Y. Negara, and D. Fahmi, "Application of health index method for transformer condition assessment," in IEEE Region 10 Annual International Conference, Proceedings/TENCON, 2015.
- [24] I. C. Durán, L. S. Rosero, A. Pavas, and O. G. Duarte, "Comparison of maintenance scheme effects on power transformer reliability performance," *Ingenieria e Investigacion*, vol. 35, pp. 73-81, 2015.
- [25] A. Naderian, S. Cress, R. Piercy, F. Wang, and J. Service, "An Approach to Determine the Health Index of Power Transformers," in Conference Record of the 2008 IEEE International Symposium on Electrical Insulation, 2008, pp. 192-196.
- [26] A. N. Jahromi, R. Piercy, S. Cress, J. R. R. Service, and W. Fan, "An approach to power transformer asset management using health index," *IEEE Electrical Insulation Magazine*, vol. 25, pp. 20-34, 2009.
- [27] F. Ortiz, I. Fernandez, A. Ortiz, C. J. Renedo, F. Delgado, and C. Fernandez, "Health indexes for power transformers: A case study," *IEEE Electrical Insulation Magazine*, vol. 32, pp. 7-17, 2016.
- [28] J. Qiu, H. Wang, D. Lin, B. He, W. Zhao, and W. Xu, "Nonparametric regression-based failure rate model for electric power equipment using lifecycle data," *IEEE Transactions on Smart Grid*, vol. 6, pp. 955-964, 2015.
- [29] F. Scatiggio, L. Calcara, and M. Pompili, "Risk prevention for HV transformers: Beyond the health index," in 34th Electrical Insulation Conference, EIC 2016, 2016, pp. 182-185.
- [30] L. Sun, Z. Ma, Y. Shang, Y. Liu, H. Yuan, and G. Wu, "Research on multi-attribute decision-making in condition evaluation for power transformer using fuzzy AHP and modified weighted averaging combination," *IET Generation, Transmission and Distribution*, vol. 10, pp. 3855-3864, 2016.
- [31] T. K. Saha, H. Ma, C. Ekanayake, D. Martin, and D. Allan, "Pattern recognition techniques for determining the health index of oil-paper insulation of in-service power transformers," *CIGRE*, 2014.
- [32] I. V. Davidenko and T. S. Kuzina, "Analysis of the modern methods of the power transformers health index calculation," in Proceedings of the 2017 IEEE Russia Section Young Researchers in Electrical and Electronic Engineering Conference, EIConRus 2017, 2017, pp. 1491-1495.
- [33] P. Lorin, L. Cheim, L. Pettersson, K. Gustafsson, and E. Tenyenhuis, "Transformer health index and probability of failure based on failure mode effects and analysis (FMEA) of a reliability centered maintenance program (RCM)," *CIGRE 2016*, 2016.
- [34] R. J. Heywood and T. McGrail, "Clarifying the link between data, diagnosis and Asset Health Indices," in IET Conference Publications, 2015.
- [35] F. Scatiggio, A. Fraioli, V. Iuliani, and M. Pompili, "Health index: the TERNAs practical approach for transformers fleet management," *CIGRE*, 2014.
- [36] K. Ibrahim, R. M. Sharkawy, H. K. Temraz, and M. M. A. Salama, "Selection criteria for oil transformer measurements to calculate the Health Index," *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 23, pp. 3397-3404, 2016.

- [37] E. Duarte, D. Falla, J. Gavin, M. Lawrence, T. McGrail, D. Miller, et al., "A practical approach to condition and risk based power transformer asset replacement," in 2010 IEEE International Symposium on Electrical Insulation, 2010, pp. 1-4.
- [38] G. Tanasescu, B. Gorgan, S. Busoi, A. Badita, and P. V. Notingher, "Monitoring and diagnosis of electrical equipment using a web software. Health index and remaining lifetime estimation," in Proceedings of the International Conference - 2016 Conference on Diagnostics in Electrical Engineering, Diagnostika 2016, 2016.
- [39] H. Malik, A. Azeem, and R. K. Jarial, "Application research based on modern-technology for transformer Health Index estimation," in International Multi-Conference on Systems, Sygnals & Devices, 2012, pp. 1-7.
- [40] L. Cheim, P. Lorin, L. Lin, P. Patel, and J. Vines, "A Novel Dynamic Fleet Wide Condition Assessment Tool of Power Transformers," presented at the CIGRE SC A2 & C4 Joint Colloquium 2013, Zurich, CH, 2013.
- [41] A. D. Ashkezari, H. Ma, C. Ekanayake, and T. K. Saha, "Multivariate analysis for correlations among different transformer oil parameters to determine transformer health index," in IEEE Power and Energy Society General Meeting, 2012.
- [42] P. Picher, J.-F. Boudreau, A. Manga, C. Rajotte, C. Tardif, G. Bizier, et al., "Use of Health Index and Reliability Data for Transformer Condition Assessment and Fleet Ranking," presented at the CIGRE, Paris, 2014.
- [43] E. Abbasi and O. P. Malik, "Failure rate estimation of power transformers using inspection data," in 2016 International Conference on Probabilistic Methods Applied to Power Systems, PMAPS 2016, 2016.
- [44] J. H. Jürgensen, L. Nordström, and P. Hilber, "Individual failure rates for transformers within a population based on diagnostic measures," Electric Power Systems Research, vol. 141, pp. 354-362, 2016.
- [45] M. Vermeer, J. Wetzer, P. Van Der Wielen, E. De Haan, and E. De Meulemeester, "Asset-management decision-support modeling, using a health and risk model," in 2015 IEEE Eindhoven PowerTech, PowerTech 2015, 2015.
- [46] A. E. B. Abu-Elanien, M. M. A. Salama, and M. Ibrahim, "Calculation of a health index for oil-immersed transformers rated under 69 kV using fuzzy logic," IEEE Transactions on Power Delivery, vol. 27, pp. 2029-2036, 2012.
- [47] K. I. Mohamadeen, R. M. Sharkawy, and M. M. Salama, "Binary cat swarm optimization versus binary particle swarm optimization for transformer health index determination," in ICET 2014 - 2nd International Conference on Engineering and Technology, 2015.
- [48] M. Vermeer, S. J. D., and J. Wetzer, "The application of health- and risk indices as a decision-support tool for utilities," presented at the The Asset Management Conference 2016, 2016.

Annex D Principal INTERnational standards for insulating liquids

The principal international standards currently being used for mineral oils, natural ester liquids, synthetic ester liquids and silicones are listed following:

Mineral Oil:

NEW MINERAL OIL	IEC 60296:2012 [71]
	Fluids for electrotechnical applications – Unused mineral insulating oils for transformers and switchgear
USED MINERAL OIL	IEC 60422:2013 [42]
	Mineral insulating oils in electrical equipment – Supervision and maintenance guidance

Natural ester liquids:

NEW NATURAL ESTER	IEC 62770:2013 [75]
	Fluids for electrotechnical applications – Unused natural esters for transformers and similar electrical equipment
USED NATURAL ESTER	IEC 62975 (under development)
	Use and maintenance guidelines of natural esters insulating liquids in electrical equipment

Synthetic ester liquids:

NEW SYNTHETIC ESTER	IEC 61099:2010 [77]
	Specifications for Unused Synthetic Organic Esters for Electrical Purposes
USED SYNTHETIC ESTER	IEC 61203:1992 [78]
	Synthetic organic esters for electrical purposes – Guide for maintenance of transformer esters in equipment

For silicone liquid (polydimethylsiloxane oil):

NEW SILICONE LIQUID	IEC 60836:2015
	Specifications for unused silicone insulating liquids for electrotechnical purposes
USED SILICONE LIQUID	IEC 60944:1988
	Guide for the maintenance of silicone transformer liquids

Note that these standards may use different test methods and therefore state different acceptance criteria.

Annex E References

This Annex provides details of documents that are relevant to the work of the CIGRE Working Group that developed this Technical Brochure. Many of these documents are referenced in the text of the brochure, however some are detailed only as additional sources of information.

Note that some, but not all, of the documents referenced in Annex C are also included in this reference list.

- [1] ISO, "Condition monitoring and diagnostics of power transformers," 2018.
- [2] CIGRE, Technical Brochure 735 - Transformer Post-Mortem Analysis, 2018.
- [3] IEEE Std, "C57.91-2011 - Guide for Loading Mineral-Oil-Immersed Transformers," 2011.
- [4] IEC, 60076-7 Loading guide for mineral-oil-immersed power transformers, 2018.
- [5] CIGRE, "Technical Brochure 445 - Guide for Transformer Maintenance," 2011.
- [6] IEEE Std, "C57.140 - Guide for Evaluation and reconditioning of Liquid Immersed Power Transformers," 2017.
- [7] V. Peralta and M. Bouzeghoub, "A framework for analysis of data freshness," in *International Workshop on Information Quality in Information Systems*, 2004.
- [8] V. Catterson and B. Sparling, "BIG DATA ... where is all the data, and is it useful?," in *TechCon North America*, 2017.
- [9] CIGRE, "Technical Brochure 630 - Guide for Transformer Intelligent Condition Monitoring (TICM)".
- [10] IEEE Std, C57.104 Guide for the Interpretation of Gases Generated in Oil-Immersed Transformers, 2008.
- [11] M. Duval, "A Review of Faults Detectable by Gas-in-Oil Analysis in Transformers," *IEEE Electrical Insulation Magazine - Vol 18, No. 3*, May/June 2002.
- [12] M. Duval, "Use of Duval Pentagons and Triangles for Interpretation of DGA in electrical Equipment," in *Proceedings from Techcon North America*, Albuquerque, 2016.
- [13] CIGRE, "Technical Brochure 228 - Guide on Economics of Transformer Management".
- [14] IEC, "61649 Weibull Analysis".
- [15] CIGRE, "Technical Brochure 706 - Guidelines for the use of Statistics and Statistical Tools on Life Data".
- [16] CIGRE, "Technical Brochure 642 Transformer Reliability Survey," 2015.
- [17] P. Picher, J.-F. Boudreau, C. Rajotte, C. Tardif, G. Bizier, N. Di Gaetano, D. Garon, B. Girard, J.-F. Hamel and S. Proulx, "A2-101 - Use of Health Index and Reliability Data for Transformer Condition Assessment and Fleet Ranking," 2014.
- [18] G. Russel, O. Krause, T. Saha, J. Marks, D. Martin and A. Alibegovic-Memisevic, "On the development of power transformer failure models: an Australian case study," IEEE PES conference, 2017., in *IEEE PES conference*, 2017.
- [19] E. Abbasi and O. P. Malik, "Failure rate estimation of power transformers using inspection data," in *International Conference on Probabilistic Methods Applied to Power Systems, PMAPS*, Beijing, China.
- [20] A. Jahromi, R. Piercy, J. Service, S. Cress and W. FAN, "An approach to power transformer asset management using health index," *IEEE Electrical Insulation Magazine*, vol. 25, pp. 20-34, 2009.

- [21] C. T. B. 494, "Furanic Compounds for Diagnosis".
- [22] CIGRE, Technical Brochure 323 - Ageing of cellulose in mineral-oil insulated transformers, 2007.
- [23] IEC, 60450 Measurement of the average viscometric degree of polymerization of new and aged cellulosic electrically insulating materials, 2004.
- [24] IEC, 60076-2 Power transformers - Part 2: Temperature rise for liquid-immersed transformers, 2011.
- [25] IEEE Std, C57.154 Design, Testing, and Application of Liquid-Immersed Distribution, Power, and Regulating Transformers Using High-Temperature Insulation Systems and Operating at Elevated Temperatures, 2012.
- [26] IEC, 60076-14 Liquid-immersed power transformers using high-temperature insulation materials, 2013.
- [27] IEEE Std, "Transactions on Power Delivery," vol. 17, no. 4, p. 991 – 995, Oct. 2002.
- [28] IEC, 60599 Mineral oil-filled electrical equipment in service - Guidance on the interpretation of dissolved and free gases analysis, 2015.
- [29] C. 12-206, "New Mechanisms of Moderate Hydrogen Gas Generation," Paris Session, 1998.
- [30] IEEE Std, C57.155 IEEE Guide for Interpretation of Gases Generated in Natural Ester and Synthetic Ester-Immersed Transformers, 2014.
- [31] N. Dominelli, "The analysis of transformer oil for degradation products from overheated solid insulation, sec. 6-21,," in *58 th Annual international Conference of Doble Clients*, 1991.
- [32] J. Jalbert, R. Gilbert, P. Tétreault, B. Morin and D. Lessard-Dézie, "Identification of a chemical indicator of the rupture of 1,4- β -glycosidic bonds of cellulose in an oil-impregnated insulating paper system," *Cellulose*, Vol. 14, pp. 295-309," 2007.
- [33] CIGRE, "Technical Brochure 393 - Thermal Performance," 2010.
- [34] E. S. W Lampe, "The Oxygen-free transformer, Reducing ageing by continuous degassing.," CIGRE Paper 12.05, 1976.
- [35] CIGRE, "WG A2.30 TB 349 Moisture Equilibrium and Moisture Migration within Transformer Insulation Systems," 2008.
- [36] T. Oommen, "Moisture Equilibrium curves - use and misuse," in *Doble Client Conference*, Boston, 2003.
- [37] CIGRE, Technical Brochure 625 - Copper Sulphide Long Term Mitigation and Risk Assessment, 2015.
- [38] CIGRE, Technical Brochure 378 - Copper Sulphide in Transformer Insulation.
- [39] CIGRE, Technical Brochure 659 - Transformer Thermal Modelling.
- [40] IEC, 60270 High-voltage test techniques - Partial discharge measurements, 2000.
- [41] IEEE Std, C57.106 Guide for Acceptance and Maintenance of Insulating Oil in Equipment, 2006.
- [42] IEC, 60422 Mineral insulating oils in electrical equipment – Supervision and maintenance guidance, Edition 4.0 ed., 2013-01.
- [43] IEC, 60599 Mineral oil-impregnated electrical equipment in service – Guide to the interpretation of dissolved and free gases analysis, Edition 2.1 ed., 2007-05.
- [44] IEC, "60156 Insulating liquids - Determination of the breakdown voltage at power frequency - Test method," 1995.

- [45] IEC, 60247 Insulating liquids - Measurement of relative permittivity, dielectric dissipation factor (tan δ) and d.c. resistivity, 2004.
- [46] IEC, "60814 Insulating liquids - Oil-impregnated paper and pressboard - Determination of water by automatic coulometric Karl Fischer titration," 1997.
- [47] IEC, "60076-1 Power transformers - Part 1: General," 2011.
- [48] R. Willoughby, "Do Transformer Windings Need Re-clamping?," in *TechCon Asia Pacific*, Sydney, 2015.
- [49] CIGRE, "Technical Brochure 673 - Guide on Transformer Transportation".
- [50] Alstom Grid, Power Transformers - Expertise, vol. Volume 2, p. Section 1.6.
- [51] W. Bartley, "Analysis of Transformer Failures," in *79th International Conference of Doble Clients*, Boston, MA, 2012.
- [52] C. W. A2.43, "Transformer Bushing Reliability," 2018 .
- [53] C. G. Garton, "Dielectric Loss in Thin Films of Insulating Liquids," *Journal of the Institution of Electrical Engineers - Part III: Communication Engineering, including the Proceedings of the Wireless Section of the Institution*, vol. Vol. 88, pp. pp. 23-40, March 1941.
- [54] M. Krüger, A. Kraetke, M. Koch, K. Rethmeier, M. Muhr and S. Ch, "New Diagnostic Tools for High Voltage Bushings," in *Cigre VI Workspot, Foz do Iguaçu, Brazil*, 2010.
- [55] G. Kopp, "Frequency Dependent Dissipation Factor Measurements for the Assessment of High Voltage Bushings," in *TLM Hannover*, 2010.
- [56] IEC, 60137 - Insulated bushings for alternating voltages above 1000 V.
- [57] IEEE Std, "C57.19.01 - Standard Performance Characteristics and Dimensions for Outdoor Apparatus Bushings".
- [58] IEC, "Technical Report 61464, "Guide for the interpretation of dissolved gas analysis (DGA) in bushings where oil is the impregnating medium of the main insulation (generally paper)", ISBN: 2- 8318- 4459-2, 1998-08.
- [59] IEC, 60214-1 Tap-changer- Part 1: Performance requirements and test methods, 2014.
- [60] IEC, 60214-2 - Tap-changer- Part 2: Application guide, 2018.
- [61] CEATI, "T067300-3042 - End of Life Decision Support Model for Transformer Load Tapchangers," 2007.
- [62] L. Allard, M. Foata and e. al, "Paper A2-202 - Vibro-Acoustic diagnostic contributing to an optimized maintenance strategy," , paper A2-202., in *CIGRÉ 2010 Session*.
- [63] CIGRE, "Electra Reliability Survey," 1988.
- [64] CIGRE, "Technical Brochure 227 - Life Management Techniques for Power Transformers".
- [65] IEEE Std, C57.12.00-2015 - General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers, 2015.
- [66] UNITED STATES DEPARTMENT OF THE INTERIOR BUREAU OF RECLAMATION, TRANSFORMER DIAGNOSTICS, FACILITIES INSTRUCTIONS, STANDARDS, AND TECHNIQUES VOLUME 3-31 ed., 2003.
- [67] CIGRE, Technical Brochure 343 - Recommendations for Condition Monitoring and Condition Assessment Facilities for Transformers.
- [68] IEC, 60099-4 Surge arresters - Part 4: Metal-oxide surge arresters without gaps for a.c. systems, 2014.

- [69] IEC, 60099-5 Surge arresters - Part 5: Selection and application recommendations, 2013.
- [70] CIGRE, Technical Brochure 436 - Experiences in Service with New Insulating Liquids.
- [71] IEC, 60296 Fluids for electrotechnical applications — Unused mineral insulating oils for transformers and switchgear, Edition 4.0 ed., 2012.
- [72] ASTM, D3487-09 Standard Specification for Mineral Insulating Oil Used in Electrical Apparatus.
- [73] IEC, 62975 (not published yet) – “Use and maintenance guidelines of natural esters insulating liquids in electrical equipment”, TBC.
- [74] IEEE Std, C57.147 Guide for Acceptance and Maintenance of Natural Ester Fluids in Transformers, 2008.
- [75] IEC, 62770 Fluids for electrotechnical applications – Unused natural esters for transformers and similar electrical equipment, Edition 1.0 ed., 2013-11.
- [76] ASTM, D6871-03 Standard Specification for Natural (Vegetable Oil) Ester Fluids Used in Electrical Apparatus, 2008.
- [77] IEC, 61099 - Insulating liquids - Specifications for unused synthetic organic esters for electrical purposes, 2010.
- [78] IEC, 61203 Synthetic organic esters for electrical purposes - Guide for maintenance of transformer esters in equipment, Edition 1.0 ed., 1992-12 .
- [79] T. V. Oommen, “EPRI Report EL-6761 - Bubble Generation During Transformer Overload EPRI Report EL-6761,” 1990.
- [80] T. V. Oommen, “Further Experimentation on Bubble Generation During Transformer Overload - EPRI Report EL-7391,” March 1992.
- [81] T. V. Oommen, E. M. Petrie and S. R. Lindgren, “Bubble Generation in Transformer Windings Under Overload Conditions- Paper 8E, Doble Client Conference, March 1995,” March 1995.
- [82] R. Gilbert, J. Jalbert, S. Duchesne, P. Tétreault, B. Morin and Y. Denos, “Kinetics of the production of chain-end groups and methanol from the depolymerization of cellulose during the aging of paper/oil systems. Part 2: Thermally upgraded insulating paper,” Cellulose, Vol. 17, pp. 253-269,” 2010.
- [83] J. Jalbert, E. Rodriguez-Celis, S. Duchesne, B. Morin, M. Ryadi and R. Gilbert, “Kinetics of the production of chain-end groups and methanol from the depolymerization of cellulose during the ageing of paper/oil systems. Part 3: Extension of the study under temperature conditions over 120oC,” Cellulose, Vol. 22, pp. 829-845,” 2015.
- [84] R. Gilbert, J. Jalbert, P. Tétreault, B. Morin and Y. Denos, “Kinetics of the production of chain-end groups and methanol from the depolymerization of cellulose during the aging of paper/oil systems. Part 1: Standard wood Kraft insulation,” Cellulose, Vol. 16, pp. 327-338,” 2009.
- [85] A. Schaut, A. Stijn and S. Eeckhoudt, “Applicability of methanol as a new marker for paper degradation in power transformers,” IEEE Trans. Dielectr. Electr. Insul., Vol. 18, No. 2, pp. 533-540,” 2011.
- [86] Y. Denos, A. Tanguy, J. Jalbert, R. Gilbert, P. Gervais and P. Guuinic, “Ageing diagnosis by chemical markers influence of core-type and shell-type technology,” CIGRE, Paris, France,, 2010.
- [87] J. Jalbert, R. Gilbert, Y. Denos and P. Gervais, “Methanol: a novel approach to power transformer asset management, IEEE Trans. Power Delivery, Vol.27, No. 2, pp. 514-520,” 2012.
- [88] H. Bertrand, S. Besner, R. Beauchemin, L. Bolduc, R. Busatta, M. Chaaban, P. Couture, J.-B. Dastous, L. Dupont, M. Duval, M. Gauvin, J. Goulet, J. Jalbert, F. Léonard, A. Lessard, B. Noirhomme, P. Picher, S. Riendeau, F. Torriano, M. Foata and C. Rajotte, “Advanced

- monitoring and modelling methods for power transformer asset management,” *European Journal of Electrical Engineering Subject Area: Engineering: Electrical and Electronic Engineering Vol. 10, Issue 5-6*, pp. 591-620, 2010.
- [89] J. Jalbert and M.-C. Lessard, “Cellulose chemical markers in transformer oil insulation. Part 1: Temperature correction factors,” *IEEE Trans. Dielectr. Electr. Insul.*, vol. 20, no. 6, pp. 2287-2291, 2013.
 - [90] J. Jalbert and M.-C. Lessard, “Cellulose chemical markers relationship with insulating paper post-mortem investigations, IEEE Trans. Dielectr. Electr. Insul.,” *IEEE Trans. Dielectr. Electr. Insul.*, vol. 22, no. 6 December, 2015.
 - [91] J. Jalbert, S. Duchesne, E.-M. Rodriguez-Celis, P. Tétreault and P. Colin, “Robust and sensitive analysis of methanol and ethanol from cellulose degradation in mineral oils,” *Journal of Chromatography A*, vol. 1256, pp. 240-245, 2012.
 - [92] E. Rodriguez-Celis, J. Jalbert, S. Duchesne, B. Noirhomme and M.-C. Lessard, Chemical markers use for diagnosis and life estimation of power transformers, a preliminary study of their origins”, CIGRE Canada Conference, CIGRÉ-083, 2012.
 - [93] E. M. Rodriguez-Celis, S. Duchesne, J. Jalbert and M. Ryadi, “Understanding ethanol versus methanol formation from insulating paper in power transformers,” *Cellulose*, vol. 22, pp. 3225-3236, 2015.
 - [94] CIGRE, “Technical Brochure 576 - Guide for IT Strategies for Asset Management of Substations - General Principles”.
 - [95] CIGRE, “Technical Brochure 453 - Glossary of Terms Used in the Electric Industry”.
 - [96] CIGRE, “Technical Brochure 420 - Generic Guidelines for Life Time Condition Assessment of HV Assets and Related Knowledge Rules”.
 - [97] CIGRE, “Technical Brochure 248 - Guide on Economics of Transformer Management”.
 - [98] IEEE Std, “C57.19.100 - Guide for Application of Power Apparatus Bushings”.
 - [99] IEEE Std, “C57.19.00 - General Requirements and Test Procedure for Power Apparatus Bushings”.
 - [100] IEEE Std, “IEEE C57.152 - Guide for Diagnostic Field Testing of Fluid-Filled Power Transformers, Regulators, and Reactors”.
 - [101] IEEE Std, “C57.146 - Guide for the Interpretation of Gases Generated in Silicone-Immersed Transformers”.
 - [102] IEEE Std, “C57.149 - Guide for the Application and Interpretation of Frequency Response Analysis for Oil-Immersed Transformers”.
 - [103] IEEE Std, “C57.143 - Guide for Application for Monitoring Equipment to Liquid-Immersed Transformers and Components”.
 - [104] IEEE Std, “IEEE C62.22 - IEEE Guide for the Application of Metal-Oxide Surge Arresters for Alternating-Current Systems.”.
 - [105] CIGRE, Technical Brochure 443 - DGA in Non-Mineral Oils and Load Tap Changers and Improved DGA Diagnosis Criteria.
 - [106] CIGRE, Technical Brochure 413 - Insulating Oil Regeneration and Dehalogenation.
 - [107] IEEE Std, C57.637 Guide for the Reclamation of Insulating Oil and Criteria for its use, 2007.
 - [108] IEEE Std, C57.121 Guide for Acceptance and Maintenance of Less Flammable Hydrocarbon Fluid in Transformers, 1988.

- [109] IEC, 60567 Oil-filled electrical equipment - Sampling of gases and analysis of free and dissolved gases - Guidance, Edition 4.0 ed., 2011-10.
- [110] IEC, 60475 Method of sampling insulating liquids, Edition 2.0 ed., 2011-12.
- [111] M. Krüger, M. Koch, A. Kraetke, S. Rätzke and J. Schnitzler, "New Tools for the Assessment of High Voltage Bushings," High Voltage Symposium, Stuttgart University, Germany, 2012.
- [112] F. Huellmandel, M. Appold, R. K. A. Kuechler and J. Titze, "Condition Assessment of High Voltage Bushings by Means of Dielectric Diagnosis with PDC, XV International Symposium on High Voltage Engineering, University of Ljubljana, Elektroinštitut Milan," in *XV International Symposium on High Voltage Engineering*, University of Ljubljana, Elektroinštitut Milan Vidmar, Ljubljana, Slovenia, August 27-31, 2007.
- [113] ABB, "Bushings Diagnostic and Conditioning," *ABB Brochure 2750 515-142en*, 2014.
- [114] W. Zaengl, "Dielectric Spectroscopy in Time and Frequency Domain for HV Power Equipment, Part I: Theoretical Considerations, DEIS," *IEEE*, vol. 19, no. 5, September/October 2003.
- [115] W. Zaengl, "Applications of Dielectric Spectroscopy in Time and Frequency Domain for HV Power Equipment", Part II, DEIS," *IEEE*, vol. 19, no. 6, November/December 2003.
- [116] I. Atanasova-Hoehlein; et al., "Assessment of Oil Corrosivity," in *CIGRÉ A2-C4 Joint Colloquium*, Zurich, 2013.
- [117] ASTM D, "D 1275 - Test method for corrosive sulphur in electrical insulating oils," 2006.
- [118] IEC, "62535 - Insulating liquids – Test method for detection of potentially corrosive sulphur in used and unused insulating oil," 2008.
- [119] BS EN, 60814 - Insulating liquids. Oil-impregnated paper and pressboard. Determination of water by automatic coulometric Karl Fischer titration, 1998.
- [120] J. Fabre and A. Pichon, "Deteriorating Processes and Products of Paper in Oil. Application to Transformers," in *Application to Transformers CIGRE*, 1960.
- [121] CEATI, "Report No. T063700-3039 - Health and risk index tool for transformers and LTCs".
- [122] P. Jarman, R. Haoaton, L. Walker, Q. Zhong and T. Ish, "Transformer life prediction using data from units removed from service and thermal modelling," in 43rd International Conference on Large High Voltage Electric Systems," in *CIGRE 2010*, Paris, 2010.
- [123] I. Gutman, A. Holmberg, C. Ahlholm, U. Åkesson and L. Jonsson, "Long-term service experience and inspection results of HV equipment made of silicone rubber insulators," in *CIGRE Symposium, Auckland, New Zealand*, Auckland, New Zealand, 2013.
- [124] W. A2.30, "TB 349 Moisture equilibrium and Moisture Migration withing Transformer Insulation Systems," CIGRE, 2008.